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# V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning

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A major challenge for modern AI is to learn to understand the world and learn to act largely by observation (LeCun, 2022). This paper explores a self-supervised approach that combines internet-scale video data with a small amount of interaction data (robot trajectories), to develop models capable of understanding, predicting, and planning in the physical world. We first pre-train an action-free joint-embedding-predictive architecture, V-JEPA 2, on a video and image dataset comprising over 1 million hours of internet video. V-JEPA 2 achieves strong performance on motion understanding (77.3 top-1 accuracy on Something-Something v2) and state-of-the-art performance on human action anticipation (39.7 recall-at-5 on Epic-Kitchens-100) surpassing previous task-specific models. Additionally, after aligning V-JEPA 2 with a large language model, we demonstrate state-of-the-art performance on multiple video question-answering tasks at the 8 billion parameter scale (e.g., 84.0 on PerceptionTest, 76.9 on TempCompass). Finally, we show how self-supervised learning can be applied to robotic planning tasks by post-training a latent action-conditioned world model, V-JEPA 2-AC, using less than 62 hours of unlabeled robot videos from the Droid dataset. We deploy V-JEPA 2-AC zero-shot on Franka arms in two different labs and enable picking and placing of objects using planning with image goals. Notably, this is achieved without collecting any data from the robots in these environments, and without any task-specific training or reward. This work demonstrates how self-supervised learning from web-scale data and a small amount of robot interaction data can yield a world model capable of planning in the physical world.

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**Code:** https://github.com/facebookresearch/vjepa2
**Blogpost:** https://ai.meta.com/blog/v-jepa-2-world-model-benchmarks

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## 1 Introduction

Humans have the ability to adapt and generalize when taking on new tasks and operating in unfamiliar environments. Several cognitive learning theories suggest that humans learn an internal model of the world by integrating low-level sensory inputs to represent and predict future states (Craik, 1967; Rao and Ballard, 1999), and they further posit that this world model shapes our perception at any given moment, playing a crucial role in informing our understanding of reality (Friston, 2010; Clark, 2013; Nortmann et al., 2015). Moreover, our ability to predict the effects of our actions on future states of the world is also essential for goal-oriented planning (Sutton and Barto, 1981, 1998; Ha and Schmidhuber, 2018; Wolpert and Ghahramani, 2000). Building artificial agents that learn a world model from sensory data, such as video, could enable them to *understand* the physical world, *predict* future states, and effectively — like humans — *plan* in new situations, resulting in systems capable of tackling tasks that have not been encountered before.

Previous works have explored the development of predictive world models from interaction data consisting



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# *V-JEPA 2*：自监督视频模型实现理解、预测与规划

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现代人工智能面临的一项重大挑战是学会理解世界并主要通过学习观察来采取行动（LeCun，2022）。本文探索了一种自监督方法，将互联网规模的视频数据与少量交互数据（机器人轨迹）相结合，以开发能够在物理世界中理解、预测和规划的模型。我们首先在包含超过100万小时互联网视频的视频和图像数据集上预训练了一个无动作联合嵌入预测架构——V-JEPA 2。V-JEPA 2在运动理解方面取得了强劲的性能（在Something-Something v2上达到77.3的top-1准确率），并在人类动作预测方面达到了最先进的性能（在Epic-Kitchens-100上达到39.7的recall-at-5），超越了此前针对特定任务的模型。此外，在将V-JEPA 2与大型语言模型对齐后，我们在80亿参数规模上展示了在多个视频问答任务上的最先进性能（例如，PerceptionTest上84.0，TempCompass上76.9）。最后，我们展示了如何通过使用来自Droid数据集的不到62小时未标注机器人视频，对一个潜在动作条件世界模型V-JEPA 2-AC进行后训练，从而将自监督学习应用于机器人规划任务。我们在两个不同实验室的Franka机械臂上零样本部署了V-JEPA 2-AC，并利用图像目标规划实现了物体的抓取和放置。值得注意的是，这一成果无需从这些环境中的机器人收集任何数据，也无需任何特定任务的训练或奖励。这项工作展示了如何从网络规模数据和少量机器人交互数据进行自监督学习，从而获得能够在物理世界中进行规划的世界模型。

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https://github.com/facebookresearch/vjepa2 博客文章：
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## *1* 引言

人类在承担新任务和在陌生环境中操作时，具有适应和泛化的能力。若干认知学习理论表明，人类通过整合低层次的感觉输入来学习和表征、预测未来状态，从而学习一个关于世界的内部模型（Craik，1967；Rao and Ballard，1999），并且他们进一步假定，这一世界模型在任何给定时刻塑造着我们的感知，在塑造我们对现实的理解方面发挥着至关重要的作用（Friston，2010；Clark，2013；Nortmann et al.，2015）。此外，我们预测自身行为对世界未来状态影响的能力，对于目标导向的规划也是至关重要的（Sutton and Barto，1981，1998；Ha and Schmidhuber，2018；Wolpert and Ghahramani，2000）。构建能够从视频等感觉数据中学习世界模型的人工智能体，可以使它们理解物理世界、预测未来状态，并像人类一样在新情境中有效地进行规划，从而产生能够处理此前未曾遇到过的任务的系统。

先前的工作已经探索了从交互数据中构建预测性世界模型，这些数据由

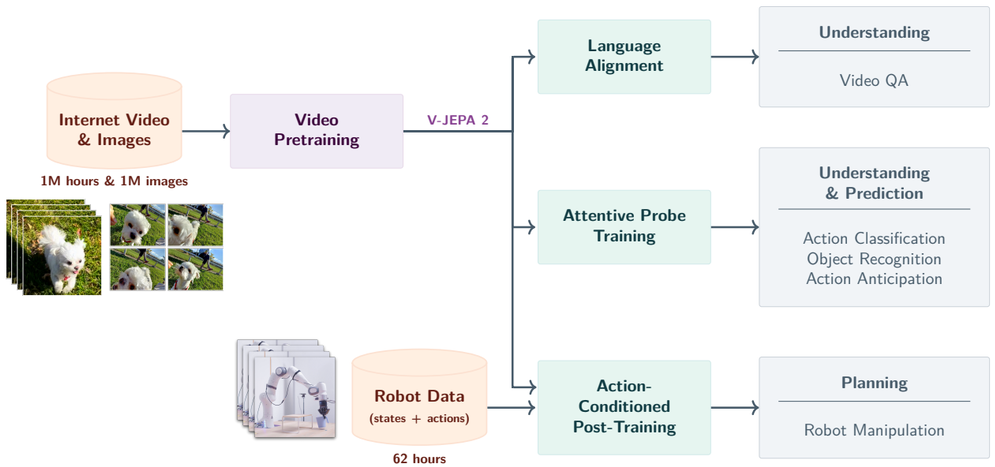



**Figure 1 V-JEPA 2 Overview.** Leveraging 1M hours of internet-scale video and 1M images, we pretrain the V-JEPA 2 video model using a visual mask denoising objective (Bardes et al., 2024; Assran et al., 2023), and leverage this model for downstream tasks such as action classification, object recognition, action anticipation, and Video Question Answering by aligning the model with an LLM backbone. After pretraining, we can also freeze the video encoder and train a new action-conditioned predictor with a small amount of robot interaction data on top of the learned representations, and leverage this action-conditioned model, V-JEPA 2-AC, for downstream robot manipulation tasks using planning within a model predictive control loop.

of state-action sequences, often also relying on explicit reward feedback from the environment to infer goals (Sutton and Barto, 1981; Fragkiadaki et al., 2015; Ha and Schmidhuber, 2018; Hafner et al., 2019b; Hansen et al., 2022). However, the limited availability of real-world interaction data constrains the scalability of these methods. To address this limitation, more recent works have leveraged both internet-scale video and interaction data towards training action-conditioned video generation models for robot control, but only demonstrate limited results in robot execution using model-based control (Hu et al., 2023; Yang et al., 2024b; Bruce et al., 2024; Agarwal et al., 2025). In particular, this line of research often emphasizes the evaluation of the faithfulness of the predictions and visual quality instead of planning capabilities, perhaps due to the computational cost of planning by generating video.

In this work, we build upon the self-supervised hypothesis as a means to learn world models that capture background knowledge of the world largely from observation. Specifically, we leverage the joint-embedding predictive architecture (JEPA) (LeCun, 2022), which learns by making predictions in a learned representation space. In contrast to approaches that focus on learning entirely from interaction data, self-supervised learning enables us to make use of internet-scale video — depicting sequences of states without direct observations of the actions — to learn to both represent video observations and learn a predictive model for world dynamics in this learned representation space. Furthermore, in contrast to approaches based on video generation, the JEPA approach focuses on learning representations for predictable aspects of a scene (e.g., the trajectory of an object in motion) while ignoring unpredictable details that generative objectives emphasize, since they make pixel-level predictions (e.g., the precise location of each blade of grass in a field, or each leaf on a tree). By scaling JEPA pretraining, we demonstrate that it yields video representations with state-of-the-art understanding and prediction capabilities, and that such representations can be leveraged as a basis for action-conditioned predictive models and enable zero-shot planning.

Our approach, V-JEPA 2, utilizes a stage-wise training procedure, beginning with action-free pre-training on internet-scale video, followed by post-training with a small amount of interaction data (see Figure 1). In the first stage, we use a mask-denoising feature prediction objective (Assran et al., 2023; Bardes et al., 2024), where the model predicts masked segments of a video in a learned representation space. We train



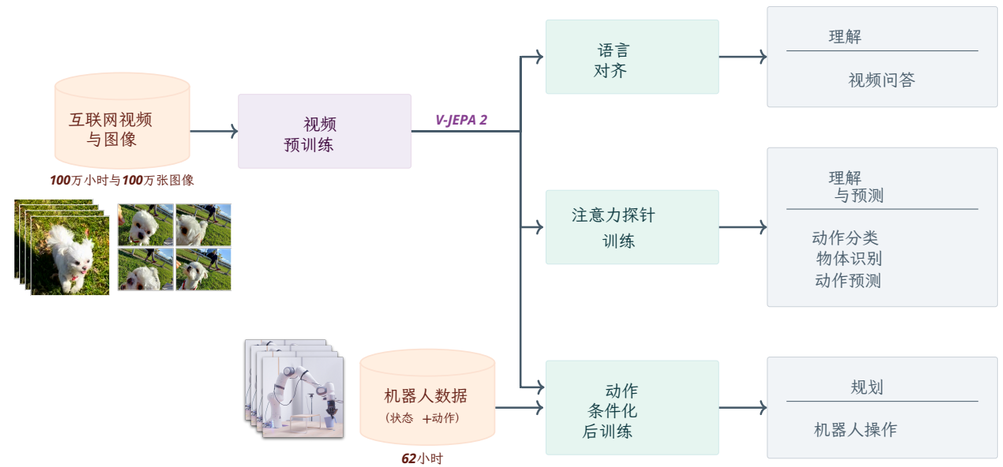



图1 V-JEPA 2概览。利用100万小时的互联网规模视频和100万张图像，我们使用视觉掩码去噪目标（Bardes et al., 2024; Assran et al., 2023）预训练V-JEPA 2视频模型，并通过将模型与LLM主干对齐，将该模型用于动作分类、物体识别、动作预测和视频问答等下游任务。预训练之后，我们还可以冻结视频编码器，并在学习到的表示之上，使用少量机器人交互数据训练一个新的动作条件预测器，并利用该动作条件模型V-JEPA 2-AC，在模型预测控制循环中通过规划来完成下游机器人操作任务。

然而，真实世界交互数据的有限可用性限制了这些方法的可扩展性。为解决这一局限，近期的工作利用互联网规模的视频和交互数据来训练动作条件视频生成模型以用于机器人控制，但在使用基于模型的控制进行机器人执行时仅展示了有限的结果（Hu et al., 2023; Yang et al., 2024b; Bruce et al., 2024; Agarwal et al., 2025）。特别是，这一研究方向往往侧重于评估预测的忠实性和视觉质量，而非规划能力，这或许是由于通过生成视频进行规划的计算成本所致。

在这项工作中，我们基于自监督假设，将其作为学习世界模型的一种手段，从而主要从观察中获取关于世界的背景知识。具体而言，我们利用联合嵌入预测架构（JEPA）（LeCun, 2022），它通过在学习到的表示空间中进行预测来学习。与专注于完全从交互数据中学习的方法相比，自监督学习使我们能够利用互联网规模的视频——这些视频描绘了状态序列，但没有对动作的直接观察——来学习表示视频观察，并在该学习到的表示空间中学习世界动态的预测模型。此外，与基于视频生成的方法相比，JEPA方法侧重于学习场景中可预测方面的表示（例如，运动物体的轨迹），同时忽略生成目标所强调的不可预测细节，因为生成目标进行像素级预测（例如，田野中每片草叶或树上每片叶子的精确位置）。通过扩展JEPA预训练，我们证明它可以产生具有最先进理解和预测能力的视频表示，并且这些表示可以作为动作条件预测模型的基础，并实现零样本规划。

我们的方法 V-JEPA 2 采用分阶段训练流程，首先在互联网规模视频上进行无动作预训练，随后使用少量交互数据进行后训练（见图 1）。在第一阶段，我们采用掩码去噪特征预测目标（Assran 等，2023；Bardes 等，2024），模型在学习到的表示空间中预测视频的被掩码片段。我们训练

the V-JEPA 2 encoder with up to 1 billion parameters and with more than 1 million hours of video. Our experiments confirm that scaling self-supervised video pretraining enhances the encoder's ability to achieve visual understanding, including broad motion and appearance recognition capabilities, through probe-based evaluations and by aligning the encoder with a language model for video question-answering (Krojer et al., 2024; Pătrăucean et al., 2023; Liu et al., 2024c; Cai et al., 2024; Shangguan et al., 2024).

Following pretraining on internet-scale video, we train an action-conditioned world model, V-JEPA 2-AC, on a small set of interaction data using the representations learned in the first stage. Our action-conditioned world model is a 300M-parameter transformer network employing a block-causal attention mechanism, which autoregressively predicts the representation of the next video frame conditioned on an action and previous states. With as little as 62 hours of unlabeled interaction data from the Droid dataset (Khazatsky et al., 2024), we demonstrate the feasibility of training a latent world model that, given sub-goals, can be leveraged to plan actions on a Franka robot arm and perform prehensile manipulation tasks from a monocular RGB camera zero-shot in a new environment.

To summarize, we show that joint-embedding predictive architectures learning from videos can be used to build a world model that enables *understanding* the physical world, *predicting* future states, and effectively *planning* in new situations; this is achieved by leveraging internet-scale video and a small amount of interaction data. Specifically:

- *Understanding — Probe-based Classification:* Scaling self-supervised video pretraining results in video representations applicable to many tasks. V-JEPA 2 excels at encoding fine-grained motion information, achieving strong performance on tasks requiring motion understanding, such as Something-Something v2, with 77.3 top-1 accuracy using an attentive probe.
- *Understanding — Video Question-Answering:* V-JEPA 2 encoder can be used to train a multi-modal large language model, to tackle video-question answering tasks. We observe state-of-the-art performance on 8B language model class on multiple benchmarks that require physical world understanding and temporal reasoning, such as MVP (44.5 paired accuracy), PerceptionTest (84.0 test set accuracy), TempCompass (76.9 multi-choice accuracy), TemporalBench (36.7 multi-binary short-QA accuracy) and TOMATO (40.3 accuracy). In particular, we show that a video encoder pre-trained *without* language supervision can be aligned with a language model and achieve state-of-the-art performance, contrary to conventional wisdom (Yuan et al., 2025; Wang et al., 2024b).
- *Prediction:* Large-scale self-supervised video pretraining enhances prediction capabilities. V-JEPA 2 achieves state-of-the-art performance on the Epic-Kitchens-100 human-action anticipation task using an attentive probe, with 39.7 recall-at-5, which is a 44% relative improvement over the previous best model.
- *Planning:* We demonstrate that V-JEPA 2-AC, obtained by post-training V-JEPA 2 with only 62 hours of unlabeled robot manipulation data from the popular Droid dataset, can be deployed in new environments to solve prehensile manipulation tasks using planning with given subgoals. Without training on any additional data from robots in our labs, and without any task-specific training or reward, the model successfully handles prehensile manipulation tasks, such as Grasp and Pick-and-Place with novel objects and in new environments.

The remainder of this paper is organized as follows. Section 2 describes the V-JEPA 2 pretraining procedure, including the key ingredients enabling scaling beyond the original V-JEPA recipe of Bardes et al. (2024). Section 3 then introduces our approach to training a task-agnostic action-conditioned world model, V-JEPA 2-AC, leveraging the pretrained V-JEPA 2 model. Section 4 demonstrates using V-JEPA 2-AC for robot control via model-based planning. Because V-JEPA 2-AC models world dynamics in a learned representation space, its capabilities fundamentally depend on the information captured in the V-JEPA 2 representation space, and so we further explore the performance of V-JEPA 2 for video understanding in Section 5 and prediction tasks in Section 6. Finally, in Section 7 we show that V-JEPA 2 can be aligned with a language model for video question answering. Section 8 discusses related work, and we conclude in Section 9.



V-JEPA 2 编码器，参数量高达 10 亿，并使用超过 100 万小时的视频进行训练。我们的实验证实，扩展自监督视频预训练可增强编码器实现视觉理解的能力，包括广泛的运动与外观识别能力，这通过基于探针的评估以及将编码器与语言模型对齐以进行视频问答来验证（Krojer 等，2024；Pătrăucean 等，2023；Liu 等，2024c；Cai 等，2024；Shangguan 等，2024）。

在互联网规模视频上完成预训练后，我们利用第一阶段学到的表征，在一小组交互数据上训练了一个动作条件世界模型 V-JEPA 2-AC。我们的动作条件世界模型是一个具有 3 亿参数的 transformer 网络，采用块因果注意力机制，以动作和先前状态为条件，自回归地预测下一视频帧的表征。仅使用来自 Droid 数据集（Khazatsky 等，2024）的 62 小时无标签交互数据，我们证明了训练潜在世界模型的可行性——该模型在给定子目标的情况下，可用于在 Franka 机械臂上规划动作，并在新环境中仅凭单目 RGB 相机零样本执行抓取式操作任务。

总之，我们证明，从视频中学习的联合嵌入预测架构可用于构建世界模型，使其能够理解物理世界、预测未来状态，并在新情境中有效规划；这是通过利用互联网规模视频和少量交互数据实现的。具体而言：

- 理解——基于探针的分类：扩展自监督视频预训练规模，可得到适用于众多任务的视频表征。V-JEPA 2 擅长编码细粒度运动信息，在需要运动理解的任务上表现强劲，例如 Something-Something v2，使用注意力探针达到 77.3 的 top-1 准确率。
- 理解——视频问答：V-JEPA 2 编码器可用于训练多模态大语言模型，以处理视频问答任务。我们在多个需要物理世界理解和时序推理的基准上，于 8B 语言模型级别观察到最先进的性能，例如 MVP（44.5 配对准确率）、PerceptionTest（84.0 测试集准确率）、TempCompass（76.9 多选准确率）、TemporalBench（36.7 多二分类短问答准确率）和 TOMATO（40.3 准确率）。特别地，我们表明，一个未经语言监督预训练的视频编码器可以与语言模型对齐并达到最先进的性能，这与传统观点相反（Yuan et al., 2025; Wang et al., 2024b）。
- 预测：大规模自监督视频预训练可增强预测能力。V-JEPA 2 使用注意力探针，在 Epic-Kitchens-100 人类动作预判任务上取得了最先进的性能，recall-at-5 达到 39.7，相比此前最佳模型相对提升了 44%。
- 规划：我们证明，V-JEPA 2-AC 仅使用来自常用 Droid 数据集的 62 小时无标注机器人操作数据进行后训练获得，便可部署到新环境中，利用给定子目标进行规划以求解抓取式操作任务。无需使用我们实验室机器人的任何额外数据进行训练，也无需任何任务特定训练或奖励，该模型便能在新环境中成功处理抓取式操作任务，例如对新物体的抓取和拾放。

本文的其余部分组织如下。第2节描述V-JEPA 2的预训练流程，包括使其能够超越Bardes等人（2024）原始V-JEPA方案进行扩展的关键要素。第3节介绍我们训练任务无关的动作条件世界模型V-JEPA 2-AC的方法，该方法利用预训练的V-JEPA 2模型。第4节展示通过基于模型的规划使用V-JEPA 2-AC进行机器人控制。由于V-JEPA 2-AC在学到的表示空间中建模世界动态，其能力从根本上取决于V-JEPA 2表示空间中所捕获的信息，因此我们进一步在第5节探讨V-JEPA 2在视频理解方面的性能，并在第6节探讨其在预测任务中的表现。最后，在第7节中我们展示V-JEPA 2可以与语言模型对齐以进行视频问答。第8节讨论相关工作，第9节进行总结。

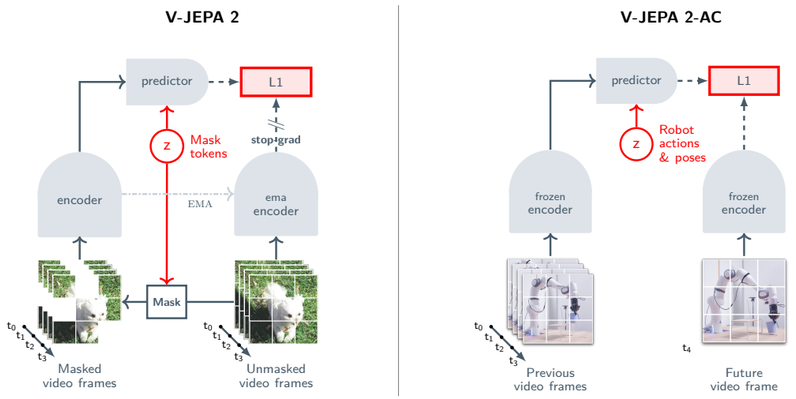



**Figure 2 Multistage training. (Left)** We first pretrain the V-JEPA 2 video encoder on internet-scale image and video data using a visual mask denoising objective (Bardes et al., 2024; Assran et al., 2023). A video clip is patchified into a sequence of tokens and a mask is applied by dropping a subset of the tokens. The encoder then processes the masked video sequence and outputs an embedding vector for each input token. Next, the outputs of the encoder are concatenated with a set of learnable mask tokens that specify the position of the masked patches, and subsequently processed by the predictor. The outputs of the predictor are then regressed to the prediction targets using an L1 loss. The prediction targets are computed by an ema-encoder, the weights of which are defined as an exponential moving average of the encoder weights. **(Right)** After pretraining, we freeze the video encoder and learn a new action-conditioned predictor, V-JEPA 2-AC, on top of the learned representation. We leverage an autoregressive feature prediction objective that involves predicting the representations of future video frames conditioned on past video frames, actions, and end-effector states. Our action-conditioned predictor uses a block-causal attention pattern such that each patch feature at a given time step can attend to the patch features, actions, and end-effector states from current and previous time steps.

## 2 V-JEPA 2: Scaling Self-Supervised Video Pretraining

We pretrain V-JEPA 2 on a visual dataset that includes over 1 million hours of video. The self-supervised training task is based on mask denoising in representation space and builds upon the V-JEPA framework (Bardes et al., 2024). In this paper, we extend the V-JEPA framework by exploring larger-scale models, increasing the size of the pretraining data, and introducing a spatial and temporal progressive resolution training strategy that enables us to efficiently pretrain models beyond short 16-frame video clips.

### 2.1 Methodology

**Mask-Denoising in Representation Space.** The V-JEPA objective aims to predict the learned representation of a video $y$ from a view $x$ of that video that has been masked, i.e., from which patches have been randomly dropped (Figure 2, left). The task meta-architecture consists of an encoder, $E_\theta(\cdot)$, which extracts video representations, and a predictor, $P_\phi(\cdot)$, which predicts the representation of masked video parts. The encoder and predictor are trained simultaneously using the objective,

$$\text{minimize}_{\theta,\phi,\Delta_y} \quad \|P_\phi(\Delta_y, E_\theta(x)) - \text{sg}(E_{\bar{\theta}}(y))\|_1, \tag{1}$$

where $\Delta_y$ is a learnable mask token that indicates the locations of the dropped patches. The loss uses a stop-gradient operation, $\text{sg}(\cdot)$, and an exponential moving average, $\bar{\theta}$, of the weights $\theta$ of the encoder network to prevent representation collapse. The loss is applied only to the predictions of the masked patches.

**Architecture.** The encoder, $E_\theta(\cdot)$, and predictor, $P_\phi(\cdot)$, are each parameterized as a vision transformer (Dosovitskiy et al., 2020) (or ViT). To encode relative position information in the vision transformer, we leverage RoPE (Rotary Position Embedding) instead of the absolute sincos position embedding used in Bardes et al.



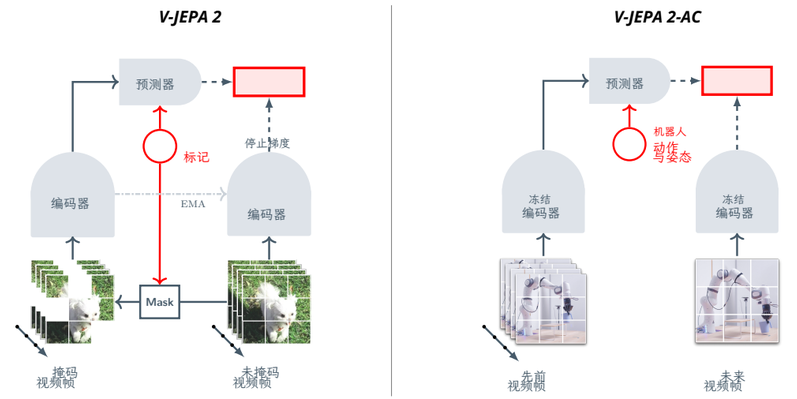



图2 多阶段训练。（左）我们首先使用视觉掩码去噪目标（Bardes et al., 2024; Assran et al., 2023）在互联网规模的图像和视频数据上预训练V-JEPA 2视频编码器。视频片段被划分为一系列token，并通过丢弃一部分token来施加掩码。编码器随后处理掩码后的视频序列，并为每个输入token输出一个嵌入向量。接下来，编码器的输出与一组指定掩码块位置的可学习掩码token拼接，随后由预测器处理。预测器的输出随后使用L1损失回归到预测目标。预测目标由ema编码器计算，其权重定义为编码器权重的指数移动平均值。（右）预训练之后，我们冻结视频编码器，并在学习到的表示之上学习一个新的动作条件预测器V-JEPA 2-AC。我们利用自回归特征预测目标，该目标涉及以过去的视频帧、动作和末端执行器状态为条件，预测未来视频帧的表示。我们的动作条件预测器使用块因果注意力模式，使得给定时间步的每个块特征可以关注当前及先前时间步的块特征、动作和末端执行器状态。

## *2 V-JEPA 2*：扩展自监督视频预训练

我们在一个包含超过100万小时视频的视觉数据集上预训练V-JEPA 2。自监督训练任务基于表示空间中的掩码去噪，并建立在V-JEPA框架（Bardes等，2024）之上。在本文中，我们扩展了V-JEPA框架，探索了更大规模的模型，增加了预训练数据的规模，并引入了空间和时间渐进分辨率训练策略，使我们能够高效地预训练超出短16帧视频片段的模型。

### *2.1* 方法

表示空间中的掩码去噪。V-JEPA 的目标是从视频的一个掩码视图 $x$ 中预测其学习到的表示 $y$，即该视图中随机丢弃了部分图像块（图 2 左侧）。任务元架构由一个编码器 $E_\theta(\cdot)$ 和一个预测器 $P_\phi(\cdot)$ 组成，前者提取视频表示，后者预测被掩码视频部分的表示。编码器和预测器使用以下目标函数同时训练，

$$\text{minimize}_{\theta,\phi,\Delta_y} \quad \|P_\phi(\Delta_y, E_\theta(x)) - \text{sg}(E_{\bar{\theta}}(y))\|_1, \tag{1}$$

其中 $\Delta_y$ 是一个可学习的掩码标记，用于指示被丢弃图像块的位置。该损失使用停止梯度操作 $\text{sg}(\cdot)$ 以及编码器网络权重 $\theta$ 的指数移动平均 $\bar{\theta}$ 来防止表示坍缩。该损失仅应用于被掩码图像块的预测。

架构。编码器 $E_\theta(\cdot)$ 和预测器 $P_\phi(\cdot)$ 分别参数化为视觉变换器（Dosovitskiy 等人，2020）（即 ViT）。为了在视觉变换器中编码相对位置信息，我们采用 RoPE（旋转位置嵌入），而非 Bardes 等人所使用的绝对 sincos 位置嵌入。

(2024). We use a 3D extension of traditional 1D-RoPE (Su et al., 2024) by partitioning the feature dimension into three approximately equal segments (for the temporal, height, and width axes) and applying the 1D rotations separately to the segment for each axis. We found that using 3D-RoPE instead of absolute sincos position embeddings (Vaswani et al., 2017) helps stabilize training for the largest models. To process a video with our transformer encoder, we first patchify it as a sequence of tubelets of size $2 \times 16 \times 16$ ($T \times H \times W$) and employ the same multiblock masking strategy as in Bardes et al. (2024).

**Key Scaling Ingredients.** In this section we introduce and study four additional key ingredients which enable scaling the V-JEPA pre-training principle to obtain our V-JEPA 2 model.

1. *Data scaling:* We increase the dataset size from 2 million to 22 million videos by leveraging and curating additional data sources.
2. *Model scaling:* We scale the encoder architecture from 300 million to over 1 billion parameters, going from a ViT-L to a ViT-g (Zhai et al., 2022).
3. *Longer training:* Adopting a warmup-constant-decay learning rate schedule simplifies hyperparameter tuning and enables us to extend training from 90 thousand up to 252 thousand iterations, effectively leveraging the additional data.
4. *Higher resolution:* We leverage the warmup-constant-decay schedule to efficiently scale to higher resolution video and longer video clips by training on shorter, lower-resolution clips during the warmup and constant phases, and then increasing resolution and/or clip-length during the final decay phase.

The remainder of this section describes each of these ingredients in further detail and also quantifies the impact of each ingredient using the evaluation protocol described next.

**Evaluation Protocol.** Our goal with model pretraining is to infuse general visual understanding into our encoder. We therefore evaluate our model and data design choices by assessing the quality of the model's learned representation on a set of six motion and appearance classification tasks: Something-Something v2 (Goyal et al., 2017), Diving-48 (Li et al., 2018), Jester (Materzynska et al., 2019), Kinetics (Kay et al., 2017), COIN (Tang et al., 2019), and ImageNet (Deng et al., 2009). We use a frozen evaluation protocol: we freeze the encoder weights and train a task-specific 4-layers attentive probe on its representation to output a predicted class. In this section, we focus mainly on the average accuracy across the six understanding tasks. Refer to Section 5 for additional details about the tasks, evaluation protocol, and results.

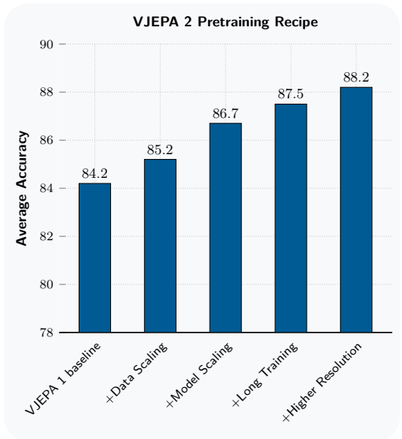



**Figure 3 Scaling Ingredients.** The effects of scaling interventions on average accuracy across 6 image and video classification tasks (SSv2, Diving-48, Jester, Kinetics, COIN, ImageNet) using a ViT-L/16 model as baseline.

## 2.2 Scaling Self-Supervised Video Learning

We first present a summary of the key findings of our scaling analysis, where we investigate the impact of the four key ingredients on downstream task average performance. Figure 3 illustrates the effects of these scaling interventions on average accuracy across 6 classification tasks, using a ViT-L/16 model pretrained on 2 million videos with the V-JEPA objective as our baseline. Increasing the dataset from 2 million to 22 million videos (VM22M) yields a 1.0-point improvement. Scaling the model from 300 million to 1 billion parameters (ViT-g/16) provides an additional 1.5-point gain. Extending training from 90K to 252K iterations contributes another 0.8-point improvement. Finally, enhancing both spatial resolution ($256 \rightarrow 384$) and temporal duration ($16 \rightarrow 64$ frames), during both pretraining and evaluation, boosts performance to 88.2%, representing a



（2024）。我们使用传统一维 RoPE（Su 等人，2024）的三维扩展，将特征维度划分为三个大致相等的段（分别对应时间、高度和宽度轴），并对每个轴的段分别应用一维旋转。我们发现，使用三维 RoPE 而非绝对 sincos 位置嵌入（Vaswani 等人，2017）有助于稳定最大模型的训练。为了使用我们的变换器编码器处理视频，我们首先将其分块为大小为 $2 \times 16 \times 16$ ($T \times H \times W$) 的管状体序列，并采用与 Bardes 等人（2024）相同的多块掩码策略。

关键扩展要素。在本节中，我们介绍并研究四项额外的关键要素，这些要素使得能够扩展 V-JEPA 预训练原则，从而获得我们的 V-JEPA 2 模型。

1. 数据扩展：我们通过利用和整理额外的数据源，将数据集规模从 200 万个视频增加到 2200 万个视频。
2. 模型扩展：我们将编码器架构从 3 亿参数扩展到超过 10 亿参数，从 ViT-L 升级到 ViT-g（Zhai 等，2022）。3. 更长训练：采用预热-恒定-衰减学习率调度简化了超参数调优，并使我们能够将训练从 9 万次迭代扩展到 25.2 万次迭代，有效利用额外数据。4. 更高分辨率：我们利用预热-恒定-衰减调度，通过在预热和恒定阶段训练更短、更低分辨率的视频片段，然后在最终衰减阶段增加分辨率和/或片段长度，从而高效扩展到更高分辨率的视频和更长的视频片段。

本节剩余部分将详细介绍上述每个组成部分，并使用接下来描述的评估协议量化每个组成部分的影响。

评估协议。我们进行模型预训练的目标是为编码器注入通用的视觉理解能力。因此，我们通过在六项运动和外观分类任务上评估模型所学表示的质量，来检验我们的模型和数据设计选择：Something-Something v2（Goyal等，2017）、Diving-48（Li等，2018）、Jester（Materzynska等，2019）、Kinetics（Kay等，2017）、COIN（Tang等，2019）和ImageNet（Deng等，2009）。我们采用冻结评估协议：冻结编码器权重，在其表示上训练一个任务特定的4层注意力探针来输出预测类别。本节主要关注六项理解任务的平均准确率。有关任务、评估协议和结果的更多细节，请参阅第5节。

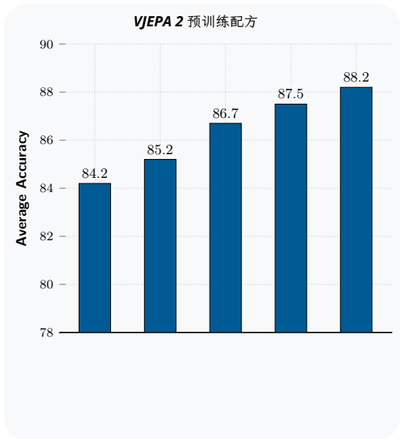



图3 扩展要素。以ViT-L/16模型为基线，扩展干预措施对6个图像和视频分类任务（SSv2、Diving-48、Jester、Kinetics、COIN、ImageNet）平均准确率的影响。

## 2.2 扩展自监督视频学习

我们首先概述扩展性分析的主要发现，其中我们研究了四个关键因素对下游任务平均性能的影响。图3展示了这些扩展干预对6个分类任务平均准确率的影响，基线是使用V-JEPA目标在200万视频上预训练的ViT-L/16模型。将数据集从200万扩展到2200万视频（VM22M）带来1.0个百分点的提升。将模型从3亿参数扩展到10亿参数

(ViT-g/16) 额外带来了 1.5 个百分点的提升。将训练从 90K 次迭代扩展到 252K 次迭代又贡献了 0.8 个百分点的改进。最后，在预训练和评估期间同时提升空间分辨率（$256 \rightarrow 384$）和时间时长（$16 \rightarrow 64$ 帧），将性能提升至 88.2%，这代表着相较于

**Table 1 VideoMix22M (VM22M) Pretraining Dataset.** To build our observation pretraining dataset, we combined four different video sources and one image dataset. We use a source-specific sampling probability during training and apply retrieval-based curation on YT1B to reduce noisy content (e.g., cartoon- or clipart-style).

| Source | Samples | Type | Total Hours | Apply Curation | Weight |
|---|---|---|---|---|---|
| SSv2 (Goyal et al., 2017) | 168K | EgoVideo | 168 | No | 0.056 |
| Kinetics (Carreira et al., 2019) | 733K | ExoVideo | 614 | No | 0.188 |
| Howto100M (Miech et al., 2019) | 1.1M | ExoVideo | 134K | No | 0.318 |
| YT-Temporal-1B (Zellers et al., 2022) | 19M | ExoVideo | 1.6M | Yes | 0.188 |
| ImageNet (Deng et al., 2009) | 1M | Images | n/a | No | 0.250 |

cumulative 4.0-point improvement over the ViT-L/16 baseline. Each individual change provides a positive impact, confirming the potential of scaling in video self-supervised learning (SSL).

## 2.3 Pretraining Dataset

Next, we describe the sources of videos and images that make up our pretraining dataset, and our approach to curating the dataset.

**Scaling Dataset Size.** We construct a large-scale video dataset by combining publicly available data sources. Using publicly-available sources in this work enables other researchers to reproduce these results. The overall dataset includes ego-centric videos from the Something-Something v2 dataset (SSv2) introduced in Goyal et al. (2017), exo-centric action videos from the Kinetics 400, 600, and 700 datasets (Kay et al., 2017; Carreira et al., 2018, 2019), YouTube tutorial videos from HowTo100M (Miech et al., 2019), and general YouTube videos from YT-Temporal-1B (Zellers et al., 2022), which we refer to as YT1B. We also include images from the ImageNet dataset (Deng et al., 2009) to increase the visual coverage of the pretraining data. To enable joint image and video pretraining, we duplicate an image temporally and treat it as a 16-frame video where all frames are identical. During training, we sample from each data source with a weighting coefficient that we determined empirically via manual tuning. The resulting dataset, which we refer to as VideoMix22M (or VM22M), consists of 22 million samples. Table 1 lists these data sources and their weights.

Figure 4 (Left) compares the performance of a ViT-L/16 pretrained on VM22M with a similar model trained on the smaller (2 million) VideoMix2M dataset from Bardes et al. (2024). Training on VM22M leads to a +1 point improvement on average performance on visual understanding tasks, compared to VM2M. Performance improvement is more prominent on appearance-based tasks such as Kinetics-400, COIN, and ImageNet, showing the importance of increasing visual coverage for those tasks.

**Data Curation.** YT1B is a large video dataset, consisting of 1.4 million video-hours, with no curation and minimal filtering compared to smaller video datasets (like Kinetics and Something-Something v2). Because uncurated and unbalanced data can hinder model performance (Assran et al., 2022; Oquab et al., 2023), we filter YT1B by adapting an existing retrieval-based curation pipeline to handle videos. Specifically, we extract scenes from YT1B videos, compute an embedding vector for each scene, and then use a cluster-based retrieval process (Oquab et al., 2023) to select video scenes according to a target distribution, which is composed of the Kinetics, Something-Something v2, COIN and EpicKitchen training datasets. We describe the details of the dataset construction procedure in Appendix A.2. Similar to Oquab et al. (2023), we ensure that none of the videos from the target validation sets are contained in the initial, uncurated data pool.

In Figure 4 (Right), we compare the average performance on visual understanding evaluations between a ViT-L model pretrained on uncurated YT-1B data and a comparable model trained on our Curated-YT-1B dataset. Training with the curated dataset yields a +1.4 point average performance improvement over the uncurated baseline. Notably, the Curated-YT-1B-trained model achieves competitive performance relative to the full VM22M dataset at the ViT-L scale. However, larger-scale models benefit more from VM22M training (see Appendix A.2), suggesting that combining Curated-YT-1B with other data sources enhances scalability.



表1 VideoMix22M（VM22M）预训练数据集。为构建我们的观测预训练数据集，我们整合了四种不同的视频来源和一个图像数据集。训练过程中，我们采用针对各来源的采样概率，并对YT1B应用基于检索的筛选，以减少噪声内容（如卡通或剪贴画风格）。

| 来源 | 样本 | Type | 总时长 | 是否应用筛选 | 权重 |
|---|---|---|---|---|---|
| SSv2 (Goyal 等，2017) | 168K | EgoVideo | 168 | No | 0.056 |
| Kinetics (Carreira 等，2019) | 733K | ExoVideo | 614 | No | 0.188 |
| Howto100M (Miech 等，2019) | 1.1M | ExoVideo | 134K | No | 0.318 |
| YT-Temporal-1B（Zellers 等，2022） | 19M | ExoVideo | 1.6M | Yes | 0.188 |
| ImageNet（Deng 等，2009） | 1M | 图像 | n/a | No | 0.250 |

ViT-L/16 基线累计 4.0 个百分点的提升。每一项单独的改动都带来了积极影响，证实了视频自监督学习（SSL）中扩展的潜力。

## *2.3* 预训练数据集

接下来，我们描述 构成我们预训练数据集的视频和图像来源，以及我们的方法来整理数据 et.

扩展数据集规模。我们通过组合公开可用的数据源来构建大规模视频数据集。在本工作中使用公开可用的数据源使其他研究人员能够复现这些结果。整体数据集包括来自 Goyal 等人（2017）提出的 Something-Something v2 数据集（SSv2）的自我中心视频、来自 Kinetics 400、600 和 700 数据集（Kay 等人，2017；Carreira 等人，2018，2019）的外中心动作视频、来自 HowTo100M（Miech 等人，2019）的 YouTube 教程视频，以及来自 YT-Temporal-1B（Zellers 等人，2022）的通用 YouTube 视频（我们称之为 YT1B）。我们还纳入了 ImageNet 数据集（Deng 等人，2009）中的图像，以增加预训练数据的视觉覆盖范围。为了实现图像和视频的联合预训练，我们将一张图像在时间上复制并视为一个 16 帧视频，其中所有帧都相同。在训练期间，我们以通过手动调优经验确定的加权系数从每个数据源中采样。最终得到的数据集（我们称之为 VideoMix22M，或 VM22M）包含 2200 万个样本。表 1 列出了这些数据源及其权重。

图4（左）比较了在VM22M上预训练的ViT-L/16与在Bardes等人（2024）较小（200万）的VideoMix2M数据集上训练的类似模型的性能。与VM2M相比，在VM22M上训练使视觉理解任务的平均性能提升了 +1个百分点。性能提升在基于外观的任务（如Kinetics-400、COIN和ImageNet）上更为显著，表明增加视觉覆盖对这些任务的重要性。

数据整理。YT1B是一个大型视频数据集，包含140万视频小时，与较小的视频数据集（如Kinetics和Something-Something v2）相比，没有经过整理且过滤极少。由于未整理和不平衡的数据可能会阻碍模型性能（Assran等人，2022；Oquab等人，2023），我们通过调整现有的基于检索的整理流程来处理视频，从而对YT1B进行过滤。具体而言，我们从YT1B视频中提取场景，为每个场景计算嵌入向量，然后使用基于聚类的检索过程（Oquab等人，2023）根据目标分布选择视频场景，该目标分布由Kinetics、Something-Something v2、COIN和EpicKitchen训练数据集组成。我们在附录A.2中描述了数据集构建过程的细节。与Oquab等人（2023）类似，我们确保目标验证集中的任何视频都不包含在初始的未整理数据池中。

在图4（右）中，我们比较了在未经整理的YT-1B数据上预训练的ViT-L模型与在我们的Curated-YT-1B数据集上训练的同类模型在视觉理解评估上的平均性能。使用整理后的数据集进行训练，相比未整理的基线模型平均性能提升了 +1.4 个百分点。值得注意的是，在ViT-L规模下，使用Curated-YT-1B训练的模型相对于完整的VM22M数据集取得了有竞争力的性能。然而，更大规模的模型从VM22M训练中获益更多（见附录A.2），这表明将Curated-YT-1B与其他数据源结合可以增强可扩展性。

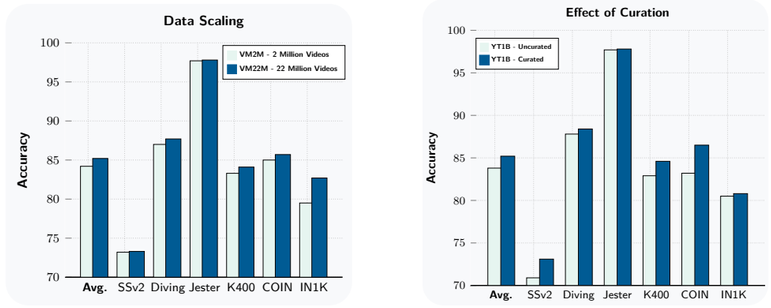






**Figure 4 Data Scaling & Curation**. We train and compare models on different data-mixes. Models are ViT-L/16 trained for 90K iterations using a cosine learning schedule following Bardes et al. (2024). **(Left)** We compare the performance of a ViT-L/16 model pretrained on the VM2M dataset and our VM22M dataset. Training on the VM22M dataset leads to a +1 point improvement in average performance. Performance improvement is more pronounced on appearance-based tasks such as Kinetics-400, COIN, and ImageNet **(Right)** We compare the performance of a ViT-L/16 model pretrained on YT1B and a model pretrained on our Curated-YT1B dataset, which leverages our cluster-based curation. Training on the curated dataset leads to a +1.4 point improvement on average performances, showing the effectiveness of data-curation.

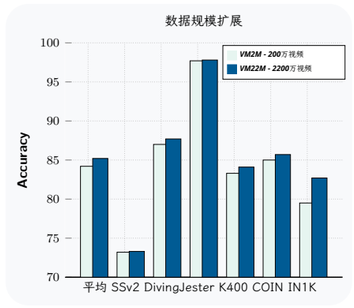
## 2.4 Pretraining Recipe

**Scaling Model Size.** To explore the scaling behavior of our model, we trained a family of encoder models with parameter counts ranging from 300 million (ViT-L) to 1 billion (ViT-g) parameters. All encoder architecture details are provided in Table 12 in the appendix. Note that each encoder uses the same predictor architecture, similar to a ViT-small. We report the average performance of these encoders on visual understanding tasks in Figure 5 (Left). Scaling the model size from 300 million (ViT-L) to 1 billion (ViT-g) parameters yields a +1.5 points average performance improvement. Both motion and appearance understanding tasks benefit from scaling, with SSv2 improving by +1.6 points and Kinetics by +1.5 points (cf.Table 4). These results confirm that self-supervised video pretraining effectively leverages larger model capacities, up to the 1B-parameter ViT-g.

**Training Schedule.** V-JEPA 2 model training employs a warmup-constant learning rate schedule followed by a cooldown phase (Zhai et al., 2022; Hägele et al., 2024). Similarly to Hägele et al. (2024), we found that this schedule performs comparably to a half-cosine schedule (Loshchilov and Hutter, 2016); it also makes exploring long training runs more cost-effective, since multiple cooldown runs can be started from different checkpoints of the constant phase. We simplified the recipe from Bardes et al. (2024) by maintaining fixed teacher EMA and weight decay coefficients instead of using ramp-up schedule, as these variations showed minimal impact on downstream understanding tasks. Figure 3 shows that extending the training schedule from 90K to 252K iterations yields a +0.8 average performance improvement with ViT-g models, validating the benefits of extended training durations. This schedule also facilitates progressive training by incrementally increasing video resolution during the cooldown phase.

**Efficient Progressive-Resolution Training.** While most previous video encoders focus on short clips of 16 frames (roughly seconds) (Bardes et al., 2024; Wang et al., 2024b, 2023), we explore training with longer clips of up to 64 frames (16 seconds) at higher spatial resolutions. However, training time increases dramatically with longer durations and higher resolutions — training our ViT-g model on $64 \times 384 \times 384$ inputs would require roughly 60 GPU-years (see Figure 5, Middle). To reduce this, we adopt a progressive resolution strategy (Touvron et al., 2019; Oquab et al., 2023) that boosts training efficiency while maintaining downstream performance. Our training process begins with a warmup phase where we train on 16-frame, $256 \times 256$-resolution videos with linear learning rate warmup over 12K iterations, followed by a main training phase with a constant learning rate for 228K iterations. Then, during the cooldown phase, we increase video duration and resolution while linearly decaying the learning rate over 12K iterations. Hence the additional





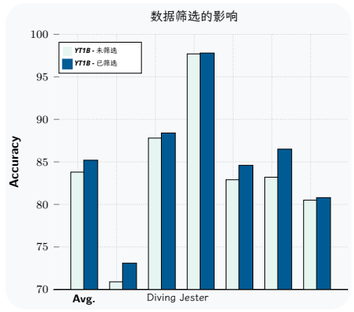




图 4 数据规模与筛选。我们在不同的数据混合上训练并比较模型。模型采用 ViT-L/16，遵循 Bardes 等人（2024）的方法，使用余弦学习率调度训练 90K 次迭代。（左）我们比较了在 VM2M 数据集和我们的 VM22M 数据集上预训练的 ViT-L/16 模型的性能。在 VM22M 数据集上训练使平均性能提升了 +1 个百分点。在基于外观的任务（如 Kinetics-400、COIN 和 ImageNet）上，性能提升更为显著。（右）我们比较了在 YT1B 上预训练的 ViT-L/16 模型与在我们的 Curated-YT1B 数据集上预训练的模型的性能，后者利用了我们基于聚类的筛选方法。在筛选后的数据集上训练使平均性能提升了 +1.4 个百分点，展示了数据筛选的有效性。

## 2.4 预训练方案

扩展模型规模。为了探索我们模型的扩展行为，我们训练了一系列编码器模型，参数量从3亿（ViT-L）到10亿（ViT-g）不等。所有编码器架构的详细信息见附录表12。注意，每个编码器使用相同的预测器架构，类似于ViT-small。我们在图5（左）中报告了这些编码器在视觉理解任务上的平均性能。将模型规模从3亿（ViT-L）扩展到10亿（ViT-g）参数，平均性能提升了 +1.5 个百分点。运动和外观理解任务均从规模扩展中获益，其中SSv2提升了 +1.6 个百分点，Kinetics提升了 +1.5 个百分点（参见表4）。这些结果证实，自监督视频预训练能够有效利用更大的模型容量，直至10亿参数的ViT-g。

训练调度。V-JEPA 2 模型训练采用预热-恒定学习率调度，随后进入冷却阶段（Zhai et al., 2022; Hägele et al., 2024）。与 Hägele et al. (2024) 类似，我们发现该调度的性能与半余弦调度（Loshchilov and Hutter, 2016）相当；它还能使探索长训练运行更具成本效益，因为可以从恒定阶段的多个检查点启动多次冷却运行。我们简化了 Bardes et al. (2024) 的方案，保持固定的教师 EMA 和权重衰减系数，而非使用渐进式调度，因为这些变化对下游理解任务的影响极小。图 3 显示，将训练调度从 90K 迭代扩展到 252K 迭代，在 ViT-g 模型上带来了 +0.8 的平均性能提升，验证了延长训练时长的收益。该调度还通过在冷却阶段逐步提高视频分辨率来促进渐进式训练。

高效的渐进分辨率训练。虽然大多数先前的视频编码器专注于16帧（约几秒）的短片段（Bardes et al., 2024; Wang et al., 2024b, 2023），我们探索了在更高空间分辨率下使用长达64帧（16秒）的较长片段进行训练。然而，训练时间会随着时长的增加和分辨率的提高而急剧增加——在 $64 \times 384 \times 384$输入上训练我们的ViT-g模型大约需要60个GPU年（见图5，中间）。为了减少这一开销，我们采用渐进分辨率策略（Touvron et al., 2019; Oquab et al., 2023），在保持下游性能的同时提升训练效率。我们的训练过程从一个预热阶段开始，在该阶段我们使用16帧、$256 \times 256$分辨率的视频进行训练，并在12K次迭代中进行线性学习率预热，随后进入主训练阶段，以恒定学习率训练228K次迭代。然后，在冷却阶段，我们增加视频时长和分辨率，同时在12K次迭代中线性衰减学习率。因此，额外的

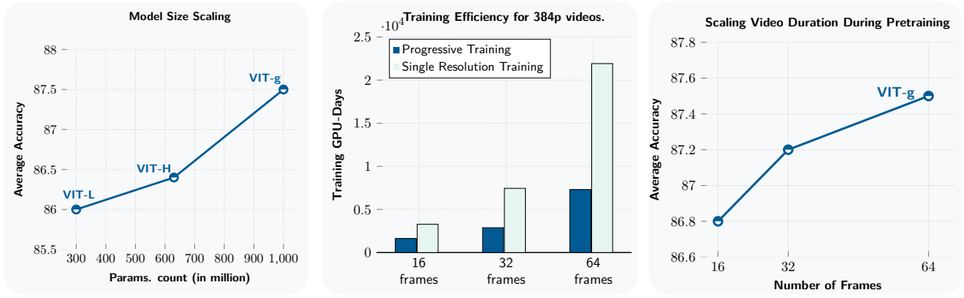



**Figure 5 Model Scaling.** We explore the impact of scaling model size and input video resolution. All models are trained on the VideoMix22M pretraining dataset. **(Left)** Average performance across six understanding tasks as a function of model scale. Models are trained with a constant learning rate until performance plateaus on downstream tasks. We then cool down the model using 64 frames at $256 \times 256$ resolution and report post-cooldown performance. Scaling the model size from 300M to 1B parameters yields a +1.7 point average improvement. **(Middle)** Training times (GPU-days) for ViT-g on A100 GPUs when training videos at $384 \times 384$ resolution with different numbers of frames per clip. We compare progressive resolution training (252K iterations at 16 frames / $256 \times 256$ resolution, followed by 12K cooldown iterations at $384 \times 384$ resolution) to the projected time for full-resolution training. Progressive training provides up to 8× speedup, significantly reducing the pretraining compute requirement. **(Right)** Effect of inscreasing video duration at cooldown on downstream performance for ViT-g. Even when only using 16-frame clips during inference/evaluation, increasing video duration during the cooldown phase of training improves average task performance by +0.7 points.

computational overhead associated with training on longer-duration, higher-resolution videos is only incurred during the final cooldown phase. This approach enables efficient high-resolution training: as shown in Figure 5 (Middle), we achieve an 8.4× reduction in GPU time for a model that can ingest 64-frame, $384 \times 384$ resolution inputs, compared to directly training such a model from scratch at full resolution throughout all phases of training. Furthermore, we still observe the benefits of a model that can process longer-duration and higher-resolution inputs as discussed next.

**Scaling temporal and spatial video resolution.** Figure 5 examines how input video resolution affects downstream task performance. When increasing clip duration from 16 to 64 frames during pretraining while maintaining a fixed 16-frame evaluation duration, we observe a +0.7 percentage point average performance improvement (Figure 5, Right). Additionally, we see that increasing the video duration and resolution during evaluation leads to a significant improvement across the tasks (refer to Table 4 and Appendix A.4.2). These results demonstrate that video self-supervised pretraining benefits from increased temporal resolution during both training and evaluation. Although we experimented with scaling to even longer video clips (128 and 256 frames), we did not observe any further improvement beyond 64 frames on this set of understanding tasks.

## 3 V-JEPA 2-AC: Learning an Action-Conditioned World Model

After pre-training, the V-JEPA 2 model can make predictions about missing part in videos. However, these predictions do not directly take into account the causal effect of actions that an agent might take. In the next stage of training, described in this section, we focus on making the model useful for planning by leveraging a small amount of interaction data. To that end, we learn a frame-causal action-conditioned predictor on top of the frozen V-JEPA 2 video encoder (Figure 2, right). We train our model on data from the Droid dataset (Khazatsky et al., 2024) consisting of data from experiments with a table-top Franka Panda robot arm collected through teleoperation. We refer to the resulting action-conditioned model as V-JEPA 2-AC, and in Section 4 we show that V-JEPA 2-AC can be used within a model-predictive control planning loop to plan actions in new environments.



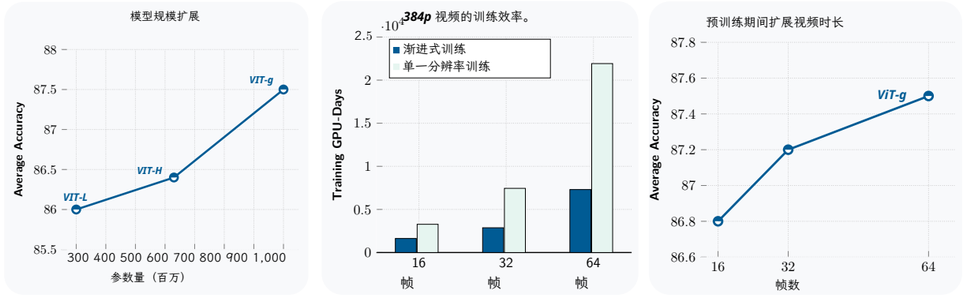



图 5 模型扩展。我们探索了模型规模和输入视频分辨率扩展的影响。所有模型均在 VideoMix22M 预训练数据集上训练。（左）六项理解任务的平均性能随模型规模的变化。模型以恒定学习率训练，直至下游任务性能趋于平稳。随后我们使用 $256 \times 256$ 分辨率的 64 帧对模型进行冷却，并报告冷却后的性能。将模型规模从 300M 扩展到 1B 参数可带来 +1.7 个百分点的平均提升。（中）ViT-g 在 A100 GPU 上以 $384 \times 384$ 分辨率训练不同每片段帧数视频时的训练时间（GPU 天数）。我们将渐进分辨率训练（16 帧 / $256 \times 256$ 分辨率下 252K 次迭代，随后在 $384 \times 384$ 分辨率下进行 12K 次冷却迭代）与全分辨率训练的预计时间进行比较。渐进训练可提供高达 8× 的加速，显著降低预训练计算需求。（右）冷却阶段增加视频时长对 ViT-g 下游性能的影响。即使在推理/评估时仅使用 16 帧片段，在训练的冷却阶段增加视频时长仍可使平均任务性能提升 +0.7 个百分点。

与在更长时长、更高分辨率视频上训练相关的计算开销仅在最后的冷却阶段才会产生。这种方法能够实现高效的高分辨率训练：如图5（中间）所示，与在所有训练阶段从头直接以全分辨率训练此类模型相比，我们实现了GPU时间的 8.4× 缩减，且模型能够接收64帧、 $384 \times 384$ 分辨率的输入。此外，我们仍然观察到了能够处理更长时长和更高分辨率输入的模型所带来的优势，这将在下文讨论。

扩展时间和空间视频分辨率。图5考察了输入视频分辨率如何影响下游任务性能。当在预训练期间将片段时长从16帧增加到64帧，同时保持固定的16帧评估时长时，我们观察到平均性能提升了 +0.7 个百分点（图5，右）。此外，我们发现评估期间增加视频时长和分辨率可在各项任务上带来显著提升（参见表4和附录A.4.2）。这些结果表明，视频自监督预训练在训练和评估期间均受益于时间分辨率的提高。尽管我们尝试扩展到更长的视频片段（128和256帧），但在这组理解任务上，我们未观察到超过64帧后的进一步提升。

## *3 V-JEPA 2-AC*：学习动作条件世界模型

经过预训练后，V-JEPA 2 模型能够对视频中缺失的部分进行预测。然而，这些预测并未直接考虑智能体可能采取的动作所带来的因果效应。在本节所述的下一训练阶段中，我们专注于利用少量交互数据使模型可用于规划。为此，我们在冻结的 V-JEPA 2 视频编码器之上学习了一个帧因果的动作条件预测器（图 2 右侧）。我们使用 Droid 数据集（Khazatsky 等，2024）中的数据来训练模型，该数据集由通过遥操作采集的桌面 Franka Panda 机械臂实验数据组成。我们将由此得到的动作条件模型称为 V-JEPA 2-AC，并在第 4 节中展示 V-JEPA 2-AC 可在模型预测控制规划循环中使用，以在新环境中规划动作。

### 3.1 Action-Conditioned World Model Training

Our goal is to take the V-JEPA 2 model after pre-training and obtain a latent world model that can be used for control of an embodied agentic system via closed-loop model-predictive control. To achieve this, we train V-JEPA 2-AC, an autoregressive model that predicts representations of future video observations conditioned on control actions and proprioceptive observations.

In this section we describe a concrete instantiation of this framework for a tabletop arm with a fixed exocentric camera, and where control actions correspond to end-effector commands. The model is trained using approximately 62 hours of unlabeled video from the raw Droid dataset, which consists of short videos, typically 3–4 seconds long, of a 7-DoF Franka Emika Panda arm equipped with a two-finger gripper. Here, *unlabeled* video refers to the fact that we do not use additional meta-data indicating any reward, what type of task was being performed in each demonstration, or whether the demonstration was successful or not in completing the task being attempted. Rather, we only use the raw video and end-effector state signals from the dataset (each video in the dataset is accompanied by meta-data indicating the end-effector state in each frame — three dimensions for position, three for orientation, and one for the gripper state).

**Model inputs.** In each iteration of training we randomly sample a mini-batch of 4 second video clips from the Droid dataset, and, for simplicity, discard any videos shorter than 4 seconds, leaving us with a smaller subset of the dataset comprising under 62 hours of video. The video clips are sampled with resolution $256 \times 256$ and a frame-rate of 4 frames-per-second (fps), yielding 16 frame clips denoted by $(x_k)_{k\in[16]}$, where each $x_k$ represents a single video frame. The robot's end-effector state in each observation is denoted by the sequence $(s_k)_{k\in[16]}$, where $s_k$ is a real-valued 7D vector defined relative to the base of the robot. The first three dimensions of $s_k$ encode the cartesian position of the end-effector, the next three dimensions encode its orientation in the form of extrinsic Euler angles, and the last dimension encodes the gripper state. We construct a sequence of actions $(a_k)_{k\in[15]}$ by computing the change in end-effector state between adjacent frames. Specifically, each action $a_k$ is a real-valued 7-dimensional vector representing the change in end-effector state between frames $k$ and $k+1$. We apply random-resize-crop augmentations to the sampled video clips with the aspect-ratio sampled in the range (0.75, 1.35).

**Loss function.** We use V-JEPA 2 encoder $E(\cdot)$ as an image encoder and encode each frame independently in a given clip to obtain a sequence of feature maps $(z_k)_{k\in[16]}$, where $z_k \coloneqq E(x_k) \in \mathbb{R}^{H\times W\times D}$ with $H \times W$ denoting the spatial resolution of the feature map, and $D$ the embedding dimension. In practice, our feature maps are encoded using the ViT-g encoder and have the shape $16 \times 16 \times 1408$. Note that the encoder is kept frozen during this post-training phase. The sequence of feature maps, end-effector states, and actions are temporally interleaved as $(a_k, s_k, z_k)_{k\in[15]}$ and processed with the transformer predictor network $P_\phi(\cdot)$ to obtain a sequence of next state representation predictions $(\hat{z}_{k+1})_{k\in[15]}$. The scalar-valued teacher-forcing loss function is finally computed as

$$\mathcal{L}_{\text{teacher-forcing}}(\phi) \coloneqq \frac{1}{T}\sum_{k=1}^{T}\|\hat{z}_{k+1} - z_{k+1}\|_1 = \frac{1}{T}\sum_{k=1}^{T}\left\|P_\phi\left((a_t, s_t, E(x_t))_{t\le k}\right) - E(x_{k+1})\right\|_1, \tag{2}$$

with $T = 15$. We also compute a two-step rollout loss to improve the model's ability to perform autoregressive rollouts at inference time. For simplicity of exposition and with slight overloading of notation, let $P_\phi(\hat{a}_{1:T}; s_k, z_k) \in \mathbb{R}^{H\times W\times D}$ denote the final predicted state representation obtained by autoregressively running V-JEPA 2-AC with an action sequence $(\hat{a}_i)_{i\in[T]}$, starting from $(s_k,\, z_k)$. We can now denote the rollout loss as:

$$\mathcal{L}_{\text{rollout}}(\phi) \coloneqq \|P_\phi(a_{1:T}, s_1, z_1) - z_{T+1}\|_1. \tag{3}$$

In practice we use $T = 2$ for computing the rollout loss, such that we only differentiate the predictor through one recurrent step.

The overall training objective is thus given by

$$L(\phi) \coloneqq \mathcal{L}_{\text{teacher-forcing}}(\phi) + \mathcal{L}_{\text{rollout}}(\phi), \tag{4}$$

and is minimized with respect to the predictor weights $\phi$. For illustrative purposes, the training procedure is depicted in Figure 6 with $T = 4$ for both the teacher forcing and rollout loss.



### *3.1* 动作条件世界模型训练

我们的目标是利用预训练后的 V-JEPA 2 模型，获得一个潜在世界模型，使其可通过闭环模型预测控制用于具身智能体系统的控制。为实现这一目标，我们训练了 V-JEPA 2-AC，这是一个自回归模型，以控制动作和本体感知观测为条件，预测未来视频观测的表征。

在本节中，我们描述了该框架的一个具体实例，适用于配备固定外部视角相机的桌面机械臂，其中控制动作对应于末端执行器指令。该模型使用来自原始Droid数据集的约62小时无标签视频进行训练，该数据集由7自由度Franka Emika Panda机械臂（配备双指夹爪）的短视频组成，时长通常为3–4秒。此处，无标签视频指的是我们不使用额外的元数据来指示任何奖励、每次演示中执行的任务类型，或演示是否成功完成了所尝试的任务。相反，我们仅使用数据集中的原始视频和末端执行器状态信号（数据集中的每个视频都附带元数据，指示每一帧的末端执行器状态——三个维度表示位置，三个表示方向，一个表示夹爪状态）。

模型输入。在训练的每次迭代中，我们从 Droid 数据集中随机采样包含 4 秒视频片段的小批量数据，为简单起见，丢弃任何短于 4 秒的视频，从而得到一个更小的数据子集，包含不足 62 小时的视频。视频片段以分辨率$256 \times 256$和每秒 4 帧（fps）的帧率采样，产生 16 帧的片段，表示为 $(x_k)^{k\in[16]}$，其中每个 $x_k$ 表示单个视频帧。每次观测中机器人的末端执行器状态由序列 $(s_k)^{k\in[16]}$ 表示，其中 $s_k$ 是相对于机器人基座定义的实值 7 维向量。$s_k$ 的前三个维度编码末端执行器的笛卡尔位置，接下来的三个维度以外在欧拉角的形式编码其姿态，最后一个维度编码夹爪状态。我们通过计算相邻帧之间末端执行器状态的变化来构建动作序列 $(a_k)^{k\in[15]}$。具体而言，每个动作 $a_k$ 是一个实值 7 维向量，表示帧 $k$ 与 $k+1$ 之间末端执行器状态的变化。我们对采样的视频片段应用随机调整大小裁剪增强，宽高比在 (0.75, 1.35) 范围内采样。

损失函数。我们使用 V-JEPA 2 编码器 $E(\cdot)$ 作为图像编码器，对给定片段中的每一帧独立编码，以获得特征图序列 $(z_k)^{k\in[16]}$，其中 $z_k \coloneqq E(x_k) \in \mathbb{R}^{H\times W\times D}$，$H \times W$ 表示特征图的空间分辨率，$D$ 表示嵌入维度。在实践中，我们的特征图使用 ViT-g 编码器编码，形状为 $16 \times 16 \times 1408$。注意，在此后训练阶段编码器保持冻结。特征图序列、末端执行器状态和动作在时间上交错为 $(a_k, s_k, z_k)^{k\in[15]}$，并由 transformer 预测器网络 $P_\phi(\cdot)$ 处理，以获得下一状态表示预测序列 $(\hat{}z_{k+1})^{k\in[15]}$。标量值的教师强制损失函数最终计算为

$$\mathcal{L}_{\text{teacher-forcing}}(\phi) \coloneqq \frac{1}{T}\sum_{k=1}^{T}\|\hat{z}_{k+1} - z_{k+1}\|_1 = \frac{1}{T}\sum_{k=1}^{T}\left\|P_\phi\left((a_t, s_t, E(x_t))_{t\le k}\right) - E(x_{k+1})\right\|_1, \tag{2}$$

其中 $T = 15$。我们还计算两步 rollout 损失，以提高模型在推理时执行自回归 rollout 的能力。为简化表述并略微重载符号，令 $P_\phi(\hat{a}_{1:T};\ s_k, z_k) \in \mathbb{R}^{H\times W\times D}$ 表示通过从 $(s_k,\, z_k)$ 开始，以动作序列 $(\hat{a}_i)^{i\in[T]}$ 自回归运行 V-JEPA 2-AC 获得的最终预测状态表示。我们现在可以将 rollout 损失表示为：

$$\mathcal{L}_{\text{rollout}}(\phi) \coloneqq \|P_\phi(a_{1:T}, s_1, z_1) - z_{T+1}\|_1. \tag{3}$$

在实践中，我们使用 $T = 2$ 来计算展开损失，这样我们只通过一个循环步骤对预测器进行微分。

因此，总体训练目标由下式给出

$$L(\phi) \coloneqq \mathcal{L}_{\text{teacher-forcing}}(\phi) + \mathcal{L}_{\text{rollout}}(\phi), \tag{4}$$

并针对预测器权重 $\phi$ 进行最小化。为便于说明，图 6 中展示了训练过程，其中 $T = 4$ 同时用于教师强制和 rollout 损失。

**Figure 6 V-JEPA 2-AC training.** V-JEPA 2-AC is trained in an autoregressive fashion, utilizing a teacher forcing loss and a rollout loss. (**Left**) In the teacher forcing loss, the predictor takes the encoding of the current frame representation as input and learns to predict the representation of the next timestep. (**Right**) The rollout loss involves feeding the predictor's output back as input, allowing the model to be trained to predict several timesteps ahead. By optimizing the sum of these two losses, V-JEPA 2-AC enhances its ability to accurately forecast the future by reducing error accumulation during rollouts.

**Architecture.** The predictor network $P_\phi(\cdot)$ is a ∼300M parameter transformer network with 24-layers, 16 heads, 1024 hidden dimension, and GELU activations. The action, end-effector state, and flattened feature maps input to the predictor are processed with separate learnable affine transformations to map them to the hidden dimension of the predictor. Similarly, the outputs of the last attention block of the predictor go through a learnable affine transformation to map them back to the embedding dimension of the encoder. We use our 3D-RoPE implementation to represent the spatiotemporal position of each video patch in the flattened feature map, while only applying the temporal rotary positional embeddings to the action and pose tokens. We use a block-causal attention pattern in the predictor so that each patch feature at a given time step can attend to the action, end-effector state, and other patch features from the same timestep, as well as those from previous time steps.

## 3.2 Inferring Actions by Planning

**Energy minimization.** Given an image of the goal state, we leverage V-JEPA 2-AC for downstream tasks by planning. Specifically, at each time step, we plan an action sequence for a fixed time horizon by minimizing a goal-conditioned energy function. We then execute the first action, observe the new state, and repeat the process. Let $s_k$ denote the current end-effector state, and $x_k$ and $x_g$ denote the current observed frame and goal image, respectively, which are separately encoded with the video encoder to obtain the feature maps $z_k$ and $z_g$. Given a planning horizon, $T$, we optimize a sequence of robot actions, $(a_i^\star)_{i\in[T]}$, by minimizing a goal-conditioned energy function,

$$\mathcal{E}(\hat{a}_{1:T};\ z_k, s_k, z_g) \coloneqq \|P(\hat{a}_{1:T}; s_k, z_k) - z_g\|_1, \tag{5}$$

such that $(a_i^\star)_{i\in[T]} \coloneqq \operatorname{argmin}_{\hat{a}_{1:T}} \mathcal{E}(\hat{a}_{1:T};\ z_k, s_k, z_g)$. As illustrated in Figure 7, the model infers an action sequence $(a_i^\star)_{i\in[T]}$ by selecting a trajectory that minimizes the L1 distance between the world model's imagined state representation $T$ steps into the future and its goal representation. In practice, we minimize (5) in each



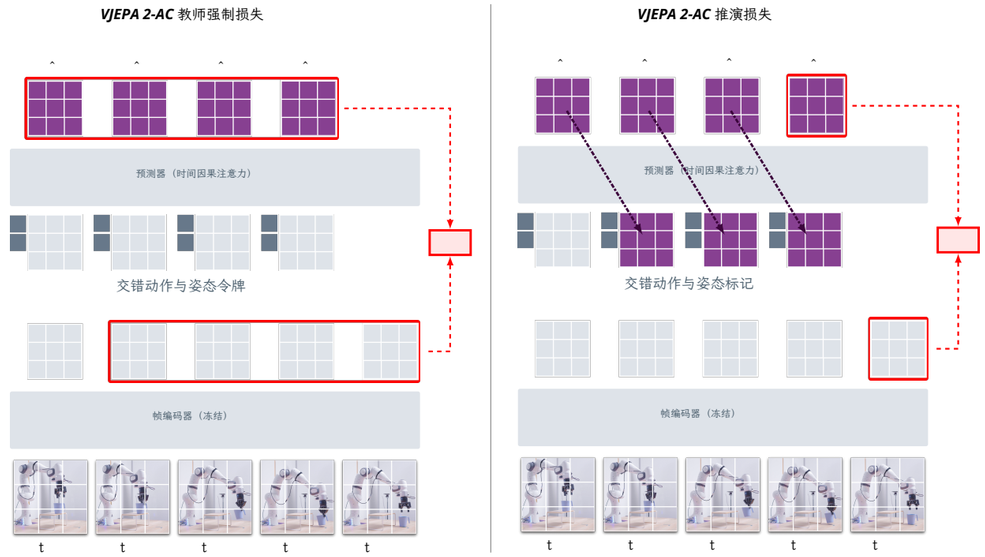



图 6 V-JEPA 2-AC 训练。V-JEPA 2-AC 以自回归方式进行训练，采用教师强制损失和展开损失。（左）在教师强制损失中，预测器将当前帧表示的编码作为输入，并学习预测下一时间步的表示。（右）展开损失涉及将预测器的输出反馈作为输入，使模型能够被训练以预测未来多个时间步。通过优化这两个损失之和，V-JEPA 2-AC 减少了展开过程中的误差累积，从而增强了准确预测未来的能力。

架构。预测器网络 $P_\phi(\cdot)$ 是一个 ∼300M 参数的 transformer 网络，具有 24 层、16 个头、1024 隐藏维度和 GELU 激活函数。输入到预测器的动作、末端执行器状态和展平特征图通过各自独立的可学习仿射变换进行处理，以将它们映射到预测器的隐藏维度。类似地，预测器最后一个注意力块的输出经过可学习仿射变换，以将它们映射回编码器的嵌入维度。我们使用我们的 3D-RoPE 实现来表示展平特征图中每个视频块的时空位置，同时仅对动作和姿态 token 应用时间旋转位置嵌入。我们在预测器中使用块因果注意力模式，使得给定时间步的每个块特征可以关注同一时间步的动作、末端执行器状态和其他块特征，以及来自先前时间步的这些特征。

## *3.2* 通过规划推断动作

能量最小化。给定目标状态的图像，我们通过规划利用 V-JEPA 2-AC 完成下游任务。具体而言，在每个时间步，我们通过最小化目标条件能量函数，为固定时间范围规划一个动作序列。然后我们执行第一个动作，观察新状态，并重复该过程。令 $s_k$ 表示当前末端执行器状态，$x_k$ 和 $x_g$ 分别表示当前观测帧和目标图像，它们分别通过视频编码器编码以获得特征图 $z_k$ 和 $z_g$。给定规划范围 $T$，我们通过最小化目标条件能量函数来优化一系列机器人动作 $(a_i^\star)_{i\in[T]}$，

$$\mathcal{E}(\hat{a}_{1:T};\ z_k, s_k, z_g) \coloneqq \|P(\hat{a}_{1:T}; s_k, z_k) - z_g\|_1, \tag{5}$$

使得 $(a_i^\star)_{i\in[T]} \coloneqq \operatorname{argmin}_{\hat{a}_{1:T}} \mathcal{E}(\hat{a}_{1:T};\ z_k, s_k, z_g)$。如图7所示，模型通过选择一条轨迹来推断动作序列 $(a_i^\star)_{i\in[T]}$，该轨迹使世界模型在未来 $T$ 步的想象状态表示与其目标表示之间的L1距离最小化。在实践中，我们在每个

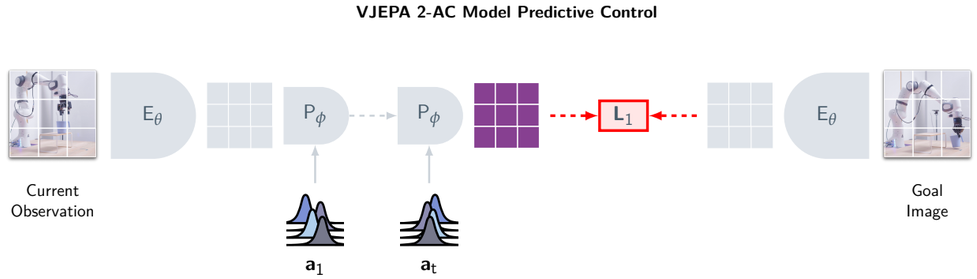



**Figure 7 Planning.** We plan an action sequence for a fixed time horizon $T$ by minimizing the L1 distance between the world model's imagined state representation $T$ steps into the future and its goal representation. The L1 loss is optimized with respect to the actions $(a_k)_{k\in[T]}$ using the cross-entropy method (Rubinstein, 1997). Specifically, in each planning step, we sample the action coordinates at each point in the planning horizon from a sequence of Gaussian distributions initialized with zero mean and unit variance. The population statistics of the top-k actions trajectories are used to update the mean and variance of the Gaussian distributions. This process is repeated for several iterations before finally returning the mean of the sequence of Gaussians as the selected action trajectory.

planning step using the Cross-Entropy Method (Rubinstein, 1997), and only execute the first action on the robot before re-planning, as in receding horizon control.

## 4 Planning: Zero-shot Robot Control

In this section we demonstrate how V-JEPA 2-AC can be used to implement basic robot skills like reaching, grasping, and pick-and-place via model-predictive control. We focus on tasks with visual goal specification and show that V-JEPA 2-AC generalizes zero-shot to new environments.

### 4.1 Experimental Setup

**Baselines.** We compare the performance of V-JEPA 2-AC with two baselines, one vision-language-action model trained with behavior cloning, and one video generation-based world model.

The first baseline is based on the Octo video-language-action model that allows for goal-image conditioning (Octo Model Team et al., 2024). We start from the open-source weights of the *octo-base-1.5* version of the model, which is pretrained on the *Open-X Embodiment* dataset containing over 1M trajectories.[1] We fine-tune the Octo model with behaviour cloning on the entire Droid dataset using hindsight relabeling (Andrychowicz et al., 2017; Ghosh et al., 2019) with image goals and end-effector states. In particular, we sample random segments of trajectories from the Droid dataset during training, and uniformly sample goal images up to 20 timesteps forward in the trajectory. We use the official open-source code for fine-tuning, including all standard Droid optimization hyperparameters, and leverage single side image view inputs at $256 \times 256$ resolution, a context of two previous frames, and a horizon of 4 future actions.

The second baseline we compare with is based on the Cosmos video generation model (Agarwal et al., 2025). We start with the open-source weights for the action-free Cosmos model (latent diffusion-7B with continuous tokenizer), which was trained on 20M hours of video, and we fine-tune the model on Droid using the officially-released action-conditioned fine-tuning code.[2] To improve performance when training on Droid, we (i) lowered the learning rate to match that used in the video-conditioned Cosmos recipe, (ii) removed the dropout in the video conditioning to improve the training dynamics, and (iii) increased the noise level by a factor of $e^2$, as we observed that the model trained with a lower noise factor struggled to leverage the information in the conditioning frame. Although the Cosmos technical report (Agarwal et al., 2025)

[1] In comparison, we train V-JEPA 2-AC on 23k trajectories from Droid, including successes and failures.
[2] https://github.com/nvidia-cosmos/cosmos-predict1



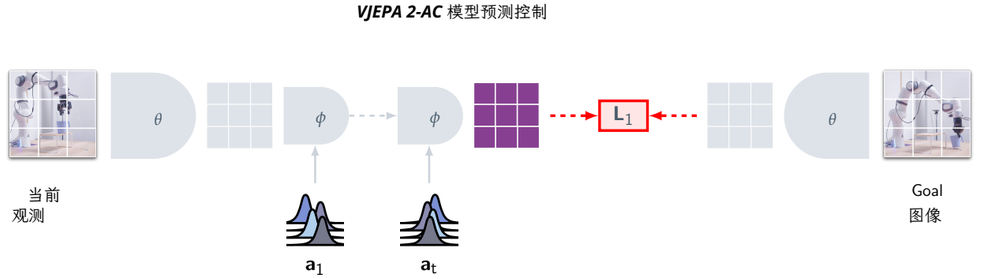



图 7 规划。我们为固定时间范围 $T$ 规划动作序列，通过最小化世界模型在未来 $T$ 步的想象状态表示与其目标表示之间的 L1 距离。L1 损失针对动作 $(a_k)_{k\in[T]}$ 使用交叉熵方法（Rubinstein, 1997）进行优化。具体而言，在每个规划步骤中，我们从一系列初始化为零均值和单位方差的高斯分布中采样规划范围内每个点的动作坐标。前 k 个动作轨迹的种群统计量用于更新高斯分布的均值和方差。该过程重复若干次迭代，最终返回高斯序列的均值作为所选动作轨迹。

规划步骤中使用交叉熵方法（Rubinstein, 1997）最小化(5)，并且如滚动时域控制那样，在重新规划之前仅执行机器人上的第一个动作。

## *4* 规划：零样本机器人控制

本节我们展示如何使用 V-JEPA 2-AC 通过模型预测控制实现基本的机器人技能，如到达、抓取和拾放。我们专注于具有视觉目标规范的任务，并展示 V-JEPA 2-AC 能够零样本泛化到新环境。

### *4.1* 实验设置

基线。我们将 V-JEPA 2-AC 的性能与两个基线进行比较，一个是通过行为克隆训练的视觉-语言-动作模型，另一个是基于视频生成的世界模型。

第一个基线基于 Octo 视频-语言-动作模型，该模型允许目标图像条件化（Octo Model Team et al., 2024）。我们从该模型 octo-base-1.5 版本的开源权重开始，该模型在包含超过100万条轨迹的 Open-X Embodiment 数据集上进行了预训练。[1] 我们使用事后重标记（Andrychowicz et al., 2017; Ghosh et al., 2019）以图像目标和末端执行器状态在整个 Droid 数据集上通过行为克隆微调 Octo 模型。具体而言，我们在训练期间从 Droid 数据集中采样随机轨迹片段，并均匀采样轨迹中最多向前20个时间步的目标图像。我们使用官方开源代码进行微调，包括所有标准 Droid 优化超参数，并利用 $256 \times 256$ 分辨率的单侧图像视图输入、两帧先前帧的上下文以及4个未来动作的预测范围。

我们对比的第二个基线基于 Cosmos 视频生成模型（Agarwal et al., 2025）。我们从无动作 Cosmos 模型（带有连续分词器的潜在扩散-7B 模型）的开源权重出发，该模型在 2000 万小时视频上训练，并使用官方发布的可动作条件微调代码在 Droid 上对该模型进行微调。[2] 为了提升在 Droid 上训练时的性能，我们（i）将学习率降低至与视频条件 Cosmos 方案中使用的学习率一致，（ii）移除了视频条件中的 dropout 以改善训练动态，以及（iii）将噪声水平提高了 $e^2$ 倍，因为我们观察到使用较低噪声因子训练的模型难以利用条件帧中的信息。尽管 Cosmos 技术报告（Agarwal et al., 2025）

[1] 相比之下，我们在来自 Droid 的 23k 条轨迹（包括成功和失败）上训练 V-JEPA 2-AC。[2] https://github.com/nvidia-cosmos/cosmos-predict1

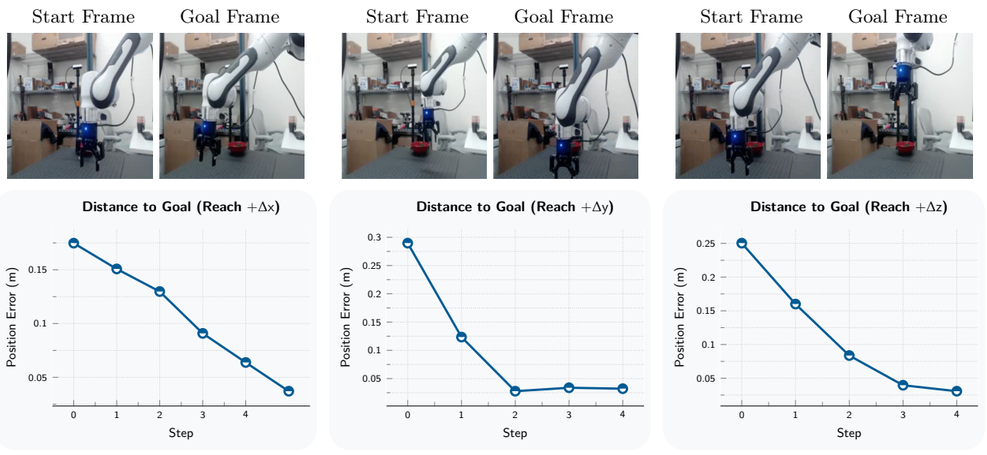



**Figure 8 Single-Goal Reaching.** Single-goal reaching involves moving the end-effector to a desired location in space based on a single goal image. This task measures for a basic understanding of actions as well as a 3D spatial understanding of the scene, including depth, from the monocular RGB camera. In each step, we use V-JEPA 2-AC to plan a sequence of actions by minimizing the L1 distance between the model's imagined future state representations and its representation of the goal frame. The first action is then executed before re-planning in the next time step. During planning, we only sample individual actions in the L1-Ball of radius 0.075 centered at the origin. Thus, the maximum achievable decrease in cartesian distance to the goal in a single step is 0.13 (∼13 cm).

mentions using world models for planning or model-predictive control as a future application, to the best of our knowledge this is the first reported attempt using Cosmos models for robot control.

**Robot deployment.** All models are deployed zero-shot on Franka Emika Panda arms with RobotiQ grippers, located in two different labs, neither of which appears in the Droid dataset. Visual input is provided through an uncalibrated low-resolution monocular RGB camera. The robots use the same exact model weights and inference code, and similar low-level controllers based on operational space control. We use blocking control for both the V-JEPA 2-AC world model and Cosmos world model (i.e., the system waits for the last commanded action to be completed before sending a new action to the controller) and experiment with both blocking and non-blocking control for Octo, and report the best performance across the two options. When planning with V-JEPA 2-AC and Cosmos, we constrain each sampled action to the L1-Ball of radius 0.075 centered at the origin, which corresponds to a maximum end-effector displacement of approximately 13 cm for each individual action, since large actions are relatively out-of-distribution for the models.

## 4.2 Results

**Single-goal reaching.** First, we evaluate on the task of single-goal reaching, which involves moving the end-effector to a desired location in space based on a single goal image. This task measures for a basic understanding of actions as well as a 3D spatial understanding of the scene (including depth) from the monocular RGB camera.

Figure 8 shows the Euclidean distance between the end-effector and its goal position during robot execution for three different single-goal reaching tasks. In all cases, the model is able to move the end-effector within less than 4 cm of its goal position, and select actions that lead to a monotonic decrease in the error. This can be seen as a form of visual servoing (Hill, 1979), wherein visual feedback from a camera is used to control a robot's motion. However, unlike classical approaches in visual servoing, V-JEPA 2-AC achieves this by training on unlabeled, real-world video data.

In Figure 9, we visualize the V-JEPA 2-AC energy landscape from equation (5) for the $\Delta y$ reaching task as a



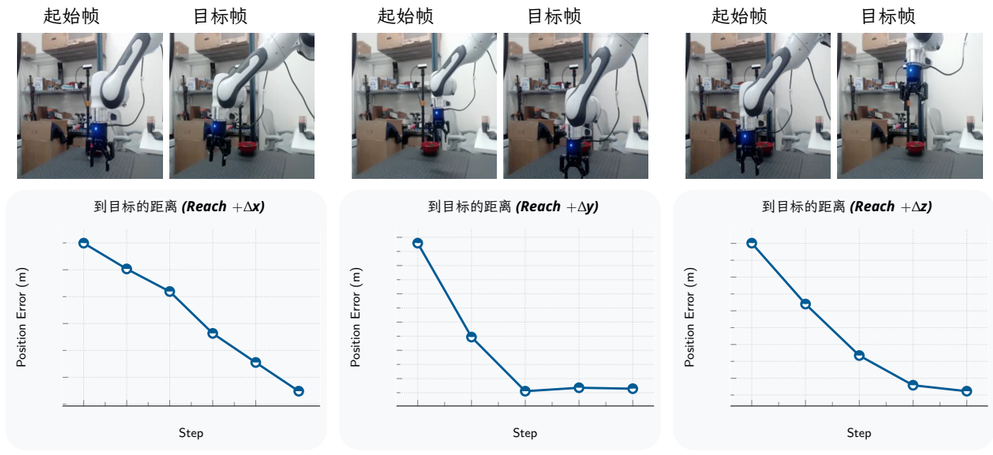



图 8 单目标到达。单目标到达是指根据单张目标图像将末端执行器移动到空间中期望位置的任务。该任务衡量模型对动作的基本理解以及对场景的三维空间理解（包括深度信息），输入来自单目 RGB 相机。在每一步中，我们使用 V-JEPA 2-AC 通过最小化模型想象未来状态表示与目标帧表示之间的 L1 距离来规划动作序列。随后执行第一个动作，并在下一时间步重新规划。在规划过程中，我们仅在以原点为中心、半径为 0.075 的 L1 球内采样单个动作。因此，单步内可实现的到目标笛卡尔距离的最大减少量为 0.13（∼13cm）。

提到了将世界模型用于规划或模型预测控制作为未来应用，但据我们所知，这是首次报道的使用 Cosmos 模型进行机器人控制的尝试。

机器人部署。所有模型均以零样本方式部署在配备 RobotiQ 夹爪的 Franka Emika Panda 机械臂上，这些机械臂位于两个不同的实验室中，且均未出现在 Droid 数据集中。视觉输入通过未经校准的低分辨率单目 RGB 相机提供。机器人使用完全相同的模型权重和推理代码，以及基于操作空间控制的类似底层控制器。我们对 V-JEPA 2-AC 世界模型和 Cosmos 世界模型均采用阻塞控制（即系统等待上一条指令动作完成后才向控制器发送新动作），并对 Octo 同时实验了阻塞控制和非阻塞控制，报告两种方案中的最佳性能。使用 V-JEPA 2-AC 和 Cosmos 进行规划时，我们将每个采样动作约束在以原点为中心、半径为 0.075 的 L1 球内，这对应每个单独动作的末端执行器最大位移约为 13 cm，因为大幅动作对模型而言相对处于分布之外。

## *4.2* 结果

单目标到达。首先，我们评估单目标到达任务，该任务涉及根据单个目标图像将末端执行器移动到空间中的期望位置。该任务衡量了对动作的基本理解，以及从单目RGB相机对场景的三维空间理解（包括深度）。

图8展示了在三个不同的单目标到达任务中，机器人执行过程中末端执行器与其目标位置之间的欧几里得距离。在所有情况下，模型能够将末端执行器移动到距其目标位置小于4厘米的范围内，并选择使误差单调递减的动作。这可以看作是一种视觉伺服（Hill, 1979），即利用相机的视觉反馈来控制机器人的运动。然而，与视觉伺服中的经典方法不同，V-JEPA 2-AC通过在没有标注的真实世界视频数据上进行训练来实现这一点。

在图 9 中，我们将公式 (5) 中 V-JEPA 2-AC 在 $\Delta y$ reaching 任务上的能量景观可视化为

function of a single cartesian-control action, sweeping $\Delta x$ and $\Delta y$ while holding $\Delta z = 0$ fixed. The energy function achieves its minimum near the ground-truth action, providing further evidence that the model has learned to reasonably infer the effect of actions without requiring precision sensing. It is also interesting to observe that the energy landscape induced by V-JEPA 2-AC is relatively smooth and locally convex, which should facilitate planning.

**Prehensile manipulation.** Next, we evaluate all models on more challenging prehensile object manipulation tasks, namely *grasp*, *reach with object*, and *pick-and-place*. Success rates are reported in Table 2 and Table 3, and averaged across 10 trials with various permutations to the task across trials (e.g., object location, starting pose, etc.). For the *grasp* and *reach with object* tasks the model is shown a single goal image. For the *pick-and-place* tasks we present two sub-goal images to the model in addition to the final goal. The first goal image shows the object being grasped, the second goal image shows the object in the vicinity of the goal position. The model first optimizes actions with respect to the first sub-goal for 4 time-steps before automatically switching to the second sub-goal for the next 10 time-steps, and finally the third goal for the last 4 time-steps. Examples of robot execution for the pick-and-place task are shown in Figure 10. Start and goal frames for all individual tasks in Lab 1 are shown in Appendix B.2. The *grasp* task requires precise control from visual feedback to correctly grip the object. The *reach with object* task requires the model to navigate while holding an object, which necessitates a basic understanding of intuitive physics to avoid dropping the object. Finally, the *pick-and-place* task tests for the ability to compose these atomic skills.

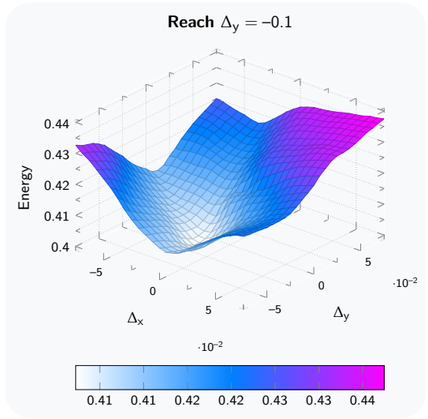



**Figure 9 V-JEPA 2-AC Energy Landscape.** Energy landscape for single-goal reaching task with respect to end-effector cartesian-control action (sweeping $\Delta x$ and $\Delta y$ while holding $\Delta z = 0$ fixed); ground truth action relating goal image to start frame is located at $(\Delta x, \Delta y) = (0, -0.1)$. We see that the energy function achieves its minimum around $(\Delta x, \Delta y) \approx (0, -0.05)$, indicating that the model has learned to reasonably infer the effect of actions without requiring precision sensing.

While all models achieve a high success-rate on *reach*, differences in performance are more apparent on tasks involving object interaction. We observe that the success-rate for all models depends on the type of object being manipulated. For instance, we find that the cup is mostly easily grasped by placing one finger inside the object and gripping around the rim, however if the control actions produced by the model are not accurate enough, the robot will miss the rim of the cup and fail to grasp the object. When manipulating the box, there are many more feasible grasping configurations, however, the model requires more precise gripper control to ensure that the fingers are open wide enough to grasp the object. We see that, for all models, the variation in success-rate with respect to the object type is due to the combination of sub-optimal actions and the unique challenges associated with manipulating each respective object. Nonetheless, we see that the V-JEPA 2-AC model achieves the highest success-rate across all tasks, highlighting the feasibility of latent planning for robot manipulation.

In Table 3, we compare planning performance when using V-JEPA 2-AC versus the Cosmos action-conditioned video generation model based on latent diffusion. In both cases we leverage the cross-entropy method (Rubinstein, 1997) for optimizing the sequence of actions using a single NVIDIA RTX 4090 GPU, and construct the energy function by encoding the goal frame in the latent space of the model, as in equation (5). With 80 samples, 10 refinement steps, and a planning horizon of 1, it takes 4 minutes to compute a single action in each planning step with Cosmos. While we achieve a high success rate of 80% on the *reach* tasks when using Cosmos, performance on object interaction tasks is weaker. Note that under a planning time of 4 minutes per action, a full pick & place trajectory requires over one hour of robot execution. By contrast, with $10\times$ more samples in each refinement step, the V-JEPA 2-AC world model requires only 16 seconds per action and leads to higher performance across all considered robot skills. We can potentially reduce the planning time for both models in future work by leveraging additional computing resources for planning, reducing the number



单个笛卡尔控制动作的函数，扫掠 $\Delta x$ 和 $\Delta y$，同时保持 $\Delta z = 0$ 固定。能量函数在真实动作附近达到最小值，这进一步证明该模型已学会在没有精确传感的情况下合理推断动作的效果。同样有趣的是，V-JEPA 2-AC 诱导的能量景观相对平滑且局部凸，这应有助于规划。

抓取式操作。接下来，我们在更具挑战性的抓取式物体操作任务上评估所有模型，即抓取、持物到达和拾放。成功率报告在表 2 和表 3 中，并在 10 次试验中取平均值，试验间对任务进行了各种排列变化（例如物体位置、起始姿态等）。对于抓取和持物到达任务，模型仅接收单张目标图像。对于拾放任务，除最终目标外，我们还向模型提供两张子目标图像。第一张目标图像显示物体被抓取，第二张目标图像显示物体位于目标位置附近。模型首先针对第一子目标优化动作 4 个时间步，然后自动切换到第二子目标执行接下来 10 个时间步，最后切换到第三目标执行最后 4 个时间步。拾放任务的机器人执行示例如图 10 所示。Lab 1 中所有独立任务的起始帧和目标帧见附录 B.2。抓取任务需要基于视觉反馈的精确控制以正确抓握物体。持物到达任务要求模型在持物状态下导航，这需要具备对直觉物理的基本理解以避免掉落物体。最后，拾放任务测试组合这些原子技能的能力。

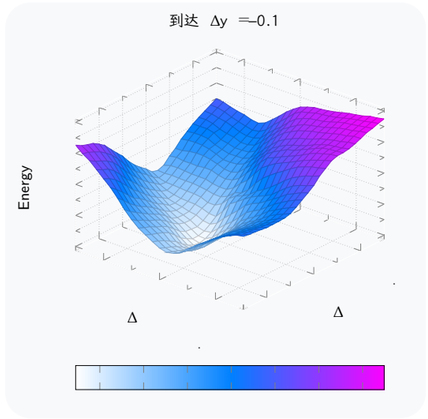



图 9 V-JEPA 2-AC 能量景观。针对单目标到达任务，末端执行器笛卡尔控制动作（扫掠 $\Delta x$ 和 $\Delta y$，同时保持 $\Delta z = 0$固定）的能量景观；将目标图像与起始帧关联的真实动作位于 $(\Delta x, \Delta y) = (0, -0.1)$。我们可以看到，能量函数在 $(\Delta x, \Delta y) \approx (0, -0.05)$ 附近达到最小值，这表明模型已学会在无需精确传感的情况下合理推断动作效果。

虽然所有模型在抵达任务上均取得了较高的成功率，但在涉及物体交互的任务上，性能差异更为明显。我们观察到，所有模型的成功率均取决于所操作物体的类型。例如，我们发现杯子最容易抓取

方式是将一根手指伸入物体内部并沿杯沿握持，然而，如果模型产生的控制动作不够精确，机器人将错过杯沿而抓取失败。在操作盒子时，可行的抓取配置要多得多，但模型需要更精确的夹爪控制以确保手指张开得足够宽以抓取物体。我们发现，对于所有模型而言，成功率随物体类型的变化是由于次优动作与操作各物体所特有的挑战共同导致的。尽管如此，我们看到V-JEPA 2-AC模型在所有任务中取得了最高的成功率，凸显了潜在规划用于机器人操作的可行性。

在表 3 中，我们比较了使用 V-JEPA 2-AC 与基于潜在扩散的 Cosmos 动作条件视频生成模型时的规划性能。在两种情况下，我们都利用交叉熵方法（Rubinstein，1997）在单块 NVIDIA RTX 4090 GPU 上优化动作序列，并按照公式（5）通过将目标帧编码到模型的潜在空间中来构建能量函数。在使用 Cosmos 时，采用 80 个样本、10 个细化步骤和规划视野为 1 的设置，每个规划步骤中计算单个动作需要 4 分钟。虽然我们在使用 Cosmos 时在 reach 任务上达到了 80% 的高成功率，但在物体交互任务上的表现较弱。请注意，在每次动作规划时间为 4 分钟的情况下，完整的 pick & place 轨迹需要超过一小时的机器人执行时间。相比之下，在每个细化步骤中使用 $10\times$ 更多样本时，V-JEPA 2-AC 世界模型每次动作仅需 16 秒，并在所有考虑的机器人技能上取得更高的性能。我们未来可以通过利用额外的计算资源进行规划来潜在地减少两种模型的规划时间，减少数量

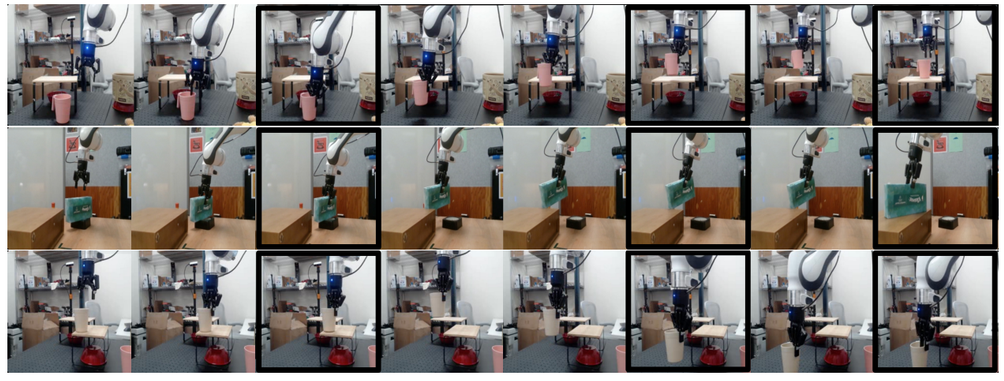


**Figure 10 Pick-&-Place.** Closed-loop robot execution of V-JEPA 2-AC for multi-goal pick-&-place tasks. Highlighted frames indicate when the model achieves a sub-goal and switches to the next goal. The first goal image shows the object being grasped, the second goal image shows the object in the vicinity of the desired location, and the third goal image shows the object placed in the desired position. The model first optimizes actions with respect to the first sub-goal for 4 time-steps before automatically switching to the second sub-goal for the next 10 time-steps, and finally the third goal for the last 4 time-steps. Robot actions are inferred through goal-conditioned planning. The V-JEPA 2-AC model is able to perform zero-shot pick-&-place tasks on two Franka arms in different labs, with various object configurations and cluttered environments.

**Table 2 Zero-Shot Robot Manipulation.** All models are deployed zero-shot on two Franka arms with RobotiQ grippers located in different labs. Given image-goals for each considered task, all models run closed loop to infer a sequence of actions to achieve the goal. Success rates are reported out of 10 trials with various permutations to the task across trials (e.g., object location, starting pose, etc.).

| | | | Grasp | | Reach w/ Obj. | | Pick-&-Place | |
|---|---|---|---|---|---|---|---|---|
| **Method** | | **Reach** | Cup | Box | Cup | Box | Cup | Box |
| Octo (Octo Model Team et al., 2024) | Lab 1 | 100% | 20% | 0% | 20% | 70% | 20% | 10% |
| | Lab 2 | 100% | 10% | 0% | 10% | 70% | 10% | 10% |
| | *Avg* | *100%* | *15%* | *0%* | *15%* | *70%* | *15%* | *10%* |
| V-JEPA 2-AC (ours) | Lab 1 | 100% | 70% | 30% | 90% | 80% | 80% | 80% |
| | Lab 2 | 100% | 60% | 20% | 60% | 70% | 80% | 50% |
| | *Avg* | *100%* | *65%* | *25%* | *75%* | *75%* | *80%* | *65%* |

of samples and refinement steps used at each time step, training a feed-froward policy in the world-models' imagination to initialize the planning problem, or potentially leveraging gradient-based planning in the case of V-JEPA 2-AC.

### 4.3 Limitations

**Sensitivity to camera positioning.** Since the V-JEPA 2-AC model is trained to predict representations of the next video frame given an end-effector Cartesian control action, without any explicit camera calibration, it must therefore implicitly infer the action coordinate axis from the monocular RGB camera input. However, in many cases, the robot base is not visible in the camera frame, and thus the problem of inferring the action coordinate axis is not well defined, leading to errors in the world model. In practice, we manually tried different camera positions before settling on one that worked well across all of our experiments. We conduct a quantitative analysis of the V-JEPA 2-AC world model's sensitivity to camera position in Appendix B.4.

**Long horizon planning.** Long horizon planning with world models is limited by a number of factors. First, autoregressive prediction suffers from error accumulation: the accuracy of the representation-space



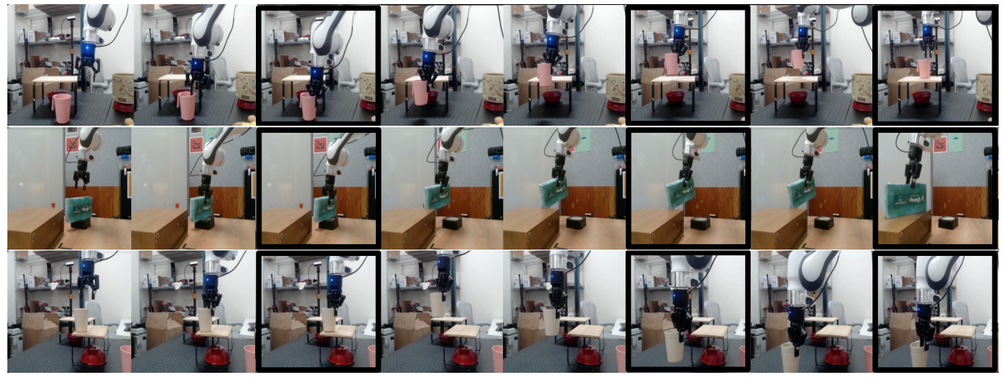


图 10 拾取与放置。V-JEPA 2-AC 在多目标拾取与放置任务中的闭环机器人执行过程。高亮帧表示模型完成一个子目标并切换到下一个目标的时刻。第一个目标图像显示物体被抓住，第二个目标图像显示物体位于期望位置附近，第三个目标图像显示物体被放置在期望位置。模型首先针对第一个子目标优化动作 4 个时间步，然后自动切换到第二个子目标并持续接下来的 10 个时间步，最后在最后 4 个时间步切换到第三个目标。机器人动作通过目标条件规划推断。V-JEPA 2-AC 模型能够在不同实验室中的两条 Franka 机械臂上执行零样本拾取与放置任务，适应各种物体配置和杂乱环境。

表 2 零样本机器人操作。所有模型均以零样本方式部署在不同实验室的两台配备 RobotiQ 夹爪的 Franka 机械臂上。对于每个考虑的任务，给定图像目标，所有模型以闭环方式运行以推断实现目标的一系列动作。成功率基于 10 次试验报告，试验间对任务进行各种排列变化（例如物体位置、起始姿态等）。

| | | | 抓取 | | 带物体接近 | | 拾取与放置 | |
|---|---|---|---|---|---|---|---|---|
| 方法 | | 接近 | Cup | Box | Cup | Box | Cup | Box |
| Octo（Octo 模型团队等，2024） | 实验室 1 | 100% | 20% | 0% | 20% | 70% | 20% | 10% |
| | 实验室 2 | 100% | 10% | 0% | 10% | 70% | 10% | 10% |
| | *Avg* | *100%* | *15%* | *0%* | *15%* | *70%* | *15%* | *10%* |
| V-JEPA 2-AC（我们的方法） | 实验室 1 | 100% | 70% | 30% | 90% | 80% | 80% | 80% |
| | 实验室 2 | 100% | 60% | 20% | 60% | 70% | 80% | 50% |
| | *Avg* | *100%* | *65%* | *25%* | *75%* | *75%* | *80%* | *65%* |

每个时间步所使用的样本数量和细化步骤、在世界模型的想象中训练前馈策略以初始化规划问题，或者在 V-JEPA2-AC 的情况下利用基于梯度的规划。

### *4.3* 局限性

对相机位置的敏感性。由于 V-JEPA 2-AC 模型经过训练，在给定末端执行器笛卡尔控制动作的情况下预测下一视频帧的表示，且没有任何显式的相机标定，因此它必须从单目 RGB 相机输入中隐式地推断动作坐标轴。然而，在许多情况下，机器人基座在相机视野中不可见，因此推断动作坐标轴的问题定义不明确，导致世界模型出现误差。在实践中，我们手动尝试了不同的相机位置，最终确定了一个在我们所有实验中均表现良好的位置。我们在附录 B.4 中对 V-JEPA 2-AC 世界模型对相机位置的敏感性进行了定量分析。

长程规划。使用世界模型进行长程规划受到若干因素的限制。首先，自回归预测存在误差累积问题：表示空间的准确性

**Table 3 Planning Performance.** Comparing closed-loop robot manipulation using MPC with V-JEPA 2-AC world model and Cosmos world model. In both cases we leverage the cross-entropy method (Rubinstein, 1997) for optimizing the sequence of actions using a single NVIDIA RTX 4090 GPU. For each robot skill, we evaluate each model across 10 tasks and average the results. With 80 samples, 10 refinement steps, and a planning horizon of 1, it takes 4 minutes to compute a single action in each planning step with Cosmos, which is an action-conditioned video generation model based on latent diffusion. Note that under a planning time of 4 minutes per action, a full pick & place trajectory takes over one hour. By contrast, with 10× more samples in each refinement step, the V-JEPA 2-AC world model requires only 16 seconds per action and leads to higher performance across all considered robot skills.

| Lab 2 | Planning Details | | | | Grasp | | Pick-&-Place | |
|---|---|---|---|---|---|---|---|---|
| **Method** | #Samples | Iter. | Horizon | Time | **Reach** | Cup | Box | Cup | Box |
| Cosmos (Agarwal et al., 2025) | 80 | 10 | 1 | 4 min. | 80% | 0% | 20% | 0% | 0% |
| V-JEPA 2-AC (ours) | 800 | 10 | 1 | 16 sec. | 100% | 60% | 20% | 80% | 50% |

predictions decreases with longer autoregressive rollouts, thereby making it more difficult to reliably plan over long horizons. Second, long-horizon planning increases the size of the search space: the number of possible action trajectories increases exponentially given a linear increase in the planning horizon, thereby making it computationally challenging to plan over long horizons. On the other hand, long-horizon planning is necessary for solving non-greedy prediction tasks, e.g., pick-and-place without image sub-goals. Future work exploring world models for long-horizon planning will enable the solution of many more complex and interesting tasks.

**Image goals.** Following many previous works in goal-conditioned robot manipulation (Finn and Levine, 2017; Lynch et al., 2020; Chebotar et al., 2021; Jang et al., 2022; Liu et al., 2022; Gupta et al., 2022), our current formulation of the optimization target assumes that we have access to visual goals. However, when deploying robots in-the-wild, it may be more natural to express goals in other forms, such as with language. Future work that aligns latent action-conditioned world models with language models will step towards more general task specification via natural language.

## 5 Understanding: Probe-based Classification

The capabilities of a representation-space world model, such as V-JEPA 2-AC discussed above, are inherently limited by the state information encoded in the learned representation space. In this section and subsequent sections, we probe the representations learned by V-JEPA 2 and compare the V-JEPA 2 encoder to other vision encoders on visual classification.

Visual classification tasks can focus either on *appearance understanding* or *motion understanding*. While appearance understanding tasks can generally be solved using information visible in a single frame of an input video clip (even when the classification labels describe actions), motion understanding tasks require several frames to correctly classify a video (Goyal et al., 2017). To ensure a balanced evaluation of both motion and appearance, we have selected three motion understanding tasks, namely Something-Something v2 (SSv2), Diving-48, and Jester, which require the model to understand human gestures and movements. For appearance understanding, we have chosen Kinetics400 (K400), COIN, and ImageNet (IN1K), which involve recognizing actions, scenes, and objects. Empirically, we show that V-JEPA 2 outperforms state-of-the-art visual encoders on motion understanding tasks, while being competitive on appearance understanding tasks.

**Attentive Probe.** We train an 4-layers attentive probe on top of the frozen encoder output using the training data from each task. Our attentive probe is composed of four transformer blocks, the last of which replaces standard self-attention with a cross-attention layer using a learnable query token. Following standard practice, several clips with a fixed number of frames are sampled from a video during inference. The classification logits are then averaged across clips. We keep the resolution similar to the one used for V-JEPA 2 pretraining. We ablate the number of layers of our attentive probe in Appendix C.2, and also provide full details on the number of clips, clip size, and other hyperparameters used in the downstream tasks.



表3 规划性能。比较使用MPC结合V-JEPA 2-AC世界模型与Cosmos世界模型进行闭环机器人操作的表现。在两种情况下，我们均利用交叉熵方法（Rubinstein, 1997）在单块NVIDIA RTX 4090 GPU上优化动作序列。对于每项机器人技能，我们在10个任务上评估每个模型并取平均值。在80个样本、10次细化步骤和规划时域为1的设置下，使用Cosmos（一种基于潜在扩散的动作条件视频生成模型）在每个规划步骤中计算单个动作需要4分钟。请注意，在每次动作规划时间为4分钟的情况下，一个完整的抓取与放置轨迹需要超过一小时。相比之下，V-JEPA 2-AC世界模型在每次细化步骤中使用 10× 更多样本，每次动作仅需16秒，并在所有考虑的机器人技能上取得更高的性能。

| 实验 2 | 规划细节 | | | | 抓取 | | 拾取与放置 | |
|---|---|---|---|---|---|---|---|---|
| 方法 | #样本数 | 迭代 | 视界 | Time | 到达 | Cup | Box | Cup | Box |
| Cosmos（Agarwal 等，2025） | 80 | 10 | 1 | 4 分钟 | 80% | 0% | 20% | 0% | 0% |
| V-JEPA 2-AC（ours） | 800 | 10 | 1 | 16秒。 | 100% | 60% | 20% | 80% | 50% |

预测能力随着自回归推演步数的增加而下降，从而使长时程的可靠规划变得更加困难。其次，长时程规划扩大了搜索空间：在规划时程线性增加的情况下，可能的动作轨迹数量呈指数增长，从而使长时程规划在计算上面临挑战。另一方面，长时程规划对于解决非贪心预测任务（例如无图像子目标的拾取与放置）是必要的。未来探索用于长时程规划的世界模型将使得解决更多复杂且有趣的任务成为可能。

图像目标。遵循目标条件机器人操作领域的许多先前工作（Finn and Levine, 2017; Lynch et al., 2020; Chebotar et al., 2021; Jang et al., 2022; Liu et al., 2022; Gupta et al., 2022），我们当前的优化目标公式假设可以获取视觉目标。然而，在真实环境中部署机器人时，以其他形式（例如语言）表达目标可能更为自然。将潜在动作条件世界模型与语言模型相对齐的未来工作，将朝着通过自然语言进行更通用任务指定的方向迈进。

## 5 理解：基于探针的分类

表示空间世界模型（如上文讨论的 V-JEPA 2-AC）的能力本质上受限于所学表示空间中编码的状态信息。在本节及后续章节中，我们通过探针方法探测 V-JEPA 2 所学到的表示，并在视觉分类任务上将 V-JEPA 2 编码器与其他视觉编码器进行比较。

视觉分类任务可以侧重于外观理解或运动理解。外观理解任务通常可以利用输入视频片段中单帧可见的信息来解决（即使分类标签描述的是动作），而运动理解任务则需要多帧信息才能正确分类视频（Goyal et al., 2017）。为确保对运动和外观两方面的平衡评估，我们选择了三个运动理解任务，即 Something-Something v2（SSv2）、Diving-48 和 Jester，这些任务要求模型理解人类手势和动作。对于外观理解，我们选择了 Kinetics400（K400）、COIN 和 ImageNet（IN1K），这些任务涉及识别动作、场景和物体。实验表明，V-JEPA 2 在运动理解任务上优于最先进的视觉编码器，同时在外观理解任务上具有竞争力。

注意力探针。我们在冻结的编码器输出之上，使用每个任务的训练数据训练一个4层注意力探针。我们的注意力探针由四个transformer块组成，其中最后一个块使用可学习的查询token将标准自注意力替换为交叉注意力层。按照标准做法，在推理过程中从视频中采样若干具有固定帧数的片段，然后对所有片段的分类logits取平均值。我们保持与V-JEPA 2预训练时相似的分辨率。我们在附录C.2中对注意力探针的层数进行了消融实验，并提供了下游任务中使用的片段数、片段大小及其他超参数的完整细节。

**Table 4 Action and Object Classification.** We report the classification performance of V-JEPA 2 models pretrained on 64 frames at resolution $256 \times 256$ for all models, except V-JEPA 2 ViT-$g_{384}$ which was pretrained at resolution $384 \times 384$, on action and object classification, and compare their performance with state-of-art image and video encoders. All models follow the same evaluation protocol except for V-JEPA 2 ViT-$g_{384}$. We use $256 \times 256$ resolution with $16 \times 2 \times 3$ inputs for SSv2 (16 frames clip, 2 temporal crops, 3 spatial crops), $16 \times 8 \times 3$ for K400, $32 \times 8 \times 3$ for COIN and $32 \times 4 \times 3$ for Diving-48 and Jester. V-JEPA 2 ViT-$g_{384}$ uses a higher resolution of $384 \times 384$ for all six tasks, and additionally uses $64 \times 2 \times 3$ inputs for SSv2 and 32x8x3 inputs for COIN. Our V-JEPA 2 ViT-g significantly outperforms other vision encoders on motion understanding tasks and is competitive on appearance tasks. It achieves the best average performance of 87.5 across all image and videos encoders. V-JEPA 2 ViT-$g_{384}$ further improves results consistently across tasks, reaching 88.2 average performance. $*$: **$PE_{core}G$** achieves an accuracy 89.8% on ImageNet using an attentive probe and input resoluton of 448px (Bolya et al., 2025). We use an input resolution of 256px and a different probe architecture in our case.

| | | | *Motion Understanding* | | | *Appearance Understanding* | | |
|---|---|---|---|---|---|---|---|---|
| **Method** | **Param.** | **Avg.** | **SSv2** | **Diving-48** | **Jester** | **K400** | **COIN** | **IN1K** |
| *Results Reported in the Literature* | | | | | | | | |
| **VideoMAEv2** (Wang et al., 2023) | 1B | – | 56.1 | – | – | 82.8 | – | 71.4 |
| **InternVideo2-1B** (Wang et al., 2024b) | 1B | – | 67.3 | – | – | 87.9 | – | – |
| **InternVideo2-6B** (Wang et al., 2024b) | 6B | – | 67.7 | – | – | 88.8 | – | – |
| **VideoPrism** (Zhao et al., 2024) | 1B | – | 68.5 | 71.3 | – | 87.6 | – | – |
| *Image Encoders Evaluated Using the Same Protocol* | | | | | | | | |
| **DINOv2** (Darcet et al., 2024) | 1.1B | 81.1 | 50.7 | 82.5 | 93.4 | 83.6 | 90.7 | 86.1 |
| **$PE_{core}G$** (Bolya et al., 2025) | 1.9B | 82.3 | 55.4 | 76.9 | 90.0 | 88.5 | **95.3** | 87.6* |
| **SigLIP2** (Tschannen et al., 2025) | 1.2B | 81.1 | 49.9 | 75.3 | 91.0 | 87.3 | 95.1 | **88.0** |
| *Video Encoders Evaluated Using the Same Protocol* | | | | | | | | |
| **V-JEPA ViT-H** (Bardes et al., 2024) | 600M | 85.2 | 74.3 | 87.9 | 97.7 | 84.5 | 87.1 | 80.0 |
| **InternVideo2$_{s2}$-1B** (Wang et al., 2024b) | 1B | 87.0 | 69.7 | 86.4 | 97.0 | **89.4** | 93.8 | 85.8 |
| **V-JEPA 2 ViT-L** | 300M | 86.0 | 73.7 | 89.0 | 97.6 | 85.1 | 86.8 | 83.5 |
| **V-JEPA 2 ViT-H** | 600M | 86.4 | 74.0 | 89.8 | 97.7 | 85.3 | 87.9 | 83.8 |
| **V-JEPA 2 ViT-g** | 1B | 87.5 | 75.3 | 90.1 | 97.7 | 86.6 | 90.7 | 84.6 |
| **V-JEPA 2 ViT-$g_{384}$** | 1B | **88.2** | **77.3** | **90.2** | **97.8** | 87.3 | 91.1 | 85.1 |

**Evaluation protocol.** We compare the performance of V-JEPA 2 on motion and appearance tasks with several other visual encoders: DINOv2 with registers (Darcet et al., 2024) is the current state-of-the-art model for self-supervised learning with images, while SigLIP2 (Tschannen et al., 2025) and the Perception Encoder $PE_{core}G$ (Bolya et al., 2025) are two state-of-the-art models for image-text contrastive pretraining. We also consider two video encoders: the self-supervised V-JEPA (Bardes et al., 2024), and InternVideo2$_{s2}$-1B (Wang et al., 2024b) which relies primarily on vision-text contrastive pretraining.

We use the same evaluation protocol for every baseline and for V-JEPA 2, learning an attentive probe on top of the frozen encoder, similar to Bardes et al. (2024). We adapt image-based models to video following the procedure used in Oquab et al. (2023), concatenating the features of each input frame. For InternVideo2$_{s2}$-1B, we use its image positional embedding for the ImageNet task, and for video tasks we interpolate its positional embedding from 4 frames to 8, producing a token count similar to V-JEPA 2. Despite using a common evaluation protocol, the baseline encoders are trained on different data (e.g., DINOv2 on LVD-142M, $PE_{core}G$ on MetaCLIP) and are thus not directly comparable. We can therefore only compare different approaches at a system level; i.e., with a consistent evaluation protocol despite differences in training protocol and data. We also include existing results from the literature using a similar frozen protocol, but with potentially different attentive head architecture. In particular, we share reported results of VideoMAEv2 (Wang et al., 2023), InternVideo-1B and 6B (Wang et al., 2024b), and VideoPrism (Zhang et al., 2024c) on the classification tasks we consider, when available. We provide complete evaluation and hyperparameters in Appendix C.1.

**Results.** Table 4 reports the classification performance of V-JEPA 2, the other encoders we evaluated, and other notable results reported in the literature. V-JEPA 2 ViT-g (at 256 resolution) significantly outperforms other vision encoders on motion understanding tasks. It achieves a top-1 accuracy of 75.3 on SSv2 compared to



表4 动作与物体分类。我们报告了V-JEPA 2模型在动作和物体分类任务上的分类性能，所有模型均在分辨率 $256 \times 256$ 下使用64帧进行预训练，但V-JEPA 2 ViT-$g_{384}$ 在分辨率$384 \times 384$下预训练，并将其性能与最先进的图像和视频编码器进行比较。除V-JEPA 2 ViT-$g_{384}$外，所有模型遵循相同的评估协议。对于SSv2，我们使用 $256 \times 256$ 分辨率和 $16 \times 2 \times 3$ 输入（16帧片段，2个时间裁剪，3个空间裁剪），K400使用 $16 \times 8 \times 3$，COIN使用 $32 \times 8 \times 3$，Diving-48和Jester使用 $32 \times 4 \times 3$。V-JEPA 2 ViT-$g_{384}$ 在所有六个任务上使用更高的 $384 \times 384$ 分辨率，并且对SSv2额外使用 $64 \times 2 \times 3$ 输入，对COIN使用32x8x3输入。我们的V-JEPA 2 ViT-g在运动理解任务上显著优于其他视觉编码器，在外观任务上具有竞争力。它在所有图像和视频编码器中取得了最佳的平均性能87.5。V-JEPA 2 ViT-$g_{384}$ 在各任务上进一步提升结果，达到88.2的平均性能。 $*$：$PE_{core}G$使用注意力探针和448px的输入分辨率在ImageNet上达到89.8%的准确率（Bolya et al., 2025）。在我们的情况下，我们使用256px的输入分辨率和不同的探针架构。

| | | | 运动理解 | | | 外观理解 | | |
|---|---|---|---|---|---|---|---|---|
| 方法 | 参数 | Avg. | SSv2 | *Diving-48* | *Jester* | K400 | COIN | IN1K |
| 文献中报告的结果 | | | | | | | | |
| VideoMAEv2（Wang 等，2023） | 1B | – | 56.1 | – | – | 82.8 | – | 71.4 |
| InternVideo2-1B（Wang 等，2024b） | 1B | – | 67.3 | – | – | 87.9 | – | – |
| InternVideo2-6B（Wang 等，2024b） | 6B | – | 67.7 | – | – | 88.8 | – | – |
| VideoPrism（Zhao 等，2024） | 1B | – | 68.5 | 71.3 | – | 87.6 | – | – |
| 使用相同协议评估的图像编码器 | | | | | | | | |
| DINOv2（Darcet 等人，2024） | 1.1B | 81.1 | 50.7 | 82.5 | 93.4 | 83.6 | 90.7 | 86.1 |
| $PE_{core}G$（Bolya 等，2025） | 1.9B | 82.3 | 55.4 | 76.9 | 90.0 | 88.5 | **95.3** | 87.6* |
| SigLIP2（Tschannen 等，2025） | 1.2B | 81.1 | 49.9 | 75.3 | 91.0 | 87.3 | 95.1 | **88.0** |
| 使用相同协议评估的视频编码器 | | | | | | | | |
| V-JEPA ViT-H（Bardes 等，2024） | 600M | 85.2 | 74.3 | 87.9 | 97.7 | 84.5 | 87.1 | 80.0 |
| InternVideo2$_{s2}$-1B（Wang 等，2024b） | 1B | 87.0 | 69.7 | 86.4 | 97.0 | **89.4** | 93.8 | 85.8 |
| ***V-JEPA 2 ViT-L*** | 300M | 86.0 | 73.7 | 89.0 | 97.6 | 85.1 | 86.8 | 83.5 |
| ***V-JEPA 2 ViT-H*** | 600M | 86.4 | 74.0 | 89.8 | 97.7 | 85.3 | 87.9 | 83.8 |
| ***V-JEPA 2 ViT-g*** | 1B | 87.5 | 75.3 | 90.1 | 97.7 | 86.6 | 90.7 | 84.6 |
| ***V-JEPA 2 ViT-$g_{384}$*** | 1B | **88.2** | **77.3** | **90.2** | **97.8** | 87.3 | 91.1 | 85.1 |

评估方案。我们在运动任务和外观任务上将 V-JEPA 2 的性能与其他若干视觉编码器进行比较：带寄存器的 DINOv2 （Darcet 等，2024）是当前图像自监督学习的最先进模型，而 SigLIP2 （Tschannen 等，2025）和感知编码器 $PE_{core}G$（Bolya 等，2025）是图像-文本对比预训练的两个最先进模型。我们还考虑了两个视频编码器：自监督的 V-JEPA（Bardes 等，2024），以及主要依赖视觉-文本对比预训练的 InternVideo2$_{s2}$-1B（Wang 等，2024b）。

我们对每个基线模型和 V-JEPA 2 采用相同的评估协议，在冻结编码器之上学习一个注意力探针，类似于 Bardes 等人（2024）的做法。我们按照 Oquab 等人（2023）使用的流程将基于图像的模型适配到视频，拼接每个输入帧的特征。对于 InternVideo2$_{s2}$-1B，我们在 ImageNet 任务中使用其图像位置嵌入，对于视频任务，我们将其位置嵌入从 4 帧插值到 8 帧，产生与 V-JEPA 2 相似的 token 数量。尽管使用相同的评估协议，基线编码器在不同的数据上训练（例如，DINOv2 在 LVD-142M 上，$PE_{core}G$ 在 MetaCLIP 上），因此不能直接比较。因此，我们只能在系统层面比较不同的方法；即在训练协议和数据存在差异的情况下，使用一致的评估协议进行比较。我们还纳入了文献中使用类似冻结协议但可能具有不同注意力头架构的现有结果。特别是，我们分享了 VideoMAEv2（Wang 等人，2023）、InternVideo-1B 和 6B（Wang 等人，2024b）以及 VideoPrism（Zhang 等人，2024c）在我们所考虑的分类任务上的已报告结果（如有）。我们在附录 C.1 中提供了完整的评估和超参数。

结果。表4报告了V-JEPA 2、我们评估的其他编码器以及文献中报告的其他显著结果的分类性能。V-JEPA 2 ViT-g（256分辨率）在运动理解任务上显著优于其他视觉编码器。它在SSv2上达到了75.3的top-1准确率，相比之下

**Table 5 Prediction: Human Action Anticipation**. Comparison with the state-of-the-art on the EK100 Action Anticipation benchmark. We report mean-class recall-at-5 for verb, noun and action on the validation set of EK100. V-JEPA 2 performance scales linearly with model size and outperforms previous state-of-the-art across all model sizes.

| Method | Param. | *Action Anticipation* | | |
|---|---|---|---|---|
| | | Verb | Noun | **Action** |
| **InAViT** (Roy et al., 2024) | 160M | 51.9 | 52.0 | 25.8 |
| **Video-LLaMA** (Zhang et al., 2023) | 7B | 52.9 | 52.0 | 26.0 |
| **PlausiVL** (Mittal et al., 2024) | 8B | 55.6 | 54.2 | 27.6 |
| *Frozen Backbone* | | | | |
| **V-JEPA 2 ViT-L** | 300M | 57.8 | 53.8 | 32.7 |
| **V-JEPA 2 ViT-H** | 600M | 59.2 | 54.6 | 36.5 |
| **V-JEPA 2 ViT-g** | 1B | 61.2 | 55.7 | 38.0 |
| **V-JEPA 2 ViT-g$_{384}$** | 1B | **63.6** | **57.1** | **39.7** |

69.7 for InternVideo and 55.4 for PE$_{Core}$G. V-JEPA 2 is also competitive on appearance tasks, reaching 84.6 on ImageNet (a +4.6 point improvement over V-JEPA). Overall, V-JEPA 2 obtains the best average performance across all six tasks, compared to other video and image encoders. The higher-resolution, longer-duration V-JEPA 2 ViT-g$_{384}$ shows further improvement across all tasks, reaching 88.2 average performance.

## 6 Prediction: Probe-based Action Anticipation

Action anticipation consists in predicting the future action given a contextual video clip leading up to some time before the action. Using the Epic-Kitchens-100 (EK100) benchmark (Damen et al., 2022), we demonstrate that V-JEPA 2 action anticipation performance increases consistently with model size. Furthermore, despite only using an attentive probe trained on top of V-JEPA 2 representations, we show that V-JEPA 2 significantly outperforms prior state-of-the-art approaches that were specifically designed for this task.

**Task.** The EK100 dataset is comprised of 100 hours of cooking activities recorded from an egocentric perspective across 45 kitchen environments. Each video in EK100 is annotated with action segments, which include a start timestamp, an end timestamp, and an action label. There are 3,568 unique action labels, each consisting of a verb and a noun category, with a total of 97 verb categories and 300 noun categories. The EK100 action anticipation task involves predicting noun, verb, and action (i.e., predicting verb and noun jointly) from a video clip, referred to as context, that occurs before the start timestamp of an action segment. The interval between the end of the context and the beginning of the action segment is the anticipation time, which is set to 1 second by default. Given that different future actions may be possible from a given context, mean-class recall-at-5 is used as the metric to measure performance (Damen et al., 2022).

**Anticipation Probe.** An attentive probe is trained on top of the frozen V-JEPA 2 encoder and predictor to anticipate future actions. Specifically, we sample a video clip that ends 1 second before an action starts. This video context is fed to the V-JEPA 2 encoder. The predictor takes the encoder representation, along with the mask tokens corresponding to the frame 1 second into the future, and predicts the representation of the future video frame. The outputs of the predictor and encoder are concatenated along the token dimension and fed to an attentive probe with a similar architecture to those used in Section 5, with the difference being that the anticipation probe's final cross-attention layer learns three query tokens (as opposed to one), and each query output is fed to a different linear classifier to predict the action category, the verb category, and the noun category respectively. A focal loss (Lin et al., 2017) is applied to each classifier independently and then summed before backpropagating through the shared attention blocks of the probe. We provide additional details and evaluation hyperparameters in Appendix D.1.

**Baselines.** We compare our model with three baselines that are trained specifically for action anticipation: InAViT (Roy et al., 2024) is a a supervised approach that leverages explicit hand-object interaction modeling,



表 5 预测：人类动作预期。在 EK100 动作预期基准上与最先进方法进行比较。我们报告了在 EK100 验证集上动词、名词和动作的平均类别 recall-at-5。V-JEPA 2 的性能随模型规模线性扩展，并在所有模型规模上超越了此前的最先进水平。

| 方法 | 参数量 | 动作预期 | | |
|---|---|---|---|---|
| | | Verb | Noun | 动作 |
| InAViT（Roy 等，2024） | 160M | 51.9 | 52.0 | 25.8 |
| Video-LLaMA（Zhang 等，2023） | 7B | 52.9 | 52.0 | 26.0 |
| PlausiVL（Mittal 等，2024） | 8B | 55.6 | 54.2 | 27.6 |
| 冻结骨干网络 | | | | |
| ***V-JEPA 2 ViT-L*** | 300M | 57.8 | 53.8 | 32.7 |
| ***V-JEPA 2 ViT-H*** | 600M | 59.2 | 54.6 | 36.5 |
| ***V-JEPA 2 ViT-g*** | 1B | 61.2 | 55.7 | 38.0 |
| ***V-JEPA 2 ViT-g$_{384}$*** | 1B | **63.6** | **57.1** | **39.7** |

InternVideo为69.7，PE$_{Core}$G为55.4。V-JEPA 2在外观任务上也具有竞争力，在ImageNet上达到84.6（相比V-JEPA提升了 +4.6 个百分点）。总体而言，与其他视频和图像编码器相比，V-JEPA 2在所有六项任务上获得了最佳的平均性能。更高分辨率、更长时长的V-JEPA 2 ViT-g$_{384}$ 在所有任务上展现出进一步提升，达到88.2的平均性能。

## *6* 预测：基于探针的动作预判

动作预判是指在动作发生前某一时刻，根据上下文视频片段预测未来动作。使用 Epic-Kitchens-100（EK100）基准（Damen 等，2022），我们证明 V-JEPA 2 的动作预判性能随模型规模增大而持续提升。此外，尽管仅使用在 V-JEPA 2 表示之上训练的注意力探针，我们仍表明 V-JEPA 2 显著优于先前专为此任务设计的最先进方法。

任务。EK100 数据集包含 100 小时的烹饪活动，以第一人称视角在 45 个厨房环境中录制。EK100 中的每个视频都标注了动作片段，包括开始时间戳、结束时间戳和动作标签。共有 3,568 个唯一动作标签，每个标签由一个动词和一个名词类别组成，总计 97 个动词类别和 300 个名词类别。EK100 动作预判任务涉及从动作片段开始时间戳之前出现的视频片段（称为上下文）中预测名词、动词和动作（即联合预测动词和名词）。上下文结束与动作片段开始之间的间隔为预判时间，默认设置为 1 秒。鉴于给定上下文可能存在不同的未来动作，采用平均类别召回率-at-5 作为衡量性能的指标（Damen 等，2022）。

预测探针。在冻结的V-JEPA 2编码器和预测器之上训练一个注意力探针，以预测未来动作。具体而言，我们采样一个在动作开始前1秒结束的视频片段。该视频上下文被输入V-JEPA 2编码器。预测器接收编码器表示，以及对应于未来1秒帧的掩码标记，并预测未来视频帧的表示。预测器和编码器的输出沿标记维度拼接，并输入到一个注意力探针中，其架构与第5节中使用的类似，不同之处在于预测探针的最终交叉注意力层学习三个查询标记（而非一个），每个查询输出被馈送到不同的线性分类器，分别预测动作类别、动词类别和名词类别。对每个分类器独立应用焦点损失（Lin等，2017），然后在通过探针的共享注意力块反向传播之前求和。我们在附录D.1中提供了额外细节和评估超参数。

基线方法。我们将我们的模型与三个专门为动作预判训练的基线方法进行比较：InAViT（Roy 等，2024）是一种监督方法，利用显式的手物交互建模，

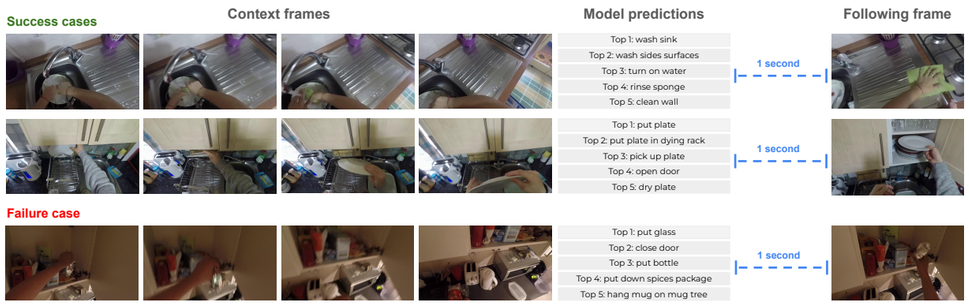



**Figure 11 Visualization of EK100 prediction.** (Left): four selected frames from the context frames. (Middle): model predictions, ordered by likelihood. (Right): following frame after the 1 second anticipation time. We show two examples where the model is successful and one example where the model fails.

and Video-LLaMA (Zhang et al., 2023) and PlausiVL (Mittal et al., 2024) are both approaches that leverage a large language model, with up to 7 billion parameters.

**Results.** Table 5 summarizes the results on the EK100 action anticipation benchmark. We compare V-JEPA 2 ViT-L, ViT-H and ViT-g encoders, increasing parameter count from 300 millions to 1 billion. All three leverage 32 frames with 8 frames per second at resolution $256 \times 256$ as video context. We also report results of ViT-$g_{384}$ which uses a resolution of $384 \times 384$. V-JEPA 2 demonstrates a linear scaling behavior with respect to model size, in terms of action prediction recall-at-5. V-JEPA 2 ViT-L with 300 million parameters achieves 32.7 recall-at-5. Increasing the size of the model to 1 billion parameters leads to a +5.3 point improvement with an action recall-at-5 of 38.0. Furthermore, V-JEPA 2 benefits from using a context with higher resolution, and V-JEPA 2 ViT-$g_{384}$ at resolution $384 \times 384$ improves recall-at-5 by an additional +1.7 points over the other models using $256 \times 256$ resolution.

V-JEPA 2 outperforms the previous state-of-the-art model PlausiVL by a significant margin, even with its 300 million parameters compared to the 8 billion parameters used in PlausiVL. In particular, V-JEPA 2 ViT-$g_{384}$ demonstrates a +12.1 points improvement over PlausiVL on action recall-at-5, corresponding to a 44% relative improvement.

In Figure 11 we visualize V-JEPA 2 predictions on three samples from the EK100 validation set, two where the model is successful and one where the model fails. For both successful examples, V-JEPA 2 not only retrieves the correct action correctly with top 1 confidence, but also proposes coherent top 2 to 5 actions, based on the given context. For example, in the top row, the correct action is "wash sink", but "turn on water" or "clean wall" would both have been valid actions given the presence of a tap and a wall. The model also predicts "rinse sponge", which is the current action being performed, probably assuming that this action could still be going on after 1 second. For the failure case, V-JEPA 2 still proposes coherent actions such as "close door" and "put down spices package", but misses the exact nature of the object: "tea package".

**Limitations.** V-JEPA 2 and the EK100 benchmark have several limitations. First, V-JEPA 2 does not fully solve EK100, there are failure cases where the model either gets the verb, the noun, or both wrong. We study the distribution of these failures in Appendix D.2. Second, we focus here on predicting actions with a 1 second anticipation time. The accuracy of V-JEPA 2 degrades when predicting at longer time horizons, see Appendix D.2. Third, the EK100 benchmark is limited to kitchen environments, with a closed well-defined vocabulary, and we do not know how well V-JEPA 2 generalizes to other environments. This limits the utility and applicability of models trained on EK100. Lastly, actions in EK100 are chosen from a fixed set of categories, making it impossible to generalize to action categories not present in the training set.



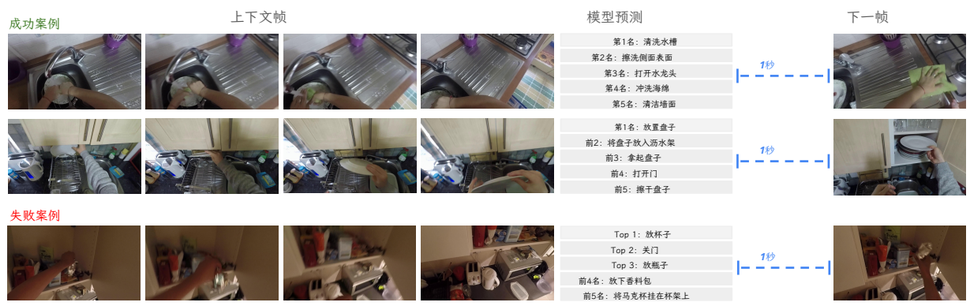



图11 EK100预测可视化。（左）：从上下文帧中选取的四帧。（中）：模型预测，按可能性排序。（右）：1秒预期时间之后的后续帧。我们展示了两个模型成功的示例和一个模型失败的示例。

而 Video-LLaMA（Zhang 等，2023）和 PlausiVL（Mittal 等，2024）都是利用大语言模型的方法，参数量高达 70 亿。

结果。表5总结了在EK100动作预测基准上的结果。我们比较了V-JEPA 2 ViT-L、ViT-H和ViT-g编码器，参数量从3亿增加到10亿。三者均利用32帧、每秒8帧、分辨率为 $256 \times 256$ 的视频上下文。我们还报告了使用分辨率 $384 \times 384$的ViT-$g_{384}$ 的结果。V-JEPA 2在动作预测recall-at-5方面展现出相对于模型规模的线性扩展行为。参数量为3亿的V-JEPA 2 ViT-L达到32.7的recall-at-5。将模型规模增大到10亿参数后，recall-at-5提升了 +5.3 个百分点，达到38.0。此外，V-JEPA 2受益于使用更高分辨率的上下文，分辨率为 $384 \times 384$ 的V-JEPA 2 ViT-$g_{384}$ 相比使用 $256 \times 256$ 分辨率的其他模型，recall-at-5进一步提升了 +1.7 个百分点。

V-JEPA 2 显著超越了先前的最先进模型 PlausiVL，即便其参数量仅为 3 亿，而 PlausiVL 使用了 80 亿参数。具体而言，V-JEPA 2 ViT-$g_{384}$ 在动作 recall-at-5 指标上相较 PlausiVL 展现出 +12.1 个百分点的提升，对应 44% 的相对改进。

在图 11 中，我们可视化了 V-JEPA 2 对 EK100 验证集三个样本的预测，其中两个样本模型预测成功，一个样本模型预测失败。对于两个成功示例，V-JEPA 2 不仅以 top-1 置信度正确检索到正确动作，还能基于给定上下文提出连贯的 top-2 至 top-5 动作。例如，在顶行中，正确动作是“洗水槽”，但考虑到水龙头和墙壁的存在，“打开水”或“清洁墙壁”也都会是合理的动作。模型还预测了“冲洗海绵”，这是当前正在执行的动作，可能是假设该动作在 1 秒后仍可能持续。对于失败案例，V-JEPA 2 仍然提出了连贯的动作，如“关门”和“放下香料包”，但错过了物体的确切性质：“茶包”。

局限性。V-JEPA 2 和 EK100 基准存在若干局限性。首先，V-JEPA 2 并未完全解决 EK100，存在模型在动词、名词或两者上均预测错误的失败案例。我们在附录 D.2 中研究了这些失败的分布。其次，我们在此聚焦于以 1 秒预见时间预测动作。当在更长的时间跨度上预测时，V-JEPA 2 的准确率会下降，见附录 D.2。第三，EK100 基准仅限于厨房环境，词汇封闭且定义明确，我们尚不清楚 V-JEPA 2 对其他环境的泛化能力如何。这限制了在 EK100 上训练的模型的实用性和适用性。最后，EK100 中的动作选自固定的类别集合，因此无法泛化到训练集中不存在的动作类别。

# 7 Understanding: Video Question Answering

In this section, we explore V-JEPA 2's ability to perform open-language video question answering (VidQA). To enable language capabilities, we train a Multimodal Large Language Model (MLLM) using V-JEPA 2 as the visual encoder in the *non-tokenized early fusion* (Wadekar et al., 2024) setup popularized by the LLaVA family of models (Li et al., 2024b). In this family of MLLMs, a visual encoder is *aligned* with a large language model by projecting the output patch embeddings of the vision encoder to the input embedding space of the LLM. The MLLM is then trained either end-to-end, or with a frozen vision encoder. The majority of the encoders used in MLLMs for VidQA are typically image encoders, which are applied independently per-frame for video inputs (Qwen Team et al., 2025; Zhang et al., 2024b). Popular instances of such encoders are CLIP (Radford et al., 2021), SigLIP (Tschannen et al., 2025), and Perception Encoder (Bolya et al., 2025), which are chosen primarily due to their semantic alignment with language, obtained by pretraining with image-caption pairs. To the best of our knowledge, our work is the first to use a video encoder that is pretrained *without* any language supervision, to train an MLLM for VidQA.

MLLM performance on downstream tasks is also highly dependent on the alignment data. In these experiments we use a dataset of 88.5 million image- and video-text pairs, similar to what was used to train PerceptionLM (Cho et al., 2025). To demonstrate the effectiveness of the V-JEPA 2 encoder, first we compare V-JEPA 2 with other state-of-the-art vision encoders in a *controlled* data setup in Section 7.2, using a subset of 18 million samples. Then, in the same controlled setup, we show that scaling the vision encoder and input resolution size both consistently improve VidQA performance in Section 7.3. Finally, we scale the alignment data, using the full 88.5 million samples to test the limits of language alignment with V-JEPA 2 in Section 7.4. Our results demonstrate that in a controlled data setup, V-JEPA 2 obtains competitive performance on open-ended VidQA tasks compared to other vision encoders. Upon scaling the alignment data, V-JEPA 2 achieves state-of-the-art performance on several VidQA benchmarks.

## 7.1 Experiment Setup

**Video Question Answering Tasks.** We evaluate on PerceptionTest (Pătrăucean et al., 2023), which assesses model performance across different skills such as memory, abstraction, physics, and semantics. Additionally, we evaluate on the MVP dataset (Krojer et al., 2024) for physical world understanding, which utilizes a minimal-video pair evaluation framework to mitigate text and appearance biases. We also evaluate on TempCompass, TemporalBench and TOMATO (Liu et al., 2024c; Cai et al., 2024; Shangguan et al., 2024) to investigate temporal understanding, and memory capabilities of models. Finally, we report results on general understanding ability using MVBench (Li et al., 2024c), which has a bias towards single-frame appearance features (Krojer et al., 2024; Cores et al., 2024), and TVBench (Cores et al., 2024), which is proposed in the literature as an alternative for general and temporal understanding, mitigating those biases.

**Visual Instruction Tuning.** To evaluate the V-JEPA 2 representations on visual-question answering tasks, we align V-JEPA 2 with an LLM using the visual instruction tuning procedure from the LLaVA framework (Liu et al., 2024a). This process involves converting the visual encoder outputs (or visual tokens) into LLM inputs using a learnable *projector* module, which is typically an MLP. We train MLLMs through a progressive three-stage process following Liu et al. (2024b): Stage 1, where we train the projector solely on image captioning data; Stage 2, where we train the full model on large-scale image question answering, and Stage 3, where we further train the model on large-scale video captioning and question answering. Through this staged training approach, the LLM incrementally improves its understanding of visual tokens. The vision encoder can either be frozen or finetuned along with the rest of the MLLM. We explore both settings as freezing the vision encoder gives a cleaner signal about the quality of the visual features, while finetuning the vision encoder yields better overall performance. Further details of the visual instruction training are described in Appendix E.

## 7.2 Comparing with Image Encoders

To isolate the contribution of vision encoders to MLLM performance and compare with V-JEPA 2, we introduce a *controlled* setup: we train individual MLLMs with different state-of-the-art encoders using the



# 7 理解：视频问答

在本节中，我们探索 V-JEPA 2 执行开放语言视频问答（VidQA）的能力。为实现语言能力，我们采用由 LLaVA 系列模型（Li et al., 2024b）推广的非 token 化早期融合（Wadekar et al., 2024）方案，以 V-JEPA 2 作为视觉编码器来训练多模态大语言模型（MLLM）。在这类 MLLM 中，通过将视觉编码器的输出 patch 嵌入投影到 LLM 的输入嵌入空间，使视觉编码器与大语言模型对齐。随后，MLLM 要么进行端到端训练，要么在冻结视觉编码器的情况下训练。用于 VidQA 的 MLLM 中所使用的大多数编码器通常是图像编码器，它们对视频输入逐帧独立应用（Qwen Team et al., 2025; Zhang et al., 2024b）。此类编码器的常见实例包括 CLIP（Radford et al., 2021）、SigLIP（Tschannen et al., 2025）和 Perception Encoder（Bolya et al., 2025），选择它们主要是因其与语言的语义对齐，这种对齐通过图像-字幕对预训练获得。据我们所知，我们的工作是首个使用未经任何语言监督预训练的视频编码器来训练用于 VidQA 的 MLLM 的工作。

MLLM 在下游任务上的性能也高度依赖于对齐数据。在这些实验中，我们使用了一个包含 8850 万个图像-文本对和视频-文本对的数据集，与用于训练 PerceptionLM（Cho et al., 2025）的数据集类似。为了展示 V-JEPA 2 编码器的有效性，我们首先在 7.2 节中使用 1800 万个样本的子集，在受控数据设置下将 V-JEPA 2 与其他最先进的视觉编码器进行比较。然后，在相同的受控设置下，我们在 7.3 节中展示，扩大视觉编码器和输入分辨率均能持续提升 VidQA 性能。最后，我们在 7.4 节中扩大对齐数据规模，使用全部 8850 万个样本，以测试 V-JEPA 2 在语言对齐方面的极限。我们的结果表明，在受控数据设置下，V-JEPA 2 在开放式 VidQA 任务上相比其他视觉编码器取得了有竞争力的性能。在扩大对齐数据后，V-JEPA 2 在多个 VidQA 基准上达到了最先进的性能。

## 7.1 实验设置

视频问答任务。我们在 PerceptionTest（Pătrăucean 等，2023）上进行评估，该基准测试模型在记忆、抽象、物理和语义等不同技能方面的表现。此外，我们在 MVP 数据集（Krojer 等，2024）上进行评估以考察物理世界理解能力，该数据集采用最小视频对评估框架来缓解文本和外观偏见。我们还分别在 TempCompass、TemporalBench 和 TOMATO（Liu 等，2024c；Cai 等，2024；Shangguan 等，2024）上进行评估，以研究模型的时间理解和记忆能力。最后，我们使用 MVBench（Li 等，2024c）报告通用理解能力的结果，该基准偏向单帧外观特征（Krojer 等，2024；Cores 等，2024），并使用 TVBench（Cores 等，2024）报告结果，该基准在文献中被提出作为通用和时间理解的替代方案，以缓解上述偏见。

视觉指令调优。为在视觉问答任务上评估 V-JEPA 2 表征，我们采用 LLaVA 框架（Liu et al., 2024a）中的视觉指令调优流程，将 V-JEPA 2 与 LLM 对齐。该过程使用可学习的投影模块（通常为 MLP）将视觉编码器输出（或视觉 token）转换为 LLM 输入。我们遵循 Liu et al. (2024b) 的方法，通过渐进式三阶段流程训练 MLLM：阶段 1，仅使用图像描述数据训练投影模块；阶段 2，在大规模图像问答数据上训练完整模型；阶段 3，进一步在大规模视频描述和问答数据上训练模型。通过这种分阶段训练方式，LLM 逐步提升对视觉 token 的理解能力。视觉编码器可以冻结，也可以与 MLLM 其余部分一同微调。我们探索了这两种设置：冻结视觉编码器能更清晰地反映视觉特征的质量，而微调视觉编码器则能获得更好的整体性能。视觉指令训练的更多细节见附录 E。

## 7.2 与图像编码器的比较

为了分离视觉编码器对 MLLM 性能的贡献并与 V-JEPA 2 进行比较，我们引入了一个受控设置：我们使用不同的最先进编码器分别训练各个 MLLM，采用

**Table 6 Comparison between off-the-shelf image encoders and V-JEPA 2 in frozen encoder setting.** All experiments use the same LLM backbone (Qwen2-7B-Instruct), data, and training setup with a **frozen** vision encoder. PerceptionTest accuracy is reported on the validation set post SFT.

| Method | Params Enc / LLM | Avg. | PerceptionTest SFT / Acc | MVP Paired-Acc | TempCompass multi-choice | TemporalBench (MBA-short QA) | TVBench Acc | TOMATO Acc | MVBench Acc |
|---|---|---|---|---|---|---|---|---|---|
| *Off-the-shelf image encoders* | | | | | | | | | |
| **DINOv2 ViT-g**$_{518}$ | 1.1B/7B | 45.7 | 67.1 | 22.4 | 62.3 | 26.8 | 47.6 | 32.0 | 61.8 |
| **SigLIP2 ViT-g**$_{384}$ | 1.1B/7B | 48.1 | **72.4** | 26.2 | 66.8 | 25.7 | 48.7 | 33.2 | 64.0 |
| **PE ViT-G/14**$_{448}$ | 1.9B/7B | 49.1 | 72.3 | 26.7 | 67.0 | 27.5 | 51.6 | 34.0 | 64.7 |
| **V-JEPA 2 ViT-g**$_{512}$ | 1B/7B | **52.3** | 72.0 | **31.1** | **69.2** | **33.3** | **55.9** | **37.0** | **67.7** |

**Table 7 Scaling Vision Encoder Size and Resolution.** We scale the vision encoder from 300 million to 1 billion parameters and input resolution from 256 pixels to 512 pixels. All experiments use the same LLM backbone (Qwen2-7B-Instruct), data, and end-to-end training (unfrozen vision encoder) setup. PerceptionTest accuracy is reported on the validation set post SFT. Increasing V-JEPA 2 encoder scale and resolution improve average performances on VidQA tasks.

| Method | Params Enc / LLM | Avg. | PerceptionTest SFT / Acc | MVP Paired-Acc | TempCompass multi-choice | TemporalBench (MBA-short QA) | TVBench Acc | TOMATO Acc | MVBench Acc |
|---|---|---|---|---|---|---|---|---|---|
| *End-to-end Evaluation* | | | | | | | | | |
| **V-JEPA 2 ViT-L**$_{256}$ | 300M/7B | 51.7 | 74.6 | 32.3 | 70.1 | 30.2 | 50.9 | 36.5 | 67.1 |
| **V-JEPA 2 ViT-H**$_{256}$ | 600M/7B | 52.0 | 74.7 | 30.6 | 70.9 | 29.8 | 54.6 | 35.1 | 68.0 |
| **V-JEPA 2 ViT-g**$_{256}$ | 1B/7B | 52.3 | 75.5 | 31.9 | 70.7 | 28.3 | 54.2 | 37.3 | 68.3 |
| **V-JEPA 2 ViT-g**$_{384}$ | 1B/7B | 54.0 | 76.5 | 33.0 | **71.7** | **33.1** | 56.5 | **39.0** | 68.5 |
| **V-JEPA 2 ViT-g**$_{512}$ | 1B/7B | **54.4** | **77.7** | **33.7** | 71.6 | 32.3 | **57.5** | 38.5 | **69.5** |

*same* LLM backbone and training setup. In this controlled setup, we use Qwen2-7B-Instruct (Yang et al., 2024a) and freeze the vision encoder. We use 18 million image and video-text aligned samples. We first compare V-JEPA 2, pretrained at resolution 512×512 with DINOv2 (Oquab et al., 2023), SigLIP-2 (Tschannen et al., 2025), and Perception Encoder (Bolya et al., 2025).

We observe that V-JEPA 2 exhibits competitive performance in the frozen setup, outperforming DINOv2, SigLIP, and Perception Encoder (PE) in all of the tested benchmarks (Table 6) except PerceptionTest where V-JEPA 2 slightly underperforms SigLIP and PE. The improvement is especially noticeable on MVP, TemporalBench, and TVBench — benchmarks that are primarily focused on temporal understanding. Additionally, since we only change the vision encoder, we provide evidence that a video encoder trained without language supervision can outperform encoders trained with language supervision, in contrast to conventional wisdom (Tong et al., 2024; Li et al., 2024b; Liu et al., 2024d; Yuan et al., 2025). The results also indicate that using a video encoder instead of an image encoder for VidQA improves spatiotemporal understanding, highlighting the need to develop better video encoders.

### 7.3 Scaling Vision Encoder Size and Input Resolution

Prior work (Fan et al., 2025) suggests that scaling the vision encoder and input resolution significantly improves VQA performance for self-supervised image encoders. Thus, we scale V-JEPA 2 from 300M to 1B parameters and the input resolution from 256 to 512 pixels, and show the results in Table 7. When increasing vision encoder capacity from 300M to 1B parameters for a fixed input resolution of 256 pixels, we observe improvements of 0.9 points on PerceptionTest, 3.3 points on TVBench, and 1.2 points on MVBench. Additionally, increasing the input resolution to 512 pixels yields further improvements across all downstream tasks, such as an improvement of 2.2 points on PerceptionTest, 4.0 points on TemporalBench, and 3.3 points on TVBench. These results suggest that further scaling the vision encoder and input resolution is a promising



表 6 现成图像编码器与 V-JEPA 2 在冻结编码器设置下的比较。所有实验使用相同的 LLM 主干（Qwen2-7B-Instruct）、数据和训练设置，且视觉编码器保持冻结。PerceptionTest 准确率报告的是 SFT 后的验证集结果。

| 方法 | 参数量 编码器 / LLM | Avg. | PerceptionTest SFT / Acc | MVP Paired-Acc | TempCompass multi-choice | TemporalBench (MBA-short QA) | TVBench Acc | TOMATO Acc | MVBench Acc |
|---|---|---|---|---|---|---|---|---|---|
| 现成图像编码器 | | | | | | | | | |
| ***DINOv2 ViT-g***$_{518}$ | 1.1B/7B | 45.7 | 67.1 | 22.4 | 62.3 | 26.8 | 47.6 | 32.0 | 61.8 |
| ***SigLIP2 ViT-g***$_{384}$ | 1.1B/7B | 48.1 | **72.4** | 26.2 | 66.8 | 25.7 | 48.7 | 33.2 | 64.0 |
| ***PE ViT-G/14***$_{448}$ | 1.9B/7B | 49.1 | 72.3 | 26.7 | 67.0 | 27.5 | 51.6 | 34.0 | 64.7 |
| ***V-JEPA 2 ViT-g***$_{512}$ | 1B/7B | **52.3** | 72.0 | **31.1** | **69.2** | **33.3** | **55.9** | **37.0** | **67.7** |

表 7 视觉编码器规模与分辨率的扩展。我们将视觉编码器从 3 亿参数扩展到 10 亿参数，并将输入分辨率从 256 像素提升至 512 像素。所有实验均使用相同的 LLM 主干（Qwen2-7B-Instruct）、数据和端到端训练（视觉编码器不冻结）设置。PerceptionTest 准确率在 SFT 后的验证集上报告。增大 V-JEPA 2 编码器规模和分辨率可提升 VidQA 任务的平均性能。

| 方法 | 参数量 编码器 / LLM | Avg. | PerceptionTest SFT / Acc | MVP Paired-Acc | TempCompass multi-choice | TemporalBench (MBA-short QA) | TVBench Acc | TOMATO Acc | MVBench Acc |
|---|---|---|---|---|---|---|---|---|---|
| 端到端评估 | | | | | | | | | |
| ***V-JEPA 2 ViT-L***$_{256}$ | 300M/7B | 51.7 | 74.6 | 32.3 | 70.1 | 30.2 | 50.9 | 36.5 | 67.1 |
| ***V-JEPA 2 ViT-H***$_{256}$ | 600M/7B | 52.0 | 74.7 | 30.6 | 70.9 | 29.8 | 54.6 | 35.1 | 68.0 |
| ***V-JEPA 2 ViT-g***$_{256}$ | 1B/7B | 52.3 | 75.5 | 31.9 | 70.7 | 28.3 | 54.2 | 37.3 | 68.3 |
| ***V-JEPA 2 ViT-g***$_{384}$ | 1B/7B | 54.0 | 76.5 | 33.0 | **71.7** | **33.1** | 56.5 | **39.0** | 68.5 |
| ***V-JEPA 2 ViT-g***$_{512}$ | 1B/7B | **54.4** | **77.7** | **33.7** | 71.6 | 32.3 | **57.5** | 38.5 | **69.5** |

相同的 LLM 主干和训练设置。在此受控设置中，我们使用 Qwen2-7B-Instruct（Yang 等，2024a）并冻结视觉编码器。我们使用 1800 万个图像和视频-文本对齐样本。我们首先将 V-JEPA 2（以分辨率 512×512 预训练）与 DINOv2（Oquab 等，2023）、SigLIP-2（Tschannen 等，2025）和 Perception Encoder（Bolya 等，2025）进行比较。

我们观察到，V-JEPA 2 在冻结设置下展现出具有竞争力的性能，在所有测试基准（表 6）中均优于 DINOv2、SigLIP 和 Perception Encoder（PE），但在 PerceptionTest 上 V-JEPA 2 略逊于 SigLIP 和 PE。这一提升在 MVP、TemporalBench 和 TVBench 上尤为明显——这些基准主要侧重于时序理解。此外，由于我们仅更换了视觉编码器，我们提供了证据表明，未经语言监督训练的视频编码器可以优于经语言监督训练的编码器，这与传统认知相反（Tong et al., 2024; Li et al., 2024b; Liu et al., 2024d; Yuan et al., 2025）。结果还表明，在 VidQA 中使用视频编码器而非图像编码器可以改善时空理解，凸显了开发更好的视频编码器的必要性。

### *7.3* 缩放视觉编码器规模与输入分辨率

先前工作（Fan et al., 2025）表明，扩大视觉编码器和输入分辨率可显著提升自监督图像编码器的VQA性能。因此，我们将V-JEPA 2从300M参数扩展至1B参数，并将输入分辨率从256像素提升至512像素，结果如表7所示。当输入分辨率固定为256像素、视觉编码器容量从300M提升至1B参数时，我们观察到PerceptionTest提升0.9分、TVBench提升3.3分、MVBench提升1.2分。此外，将输入分辨率提升至512像素可在所有下游任务上带来进一步提升，例如PerceptionTest提升2.2分、TemporalBench提升4.0分、TVBench提升3.3分。这些结果表明，进一步扩展视觉编码器和输入分辨率是提升VidQA性能的一个有前景的

**Table 8 Comparison with state-of-the-art.** We use the full 88.5M-sample alignment dataset and train using the same methodology as PLM 8B Cho et al. (2025), using a Llama 3.1 backbone. We observe significant improvements in downstream evaluations, obtaining state-of-the-art results in the 8B model class. PerceptionTest accuracy is reported on the test set with SFT for V-JEPA 2; all other results are zero-shot.

| Method | Params Enc / LLM | Avg. | PerceptionTest Test Acc | MVP Paired-Acc | TempCompass multi-choice | TemporalBench (MBA-short QA) | TOMATO Acc | TVBench Acc | MVBench Acc |
|---|---|---|---|---|---|---|---|---|---|
| *≤ 8B Video Language Models Results Reported in the Literature* | | | | | | | | | |
| **InternVL-2.5** (Chen et al., 2024) | 300M/7B | 52.1 | 68.9 | 39.9 | 68.3 | 24.3 | 29.4 | 61.6 | 72.6 |
| **Qwen2VL** (Wang et al., 2024a) | 675M/7B | 47.0 | 66.9 | 29.2 | 67.9 | 20.4 | 31.5 | 46.0 | 67.0 |
| **Qwen2.5VL** (Qwen Team et al., 2025) | 1B/7B | 49.7 | 70.5 | 36.7 | 71.7 | 24.5 | 24.6 | 50.5 | 69.6 |
| **PLM 8B** (Cho et al., 2025) | 1B/8B | 56.7 | 82.7 | 39.7 | 72.7 | 28.3 | 33.2 | **63.5** | **77.1** |
| **V-JEPA 2 ViT-g$_{384}$ LLama 3.1 8B** | 1B/8B | **59.5** | **84.0** | **44.5** | **76.9** | **36.7** | **40.3** | 60.6 | 73.5 |

direction for improving VidQA performance.

## 7.4 Improving the State-of-the-art by Scaling Data

After developing a better understanding of the capabilities of V-JEPA 2 for training an MLLM in the controlled setup, we study the effect of increasing alignment dataset size to improve the state-of-the-art of VidQA. Step changes on downstream task performance are often achieved by increasing the scale of the training data, as observed by Cho et al. (2025). To that end, we increase the scale of MLLM training data from 18 million to the full 88.5 million (4.7×). While increasing the model resolution helps in downstream performance, it comes with the challenge of accommodating a large number of visual tokens in the LLM input. We therefore choose V-JEPA 2 ViT-g$_{384}$, leading to 288 visual tokens per frame. We follow the same recipe as Cho et al. (2025) to train V-JEPA 2 ViT-g$_{384}$, using Llama 3.1 as the backbone. To simplify the training process, we use an MLP projector without pooling. Details on the scaling training setup are described in Appendix E.

Scaling the data uniformly improves the downstream benchmark performance, resulting in state-of-the-art results (Table 8) on multiple benchmarks — PerceptionTest, MVP, TempCompass, TemporalBench and TOMATO. Compared to the current state-of-the-art PerceptionLM 8B (Cho et al., 2025), we observe an increase of 1.3 points on accuracy for PerceptionTest test set, 4.8 points on paired accuracy for MVP, 4.2 points on accuracy for TempCompass, 8.4 points on Multi-binary accuracy for short-QA segment for TemporalBench and 7.1 points on accuracy for TOMATO. V-JEPA 2 does not outperform PerceptionLM on TVBench and MVBench, however it still significantly outperforms other related baselines (InternVL 2.5, Qwen2VL and Qwen2.5VL). These results underscore the need to scale training data for vision-language alignment and provide evidence that an encoder pretrained without language supervision, such as V-JEPA 2, can achieve state-of-the-art results with sufficient scale.

# 8 Related Work

**World models and planning.** As early as the work of Sutton and Barto (1981) and Chatila and Laumond (1985), AI researchers have sought to build agents that use internal models of the world — modeling both dynamics of the world, as well as mapping the static environment — to enable efficient planning and control. Previous work has investigated world models in simulated tasks (Fragkiadaki et al., 2015; Ha and Schmidhuber, 2018; Hafner et al., 2019b,a; Hansen et al., 2022, 2023; Hafner et al., 2023; Schrittwieser et al., 2020; Samsami et al., 2024), as well as real-world locomotion and manipulation tasks (Lee et al., 2020; Nagabandi et al., 2020; Finn et al., 2016; Ebert et al., 2017, 2018; Yen-Chen et al., 2020). World model approaches either learn predictive models directly in pixel-space (Finn et al., 2016; Ebert et al., 2017, 2018; Yen-Chen et al., 2020), in a learned representation space (Watter et al., 2015; Agrawal et al., 2016; Ha and Schmidhuber, 2018; Hafner et al., 2019b; Nair et al., 2022; Wu et al., 2023b; Tomar et al., 2024; Hu et al., 2024; Lancaster et al., 2024), or utilizing more structured representation spaces such as keypoint representations (Manuelli et al., 2020; Das



表 8 与最先进方法的比较。我们使用完整的 8850 万样本对齐数据集，并采用与 PLM 8B Cho et al. (2025) 相同的方法进行训练，使用 Llama 3.1 作为骨干网络。我们观察到下游评估有显著提升，在 8B 模型类别中取得了最先进的结果。PerceptionTest 准确率是在 V-JEPA 2 使用 SFT 的测试集上报告的；所有其他结果均为零样本。

| 方法 | 参数量 编码器/LLM | Avg. | PerceptionTest Test Acc | MVP Paired-Acc | TempCompass multi-choice | TemporalBench (MBA-short QA) | TOMATO Acc | TVBench Acc | MVBench Acc |
|---|---|---|---|---|---|---|---|---|---|
| ≤ 8B 视频语言模型文献中报告的结果 | | | | | | | | | |
| InternVL-2.5（Chen 等，2024） | 300M/7B | 52.1 | 68.9 | 39.9 | 68.3 | 24.3 | 29.4 | 61.6 | 72.6 |
| Qwen2VL（Wang 等，2024a） | 675M/7B | 47.0 | 66.9 | 29.2 | 67.9 | 20.4 | 31.5 | 46.0 | 67.0 |
| Qwen2.5VL（Qwen 团队等，2025） | 1B/7B | 49.7 | 70.5 | 36.7 | 71.7 | 24.5 | 24.6 | 50.5 | 69.6 |
| PLM 8B（Cho 等，2025） | 1B/8B | 56.7 | 82.7 | 39.7 | 72.7 | 28.3 | 33.2 | **63.5** | **77.1** |
| ***V-JEPA 2 ViT-g$_{384}$ LLama 3.1 8B*** | 1B/8B | **59.5** | **84.0** | **44.5** | **76.9** | **36.7** | **40.3** | 60.6 | 73.5 |

改进VidQA性能的方向。

## *7.4* 通过扩展数据提升最先进水平

在更好地理解了 V-JEPA 2 在受控设置下训练 MLLM 的能力之后，我们研究了增加对齐数据集规模以提升 VidQA 最先进水平的效果。正如 Cho 等人（2025）所观察到的，下游任务性能的阶跃式提升通常通过增加训练数据的规模来实现。为此，我们将 MLLM 训练数据的规模从 1800 万增加到完整的 8850 万（4.7×）。虽然提高模型分辨率有助于下游性能，但随之而来的挑战是在 LLM 输入中容纳大量视觉 token。因此，我们选择 V-JEPA 2 ViT-g$_{384}$, ，每帧产生 288 个视觉 token。我们遵循 Cho 等人（2025）的相同方案，使用 Llama 3.1 作为骨干来训练 V-JEPA 2 ViT-g$_{384}$, 。为简化训练过程，我们使用不带池化的 MLP 投影器。有关扩展训练设置的详细信息在附录 E 中描述。

均匀扩展数据能够提升下游基准性能，从而在多个基准上取得最先进的结果（表8）——PerceptionTest、MVP、TempCompass、TemporalBench 和 TOMATO。与当前最先进的 PerceptionLM 8B（Cho 等，2025）相比，我们观察到 PerceptionTest 测试集准确率提升 1.3 分，MVP 配对准确率提升 4.8 分，TempCompass 准确率提升 4.2 分，TemporalBench 短问答片段多二分类准确率提升 8.4 分，TOMATO 准确率提升 7.1 分。V-JEPA 2 在 TVBench 和 MVBench 上未能超越 PerceptionLM，但仍显著优于其他相关基线（InternVL 2.5、Qwen2VL 和 Qwen2.5VL）。这些结果凸显了扩展训练数据以实现视觉-语言对齐的必要性，并提供了证据表明，像 V-JEPA 2 这样在没有语言监督下预训练的编码器，在足够规模下能够取得最先进的结果。

# *8* 相关工作

世界模型与规划。早在 Sutton 和 Barto（1981）以及 Chatila 和 Laumond（1985）的工作中，AI 研究者就试图构建使用世界内部模型的智能体——既对世界的动态进行建模，也对静态环境进行映射——以实现高效的规划与控制。先前的工作在模拟任务中研究了世界模型（Fragkiadaki 等，2015；Ha 和 Schmidhuber，2018；Hafner 等，2019b,a；Hansen 等，2022，2023；Hafner 等，2023；Schrittwieser 等，2020；Samsami 等，2024），也在真实世界的运动与操作任务中进行了研究（Lee 等，2020；Nagabandi 等，2020；Finn 等，2016；Ebert 等，2017，2018；Yen-Chen 等，2020）。世界模型方法要么直接在像素空间中学习预测模型（Finn 等，2016；Ebert 等，2017，2018；Yen-Chen 等，2020），要么在学习到的表示空间中学习（Watter 等，2015；Agrawal 等，2016；Ha 和 Schmidhuber，2018；Hafner 等，2019b；Nair 等，2022；Wu 等，2023b；Tomar 等，2024；Hu 等，2024；Lancaster 等，2024），或者利用更具结构化的表示空间，例如关键点表示（Manuelli 等，2020；Das

et al., 2020). Previous approaches that have demonstrated real world performance on robotics tasks have trained task-specific world models, and they rely on interaction data from the environment in which the robot is deployed. Evaluation is focused on demonstrating performance of world modeling approaches within the explored task space, instead of generalization to new environments or unseen objects. In this work we train a task-agnostic world model, and demonstrate generalization to new environments and objects.

Some recent works leverage both internet-scale video and interaction data towards training general purpose (task-agnostic) action-conditioned video generation models for autonomous robots (Bruce et al., 2024; Agarwal et al., 2025; Russell et al., 2025). However, thus far these approaches only demonstrate the ability to generate visually valid-looking plans given actions of the robot, but they have not demonstrated the ability to use those models to actually control the robot.

Other works have explored the integration of generative modeling into policy learning (Du et al., 2024; Wu et al., 2023a; Zhao et al., 2025; Zhu et al., 2025; Du et al., 2023; Zheng et al., 2025; Rajasegaran et al., 2025). Differently from this line of work, our goal is to leverage a world model through model-predictive control instead of policy learning to avoid the imitation learning phase that requires expert trajectories. Both approaches are orthogonal and could be combined in future works. Closest to our work, Zhou et al. (2024); Sobal et al. (2025) show that you can learn a world model stage-wise or end-to-end and use it to solve planning tasks zero-shot. While those previous works focus on small-scale planning evaluation, we show that similar principles can be scaled and used to solve real-world robotic tasks.

**Vision-Language-Action models for Robotic Control.** Recent imitation learning approaches in real-world robotic control have made significant progress towards learning policies that show increasingly good generalization capabilities. This is achieved by leveraging video-languange models that have been pre-trained on internet scale video and text data, which are then fine-tuned (or adapted) to also predict actions by using behavior cloning from expert demonstrations (Driess et al., 2023; Brohan et al., 2023; Black et al., 2024; Kim et al., 2024; Bjorck et al., 2025; Black et al., 2025). Although these approaches show promising generalization results, it is unclear whether they will be able to learn to predict behaviors that were not demonstrated in the training data since they lack an explicit predictive model of the world and do leverage inference-time computation for planning. They require high-quality large scale teleoperation data, and can only utilize successful trajectories. In contrast, we focus on leveraging any interaction data whether it comes from a successful or failed interaction with the environment.

**Vision Foundation Models.** Video foundation models in computer vision have shown that large-scale observation datasets comprised of images and/or videos can be leveraged to learn generalist vision encoders that perform well along a wide range of downstream tasks using self-supervised learning approaches from images (Grill et al., 2020; Assran et al., 2023; Oquab et al., 2023; Fan et al., 2025), videos (Bardes et al., 2024; Carreira et al., 2024; Wang et al., 2023; Rajasegaran et al., 2025), with weak language supervision (Wang et al., 2024b; Bolya et al., 2025), or a combination thereof (Tschannen et al., 2025; Fini et al., 2024). Previous works, however, tend to focus on understanding evaluation using probe-based evaluation or visual question answering tasks after aligning with a large-language model. While such tasks have served to drive progress, it remains an important goal of a visual system to enable an agent to interact with the physical world (Gibson, 1979). Beyond results on visual understanding tasks, we investigate how large-scale self-supervised learning from video can enable solving planning tasks in new environments in a zero-shot manner.



等，2020）。先前在机器人任务上展示了真实世界性能的方法训练的是任务特定的世界模型，并且依赖于机器人所部署环境中的交互数据。评估的重点是展示世界建模方法在所探索任务空间内的性能，而非对新环境或未见物体的泛化能力。在本工作中，我们训练了一个任务无关的世界模型，并展示了其对新环境和物体的泛化能力。

近期一些工作利用互联网规模的视频和交互数据，面向自主机器人训练通用（任务无关的）动作条件视频生成模型（Bruce et al., 2024; Agarwal et al., 2025; Russell et al., 2025）。然而，迄今为止这些方法仅展示了在给定机器人动作的情况下生成视觉上合理的计划的能力，但尚未证明能够利用这些模型实际控制机器人。

其他工作探索了将生成建模集成到策略学习中（Du et al., 2024; Wu et al., 2023a; Zhao et al., 2025; Zhu et al., 2025; Du et al., 2023; Zheng et al., 2025; Rajasegaran et al., 2025）。与这一研究方向不同，我们的目标是通过模型预测控制而非策略学习来利用世界模型，以避免需要专家轨迹的模仿学习阶段。两种方法是正交的，未来可以结合。与我们的工作最接近的是，Zhou et al. (2024); Sobal et al. (2025) 表明可以分阶段或端到端地学习世界模型，并将其用于零样本求解规划任务。虽然这些先前的工作聚焦于小规模规划评估，但我们表明类似的原则可以扩展并用于求解真实世界的机器人任务。

用于机器人控制的视觉-语言-动作模型。近期在真实世界机器人控制中的模仿学习方法在策略学习方面取得了显著进展，展现出越来越好的泛化能力。这是通过利用在互联网规模视频和文本数据上预训练的视觉-语言模型实现的，这些模型随后通过使用专家演示的行为克隆进行微调（或适配）以预测动作（Driess et al., 2023; Brohan et al., 2023; Black et al., 2024; Kim et al., 2024; Bjorck et al., 2025; Black et al., 2025）。尽管这些方法展现出有前景的泛化结果，但尚不清楚它们是否能够学会预测训练数据中未曾演示过的行为，因为它们缺乏显式的世界预测模型，也未利用推理时的计算来进行规划。它们需要高质量的大规模遥操作数据，并且只能利用成功的轨迹。相比之下，我们专注于利用任何交互数据，无论其来自与环境的成功交互还是失败交互。

视觉基础模型。计算机视觉中的视频基础模型已经表明，由图像和/或视频构成的大规模观测数据集，可以被用来学习通用视觉编码器，这些编码器通过基于图像的自监督学习方法（Grill et al., 2020; Assran et al., 2023; Oquab et al., 2023; Fan et al., 2025）、基于视频的自监督学习方法（Bardes et al., 2024; Carreira et al., 2024; Wang et al., 2023; Rajasegaran et al., 2025）、带有弱语言监督的方法（Wang et al., 2024b; Bolya et al., 2025），或上述方法的组合（Tschannen et al., 2025; Fini et al., 2024），在广泛的下游任务上表现良好。然而，以往的工作往往侧重于使用基于探针的评估或在与大语言模型对齐后的视觉问答任务来进行理解评估。尽管这类任务推动了进展，但使智能体能够与物理世界交互仍然是视觉系统的一个重要目标（Gibson, 1979）。除了视觉理解任务上的结果之外，我们还研究了从视频中进行大规模自监督学习如何能够以零样本方式解决新环境中的规划任务。

## 9 Conclusion

This study demonstrates how joint-embedding predictive architectures, learning in a self-supervised manner from web-scale data and a small amount of robot interaction data, can yield a world model capable of understanding, predicting, and planning in the physical world. V-JEPA 2 achieves state-of-art performances on action classification requiring motion understanding and human action anticipation. V-JEPA 2 also outperforms previous vision encoders on video questions-answering tasks when aligned with a large-language model. Additionally, post-training an action-conditioned world model, V-JEPA 2-AC, using V-JEPA 2's representation, enables successful zero-shot prehensile manipulation tasks, such as Pick-and-Place, with real-world robots. These findings indicate V-JEPA 2 is a step towards developing advanced AI systems that can effectively perceive and act in their environment.

**Future work.** There are several important avenues for future work to address limitations of V-JEPA 2. First, in this work we have focused on tasks requiring predictions up to roughly 16 seconds into the future. This enables planning for simpler manipulation tasks, like grasp and reach-with-object, from a single goal image. However, to extend this to longer-horizon tasks such as pick-and-place or even more complex tasks, without requiring sub-goals will require further innovations in modeling. Developing approaches for hierarchical models capable of making predictions across multiple spatial and temporal scales, at different levels of abstraction, is a promising direction.

Second, as mentioned in Section 4, V-JEPA 2-AC currently relies upon tasks specified as image goals. Although this may be natural for some tasks, there are other situations where language-based goal specification may be preferable. Extending the V-JEPA 2-AC to accept language-based goals, e.g., by having a model that can embed language-based goals into the V-JEPA 2-AC representation space, is another important direction for future work. The results described in Section 7, aligning V-JEPA 2 with a language model, may serve as a starting point.

Finally, in this work we scaled V-JEPA 2 models up to a modest 1B parameters. The results in Section 2 demonstrated consistent performance improvements while scaling to this level. Previous work has investigated scaling vision encoders to as large as 20B parameters (Zhai et al., 2022; Carreira et al., 2024). Additional work is needed in this direction to develop scalable pre-training recipes that lead to sustained performance improvements with scale.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## 9 结论

本研究展示了联合嵌入预测架构如何通过从网络规模数据和少量机器人交互数据中以自监督方式进行学习，产生一个能够在物理世界中理解、预测和规划的世界模型。V-JEPA 2 在需要运动理解的动作分类和人类动作预测任务上达到了最先进的性能。当与大语言模型对齐时，V-JEPA 2 在视频问答任务上也优于以往的视觉编码器。此外，利用 V-JEPA 2 的表征对动作条件世界模型 V-JEPA 2-AC 进行后训练，使真实世界机器人能够成功完成零样本抓取操作任务，例如拾取与放置。这些发现表明，V-JEPA 2 是朝着开发能够在环境中有效感知和行动的高级 AI 系统迈出的一步。

未来工作。未来有若干重要方向可以解决 V-JEPA 2 的局限性。首先，在本工作中，我们聚焦于需要预测未来大约 16 秒的任务。这使得能够从单张目标图像出发，为较简单的操作任务（如抓取和带物体伸取）进行规划。然而，要将其扩展到更长时域的任务（如拾取放置，甚至更复杂的任务），而不需要子目标，则需要在建模方面进一步的创新。开发能够跨多个空间和时间尺度、在不同抽象层次上进行预测的分层模型方法，是一个有前景的方向。

其次，如第 4 节所述，V-JEPA 2-AC 目前依赖于以图像目标指定的任务。尽管这对某些任务可能是自然的，但在其他情况下，基于语言的目标指定可能更为可取。扩展 V-JEPA 2-AC 以接受基于语言的目标——例如，通过让一个模型能够将基于语言的目标嵌入到 V-JEPA 2-AC 的表示空间中——是未来工作的另一个重要方向。第 7 节中描述的将 V-JEPA 2 与语言模型对齐的结果，可以作为起点。

最后，在这项工作中，我们将 V-JEPA 2 模型扩展到了适度的 10 亿参数规模。第 2 节的结果表明，在扩展至这一规模的过程中，性能持续提升。先前的工作已将视觉编码器扩展至高达 200 亿参数（Zhai 等，2022；Carreira 等，2024）。这一方向仍需开展更多工作，以开发出能够随规模持续带来性能提升的可扩展预训练方案。


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# Appendix

## A V-JEPA 2 Pretraining

### A.1 Pretraining Hyperparameters

As detailed in Section 2.4, our training pipeline consisted of two phases: 1) a constant learning rate phase and 2) a cooldown phase. For all models, we trained in the first phase until we observed plateauing or diminishing performance on the IN1K, COIN, and SSv2 tasks. At this point, we initiated the cooldown phase.

**Table 9 Pretraining Hyperparameters.** Common parameters for pretraining large computer vision models. We report these parameters for both the primary training phase and the cooldown phase.

| Parameter | Primary Phase | Cooldown Phase |
|---|---|---|
| Number of frames | 16 | 64 |
| Frames per Second | 4.0 | 4.0 |
| Crop Size | 256 | [256, 384, 512] |
| Random Resize Aspect Ratio | [0.75 1.35] | [0.75, 1.35] |
| Random Resize Scale | [0.3, 1.0] | [0.3, 1.0] |
| Steps | Variable | 12000 |
| Warmup Steps | 12000 | N/A |
| Batch Size (global) | 3072 | 3072 |
| Starting Learning Rate | 1e-4 | 5.25e-4 |
| Final Learning Rate | 5.25e-4 | 1e-6 |
| Weight Decay | 0.04 | 0.04 |
| EMA | 0.99925 | 0.99925 |
| Spatial Mask Scale | [0.15, 0.7] | [0.15, 0.7] |
| Temporal Mask Scale | [1.0, 1.0] | [1.0, 1.0] |
| Mask Aspect Ratio | [0.75 1.5] | [0.75, 1.5] |
| Tubelet Size | 2 | 2 |
| Patch Size | 16 | 16 |

Training in the first phase began with a learning rate warmup for 12,000 steps followed by a constant learning rate for the rest of the phase. We checked evaluations every 60,000 steps. The cooldown phase began with a learning rate at 5.25e-4, which was linearly ramped down to the final learning rate. Throughout both phases, all other hyperparameters were kept constant.

In the cooldown phase we increased the number of frames per clip while keeping the frames-per-second constant, as we saw a substantial benefit from feeding the model more frames (see Figure 5). In addition, we also increased the crop size of the model in this phase, which gave a substantial benefit to tasks like IN1K, which goes from 84.6 at a 256 crop to 85.1 at a 384 crop. Hyperparameters for both phases are summarized in Table 9.

Throughout the Appendix, we refer to a "abbreviated" training recipe that corresponds to a 90,000-step training following the procedure of Bardes et al. (2024). There are a few key differences with the abbreviated recipe. The first is the learning rate: the abbreviated recipe begins with a linear warmup followed by a cosine decay. The second are the schedules for weight decay and EMA, which are linearly ramped from a starting to a final value. The last is the total number of steps, which is restricted to 90,000. We use the abbreviated schedule for several of our ablations on data mixtures, as this allows us to interrogate the effects of data curation on a shorter compute budget.

### A.2 Pretraining data

We began curation of YT1B by applying scene extraction via the PySceneDetect library,[3] which splits videos into clips at scene transitions. We discard scenes shorter than 4 seconds, retaining 316 million scenes. The

[3] https://github.com/Breakthrough/PySceneDetect



# 附录

## *A V-JEPA2* 预训练

### *A.1* 预训练超参数

如第 2.4 节所述，我们的训练流程包含两个阶段：1) 恒定学习率阶段和 2) 冷却阶段。对于所有模型，我们在第一阶段持续训练，直到观察到在 IN1K、COIN 和 SSv2 任务上的性能趋于平稳或下降。此时，我们启动冷却阶段。

表 9 预训练超参数。用于预训练大型计算机视觉模型的通用参数。我们报告了主要训练阶段和冷却阶段的这些参数。

| 参数 | 主阶段 | 冷却阶段 |
|---|---|---|
| 帧数 | 16 | 64 |
| 每秒帧数 | 4.0 | 4.0 |
| 裁剪尺寸 | 256 | [256, 384, 512] |
| 随机缩放宽高比 | [0.75 1.35] | [0.75, 1.35] |
| 随机缩放比例 | [0.3, 1.0] | [0.3, 1.0] |
| 步数 | 变量 | 12000 |
| 预热步数 | 12000 | N/A |
| 批大小（全局） | 3072 | 3072 |
| 初始学习率 | 1e-4 | 5.25e-4 |
| 最终学习率 | 5.25e-4 | 1e-6 |
| 权重衰减 | 0.04 | 0.04 |
| EMA | 0.99925 | 0.99925 |
| 空间掩码比例 | [0.15, 0.7] | [0.15, 0.7] |
| 时间掩码比例 | [1.0, 1.0] | [1.0, 1.0] |
| 掩码宽高比 | [0.75 1.5] | [0.75, 1.5] |
| 管状块尺寸 | 2 | 2 |
| 补丁大小 | 16 | 16 |

第一阶段的训练以学习率预热开始，持续12,000步，随后在剩余阶段保持恒定学习率。我们每60,000步进行一次评估检查。冷却阶段以5.25e-4的学习率开始，并线性衰减至最终学习率。在这两个阶段中，所有其他超参数均保持不变。

在冷却阶段，我们增加了每个片段的帧数，同时保持帧率不变，因为我们发现向模型输入更多帧能带来显著收益（见图5）。此外，我们还在这一阶段增加了模型的裁剪尺寸，这对IN1K等任务带来了显著收益，从256裁剪下的84.6提升到384裁剪下的85.1。两个阶段的超参数总结在表9中。

在附录中，我们引用了一种"简化版"训练配方，对应于遵循Bardes等人（2024）流程的90,000步训练。与简化版配方相比有几个关键差异。第一个是学习率：简化版配方以线性预热开始，随后进行余弦衰减。第二个是权重衰减和EMA的调度策略，它们从起始值到最终值线性递增。最后是总步数，限制为90,000步。我们在多项数据混合消融实验中使用了简化版调度，因为这使我们能够在更短的计算预算下探究数据整理的效果。

### *A.2* 预训练数据

我们通过使用 PySceneDetect 库进行场景提取来开始对 YT1B 的整理，[3] 该库在场景转换处将视频分割成片段。我们丢弃短于 4 秒的场景，保留 3.16 亿个场景。

[3] https://github.com/Breakthrough/PySceneDetec t

**Table 10 Abbreviated Pretraining Hyperparameters.** Common parameters for pretraining large computer vision models, targeting our abbreviated recipe.

| Parameter | Abbreviated Recipe |
|---|---|
| Number of frames | 16 |
| Frames per Second | 4.0 |
| Crop Size | 256 |
| Random Resize Aspect Ratio | [0.75 1.35] |
| Random Resize Scale | [0.3, 1.0] |
| Steps | 90000 |
| Warmup Steps | 12000 |
| Batch Size (global) | 3072 |
| Starting Learning Rate | 2e-4 |
| Larning Rate | 6.25e-4 |
| Final Learning Rate | 1e-6 |
| Starting Weight Decay | 0.04 |
| Final Weight Decay | 0.4 |
| Starting EMA | 0.999 |
| Final EMA | 1.0 |
| Spatial Mask Scale | [0.15, 0.7] |
| Temporal Mask Scale | [1.0, 1.0] |
| Mask Aspect Ratio | [0.75 1.5] |
| Tubelet Size | 2 |
| Patch Size | 16 |

DINOv2 ViT-L model is then applied on the middle frame of each clip to extract scene embeddings. YT1B embeddings are then clustered into 1.5 million clusters, using the same clustering strategy as Oquab et al. (2023). Embeddings are also extracted in the same manner for all videos in the target distribution, then assigned to the closest YT1B cluster. We only keep those clusters to which at least one target video was assigned—about 210k clusters out of the original 1.5 million. The retained clusters contain 115 million scenes. Cluster-based retrieval matches the support, but not the weighting, of the target distribution. We use a weighted sampling scheme to rebalance the data to better match the target distribution. W sample from the clusters using a weighted sampling strategy: $w_c = \sum_{d=1}^{D} w_d \times \frac{N_{d,c}}{N_d}$ where $w_c$ is the weighting coefficient for the $c$th cluster, $w_d$ is the weighting coefficient for the $d$th target dataset (from Table 11, $N_{d,c}$ is the number of samples from the $d$th dataset in the $c$th cluster, $N_d$ is the total number of samples in the $d$th dataset, and $D$ is the total count of target datasets. We assigned the retrieval weights approximately based on how many scenes were retrieved by each target dataset, with some extra weighting assigned to EpicKitchen. This gave a final curated dataset with statistics more closely matching those of handcrafted datasets from the literature. We found that in isolation, using curated YT1B in place of its uncurated counterpart gave much better results on downstream understanding tasks (see Figure 4).

**Table 11 Data Curation Statistics.** We summarized the number of extracted scenes and hours of videos across clusters extracted from YT1B. The final line includes duplicates among retrievals of K710, SSv2, COIN, and EpicKitchen.

| Retrieval Target | Cluster Count | Number of Scenes | Retrieval Weight |
|---|---|---|---|
| Uncurated YT1B | 1.5M | 316M | |
| K710 | 170k | 100M | 0.7 |
| SSv2 | 41k | 19M | 0.125 |
| COIN | 37k | 21M | 0.125 |
| EpicKitchen | 4k | 13k | 0.05 |
| Final Curated (includes duplicates) | 210k | 115M | |

The overall statistics from how many clusters and scenes were retrieved with this strategy are summarized in Table 11. The overall dataset is weighted towards clusters retrieved with K710. This, combined with its retrieval weight of 0.7, gives the overall curated dataset a heavy Kinetics weighting that we saw reflected in



表 ***10*** 简略版 预训练超参数。用于预训练大型计算机视觉模型的通用参数，针对我们的 n
模型，针对我们的 简略配方。

| 参数 | 简略配方 |
|---|---|
| 帧数 | 16 |
| 每秒帧数 | 4.0 |
| 裁剪尺寸 | 256 |
| 随机缩放宽高比 | [0.75 1.35] |
| 随机缩放比例 | [0.3，1.0] |
| 步数 | 90000 |
| 预热步数 | 12000 |
| 批量大小（全局） | 3072 |
| 初始学习率 | 2e-4 |
| 学习率 | 6.25e-4 |
| 最终学习率 | 1e-6 |
| 初始权重衰减 | 0.04 |
| 最终权重衰减 | 0.4 |
| 初始EMA | 0.999 |
| 最终EMA | 1.0 |
| 空间掩码尺度 | [0.15，0.7] |
| 时间掩码尺度 | [1.0，1.0] |
| 掩码宽高比 | [0.75 1.5] |
| Tubelet 大小 | 2 |
| Patch 大小 | 16 |

然后，将 DINOv2 ViT-L 模型应用于每个片段的中间帧以提取场景嵌入。随后，使用与 Oquab 等人（2023）相同的聚类策略，将 YT1B 嵌入聚类为 150 万个簇。对于目标分布中的所有视频，也以相同方式提取嵌入，然后将其分配到最近的 YT1B 簇。我们仅保留至少有一个目标视频被分配到的那些簇——在原始 150 万个簇中约有 21 万个簇。保留的簇包含 1.15 亿个场景。

基于聚类的检索与目标分布的支撑集相匹配，但未匹配其权重。我们采用加权采样方案来重新平衡数据，以更好地匹配目标分布。我们从聚类中进行加权采样：$w_c = \sum_{d=1}^{D} \frac{w_d}{N_d} \times {}^{N_d}, {}_{N_d}{}^{c}$ 其中 $w_c$ 是第 $c$ 个聚类的加权系数，$w_d$ 是第 $d$ 个目标数据集的加权系数（来自表 11，$\frac{}{N_d}$，$_c$ 是第 $c$ 个聚类中来自第 $d$ 个数据集的样本数，$N_d$ 是第 $d$ 个数据集中的样本总数，$D$ 是目标数据集的总数。我们根据每个目标数据集检索到的场景数量来近似分配检索权重，并对 EpicKitchen 赋予额外权重。由此得到的最终精选数据集，其统计特性更接近文献中手工构建的数据集。我们发现，在单独使用时，用精选的 YT1B 替代其未精选版本，在下游理解任务上取得了更好的结果（见图 4）。

表 11 数据整理统计。我们汇总了从 YT1B 中提取的各聚类所对应的提取场景数量和视频小时数。最后一行包含 K710、SSv2、COIN 和 EpicKitchen 检索结果之间的重复项。

| 检索目标 | 聚类数量 | 场景数量 | 检索权重 |
|---|---|---|---|
| 未整理的 YT1B | 1.5M | 316M | |
| K710 | 170k | 100M | 0.7 |
| SSv2 | 41k | 19M | 0.125 |
| COIN | 37k | 21M | 0.125 |
| EpicKitchen | 4k | 13k | 0.05 |
| 最终整理版（含重复项） | 210k | 115M | |

通过该策略检索到多少个聚类和场景的总体统计数据汇总在表11中。整体数据集偏向于通过K710检索到的聚类。这一点，加上其0.7的检索权重，使得整体精选数据集具有较重的Kinetics权重，我们在

K400 performance for our ablation experiments (see Appendix A.4.1). As shown in Table 1 in the main body, we combined this Curated YT1B with SSv2, Kinetics, HowTo100M, and ImageNet to create our final VM22M dataset.

## A.3 Scaling Model Size

Details of the model architecture are shown in Table 12. All models are parameterized as vision transformers Dosovitskiy et al. (2020), using the standard $16 \times 16$ patch size. When scaling model size, we increase the encoder from a ViT-L (300M parameters) to a ViT-g (1B parameters), while the predictor size is kept fixed across all pre-training experiments.

**Table 12 Model architecture details.** Family of encoders and predictor architectures used during V-JEPA 2 pretraining, with some of the major parameters.

| Model | Params | Width | Depth | Heads | MLP | Embedder |
|---|---|---|---|---|---|---|
| *Encoders:* $E_\theta(\cdot)$ | | | | | | |
| ViT-L | 300M | 1024 | 24 | 16 | 4096 | $2 \times 16 \times 16$ strided conv |
| ViT-H | 600M | 1280 | 32 | 16 | 5120 | $2 \times 16 \times 16$ strided conv |
| ViT-g | 1B | 1408 | 40 | 22 | 6144 | $2 \times 16 \times 16$ strided conv |
| *Predictor:* $P_\phi(\cdot)$ | | | | | | |
| ViT-s | 22M | 384 | 12 | 12 | 1536 | N.A. |

## A.4 Additional Results

### A.4.1 Effect of Data Curation

Table 13 shows the results of data curation on a subset of downstream classification tasks. For this table, we trained models at the ViT-L and ViT-g scale using the abbreviated training recipe of the original V-JEPA (Bardes et al., 2024). When training smaller scale models (ViT-L), training a model on the curated variant of YT1B leads to across-the-board improvements over the uncurated variant. However, when moving to a mixed data setting (i.e., adding images and hand-selected videos), performance actually drops for a subset of tasks when using curated data, with performance on SSv2 at 72.8 for VM22M (Mixed+Curated YT1B) vs. 73.3 for Mixed+Uncurated YT1B. In some cases, the model trained with Curated YT1B alone is better than the one with mixed data, such as on the COIN (86.5 vs. 86.25) and K400 (84.6 vs. 83.7) evaluation tasks. This result is somewhat surprising, as despite including the K710 training data in the Mixed setting, we find that it does not improve performance over Curated YT1B for the K400 evaluation task.

**Table 13 Effects of Data Curation on Video Understanding.** Results are reported at both the ViT-L and ViT-g model scales. Models at both scales were pretrained using the abbreviated schedule of Bardes et al. (2024).

| Training Data | IN1K | COIN | SSv2 | K400 |
|---|---|---|---|---|
| *ViT-L* | | | | |
| Uncurated YT1B | 80.6 | 83.2 | 70.9 | 82.9 |
| Curated YT1B | 80.8 | **86.5** | 73.1 | **84.6** |
| Mixed+Uncurated YT1B | **82.9** | 86.25 | **73.3** | 83.0 |
| VM22M | **82.9** | 86.0 | 72.8 | 83.7 |
| *ViT-g* | | | | |
| Uncurated YT1B | 81.8 | 86.4 | 73.6 | 85.1 |
| Curated YT1B | 81.7 | 88.4 | 74.8 | **86.5** |
| Mixed+Uncurated YT1B | 83.7 | 88.5 | 75.5 | 85.9 |
| VM22M | **83.9** | **89.2** | **75.6** | 86.2 |

However, this behavior is not constant across scales. At the ViT-g scale, VM22M (Mixed+Curated YT1B) outperforms Mixed+Uncurated YT1B on all tasks.



消融实验的K400性能中看到了这一反映（见附录A.4.1）。如正文表1所示，我们将此精选YT1B与SSv2、Kinetics、HowTo100M和ImageNet结合，创建了最终的VM22M数据集。

## *A.3* 模型规模扩展

模型架构的详细信息见表 12。所有模型均参数化为视觉变换器 Dosovitskiy *et al.* (2020)，使用标准 $16 \times 16$ patch 大小。在扩展模型规模时，我们将编码器从 ViT-L（300M 参数）增大到 ViT-g（1B 参数），而预测器大小在所有预训练实验中保持固定。

表 12 模型架构详情。V-JEPA 2 预训练期间使用的编码器族和预测器架构，以及部分主要参数。

| 模型 | 参数 | 宽度 | 深度 | 注意力头 | MLP | 嵌入器 |
|---|---|---|---|---|---|---|
| 编码器： $E_\theta(\cdot)$ | | | | | | |
| ViT-L | 300M | 1024 | 24 | 16 | 4096 | $2 \times 16 \times 16$ 步长卷积 |
| ViT-H | 600M | 1280 | 32 | 16 | 5120 | $2 \times 16 \times 16$ 步长卷积 |
| ViT-g | 1B | 1408 | 40 | 22 | 6144 | $2 \times 16 \times 16$ 步长卷积 |
| 预测器： $P_\phi(\cdot)$ | | | | | | |
| ViT-s | 22M | 384 | 12 | 12 | 1536 | N.A. |

## *A.4* 补充结果

### *A.4.1* 数据整理的效果

表13展示了对部分下游分类任务进行数据整理的结果。对于此表，我们使用原始V-JEPA（Bardes等，2024）的简化训练配方，在ViT-L和ViT-g规模上训练了模型。当训练较小规模的模型（ViT-L）时，在YT1B的整理版本上训练模型相比于未整理版本带来了全面的改进。然而，当转向混合数据设置（即添加图像和精选视频）时，使用整理数据实际上会导致部分任务的性能下降，例如SSv2上的性能为72.8（VM22M（混合+整理YT1B））对比73.3（混合+未整理YT1B）。在某些情况下，仅使用整理YT1B训练的模型比使用混合数据的模型更好，例如在COIN（86.5 vs. 86.25）和K400（84.6 vs. 83.7）评估任务上。这一结果有些令人惊讶，因为尽管在混合设置中包含了K710训练数据，但我们发现对于K400评估任务，它并未比整理YT1B带来性能提升。

表 13 数据整理对视频理解的影响。结果在 ViT-L 和 ViT-g 两种模型规模下均有报告。两种规模的模型均使用 Bardes 等人（2024）的简化训练计划进行预训练。

| 训练数据 | IN1K | COIN | SSv2 | K400 |
|---|---|---|---|---|
| ViT-L | | | | |
| 未整理的 YT1B | 80.6 | 83.2 | 70.9 | 82.9 |
| 整理后的 YT1B | 80.8 | **86.5** | 73.1 | **84.6** |
| 混合+未整理的 YT1B | **82.9** | 86.25 | **73.3** | 83.0 |
| VM22M | **82.9** | 86.0 | 72.8 | 83.7 |
| ViT-g | | | | |
| 未筛选的YT1B | 81.8 | 86.4 | 73.6 | 85.1 |
| 筛选后的YT1B | 81.7 | 88.4 | 74.8 | **86.5** |
| 混合+未筛选的YT1B | 83.7 | 88.5 | 75.5 | 85.9 |
| VM22M | **83.9** | **89.2** | **75.6** | 86.2 |

然而，这种行为在不同规模下并不一致。在 ViT-g 规模下，VM22M（Mixed+Curated YT1B）在所有任务上都优于 Mixed+Uncurated YT1B。

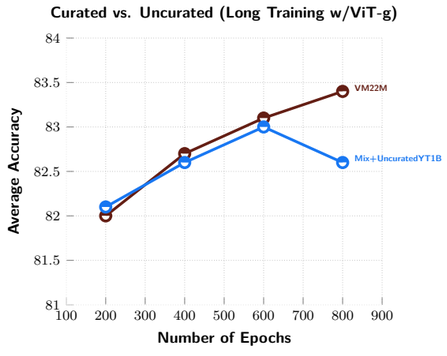



**Figure 12 Effect of data curation for V-JEPA 2 pre-training**. We show model performance averaged across the IN1K, COIN, SSv2, and K400 tasks as a function of pre-training "epochs" (equivalent to 300 optimization steps). Models trained with and without uncurated data achieve similar performance until epoch 600, at which point the performance of the model trained with uncurated YT1B beings degrading.

When following these tasks with the long training schedule, we continue to see differences between VM22M and Mixed+Uncurated YT1B at the ViT-g model scale, as shown in Figure 12, which compares the performance of the models while averaging across the IN1K, COIN, SSv2, and K400 image understanding tasks. Initially, the two models improve at roughly the same rate, but their performance diverges after epoch 600 where the model using uncurated data fails to continue improving.

### A.4.2 Effect of Long Training Schedule and cooldown

In Table 14 we demonstrate the effects of the two-stage training process. When comparing to the ViT-g results in Table 13, we see that the abbreviated schedule is superior to the constant learning rate schedule prior to the cooldown phase. The primary benefits are achieved during the cooldown phase, which uses 64 frames for pretraining in combination with a ramped down learning rate. This leads to a large benefit of over a full point across all evaluations.

**Table 14 Effects of Long Training and cooldown.** Results are reported at ViT-g model with cooldown at different resolutions.

| Training Stage | IN1K | COIN | SSv2 | K400 |
|---|---|---|---|---|
| Phase 1 (epoch 800, no cooldown) | 83.8 | 89.1 | 75.1 | 85.8 |
| Phase 2 (annealed, $256 \times 256$ resolution) | 84.6 | **90.7** | 75.3 | 86.6 |
| Phase 2 (annealed, $384 \times 384$ resolution) | **85.1** | 90.2 | **76.5** | **87.3** |

### A.4.3 Effect of Video Length at Evaluation.

Figure 13 examines how input video duration affects downstream task performance during evaluation. Using a model pretrained on 64-frame clips, we observe a +9.7 percentage point average improvement when increasing the video duration from 16 to 64 frames during evaluation. Note that this ablation uses a single clip evaluation protocol (i.e. we sample only one clip per video) instead of the standard multiclip evaluations due to memory constraints.



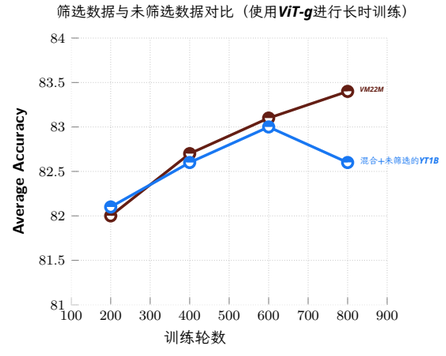



图12 V-JEPA 2预训练中数据筛选的影响。我们展示了模型在IN1K、COIN、SSv2和K400任务上的平均性能随预训练"轮数"（相当于300个优化步）的变化。使用和不使用未筛选数据训练的模型在第600轮之前性能相似，此后使用未筛选YT1B训练的模型性能开始下降。

在使用长训练计划执行这些任务时，我们在 ViT-g 模型规模下继续观察到 VM22M 与 Mixed+Uncurated YT1B 之间的差异，如图 12 所示，该图比较了模型在 IN1K、COIN、SSv2 和 K400 图像理解任务上的平均性能。最初，两个模型的提升速度大致相同，但在第 600 个 epoch 之后，它们的性能出现分化，使用未筛选数据的模型未能继续提升。

### *A.4.2* 长时训练计划和冷却阶段的影响

在表14中，我们展示了双阶段训练过程的效果。与表13中的ViT-g结果相比，我们可以看到，在冷却阶段之前，缩短的训练计划优于恒定学习率计划。主要收益在冷却阶段实现，该阶段使用64帧进行预训练，并结合逐渐降低的学习率。这在所有评估中带来了超过一整点的大幅收益。

表*14* 长时间训练的影响　训练与冷却。结果报告于ViT-g模型，冷却阶段采用不同分辨率

| 训练阶段 | IN1K | COIN | SSv2 | K400 |
|---|---|---|---|---|
| 阶段1（第800轮，无冷却） | 83.8 | 89.1 | 75.1 | 85.8 |
| 阶段2（退火，$256 \times 256$ 分辨率） | 84.6 | **90.7** | 75.3 | 86.6 |
| 阶段 2（退火，$384 \times 384$ 分辨率） | **85.1** | 90.2 | **76.5** | **87.3** |

### *A.4.3* 评估时视频长度的影响。

图13考察了输入视频时长如何影响评估时的下游任务性能。使用在64帧片段上预训练的模型，我们观察到在评估时将视频时长从16帧增加到64帧时，平均提升了 +9.7 个百分点。请注意，由于内存限制，此消融实验采用单片段评估协议（即每个视频仅采样一个片段），而非标准的多片段评估。

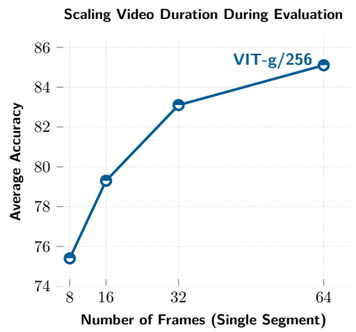



**Figure 13 Effect of video duration during evaluation.** Task performance further improves by running inference on longer video clips. All evaluations use ViT-g models that were annealed with 64 frames at resolution $256 \times 256$. Due to memory constraints, results are reported using a single clip evaluation protocol. Increasing the number of frames processed at inference time boosts average performance by up to +9.7 points.

# B V-JEPA 2-AC Post-training

## B.1 Post-Training Hyperparameters

The V-JEPA 2-AC model is trained with the AdamW (Loshchilov and Hutter, 2017) optimizer using a warmup-constant-decay learning-rate schedule, and a constant weight-decay of 0.04. We linearly warmup the learning rate from $7.5 \times 10^{-5}$ to $4.25 \times 10^{-4}$ over 4500 iterations, then hold it constant for 85500 iterations, and finally decay it to 0 over 4500 iterations. We use a batch size of 256 comprising 4 second video clips sampled randomly from trajectories in the Droid raw dataset at a frame rate of 4 fps. We train on the left extrinsic camera views from Droid — one could also train on videos from right camera views, however we found that training on both left and right camera views, without additionally conditioning on the camera position, degraded performance. For simplicity, we discard any videos shorter than 4 seconds, leaving us with less than 62 hours of video for training. We apply random-resize-crop augmentations to the sampled video clips with the aspect-ratio sampled in the range (0.75, 1.35).

## B.2 Robot Task Definitions

Figure 14 shows examples of start and goal frames for prehensile manipulation task with a cup in Lab 1. For the *grasp* and *reach with object* tasks the model is shown a single goal image. For the *pick-and-place* tasks we present two sub-goal images to the model in addition to the final goal. The first goal image shows the object being grasped, the second goal image shows the object in the vicinity of the goal position. The model first optimizes actions with respect to the first sub-goal for 4 time-steps before automatically switching to the second sub-goal for the next 10 time-steps, and finally the third goal for the last 4 time-steps. When planning with V-JEPA 2-AC, we use 800 samples, 10 refinement steps based on the top 10 samples from the previous iteration, and a planning horizon of 1. Since all considered tasks are relatively greedy, we found a short planning horizon to be sufficient for our setup. While longer planning horizons also worked reasonably well, they require more planning time.

## B.3 Visualizing World Model Predictions

To visualize the model's predictions, we train a frame decoder on the Droid dataset that maps the V-JEPA 2 representations to human-interpretable pixels. Specifically, we process 4 frame clips with the frozen V-JEPA 2 video encoder, decode each frame separately using our decoder network, and then update the weights of the decoder using a mean-squared error (L2) pixel reconstruction loss. The decoder is a feedforward network (fully deterministic regression model that does not use any sampling internally) with output dimension $256 \times 256 \times 3$, parameterized as a ViT-L. We train the decoder for 150000 optimization steps using AdamW with a fixed



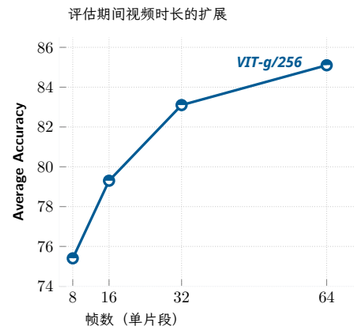



图 13 评估期间视频时长的影响。通过在更长的视频片段上运行推理，任务性能进一步提升。所有评估均使用 ViT-g 模型，该模型以分辨率 $256 \times 256$ 使用 64 帧进行退火处理。由于内存限制，结果采用单片段评估协议报告。增加推理时处理的帧数可将平均性能提升最多 +9.7 个点。

# *B V-JEPA 2-AC* 后训练

## *B.1* 后训练超参数

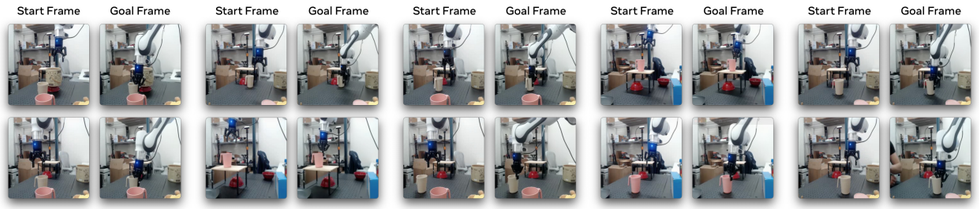
V-JEPA 2-AC模型使用AdamW（Loshchilov and Hutter, 2017）优化器进行训练，采用预热-恒定-衰减的学习率调度，权重衰减恒定为0.04。我们将学习率在4500次迭代内从 $7.5 \times 10^{-5}$ 线性预热至 $4.25 \times 10^{-4}$，然后在85500次迭代中保持恒定，最后在4500次迭代内衰减至0。我们使用批次大小为256，由从Droid原始数据集轨迹中随机采样的4秒视频片段组成，帧率为4 fps。我们使用Droid的左侧外部相机视角进行训练——也可以使用右侧相机视角的视频进行训练，但我们发现同时使用左右相机视角训练而不额外以相机位置为条件会降低性能。为简单起见，我们丢弃任何短于4秒的视频，剩余训练视频不足62小时。我们对采样的视频片段应用随机调整大小裁剪增强，长宽比在(0.75, 1.35)范围内采样。

## *B.2* 机器人任务定义

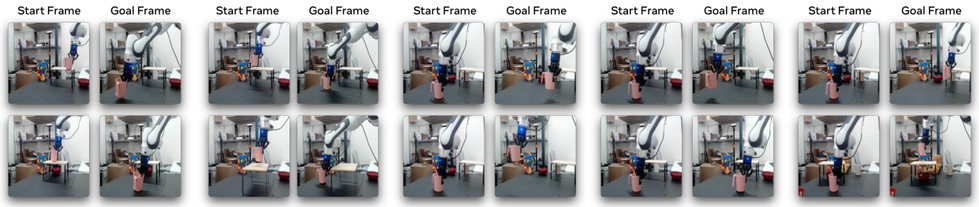
图 14 展示了 Lab 1 中杯子抓取操作任务的起始帧和目标帧示例。对于带物体的抓取和到达任务，模型仅接收单张目标图像。对于拾取和放置任务，除最终目标外，我们还向模型提供两张子目标图像。第一张目标图像显示物体被抓取的状态，第二张目标图像显示物体位于目标位置附近的状态。模型首先针对第一子目标优化动作 4 个时间步，然后自动切换到第二子目标进行接下来 10 个时间步的优化，最后在最后 4 个时间步切换到第三目标。使用 V-JEPA 2-AC 进行规划时，我们使用 800 个样本、基于前一次迭代中前 10 个样本的 10 步细化，以及 1 的规划视野。由于所有考虑的任务都相对贪心，我们发现较短的规划视野足以满足我们的设置。虽然较长的规划视野也能较好地工作，但需要更多的规划时间。

## *B.3* 可视化世界模型预测

为了可视化模型的预测结果，我们在 Droid 数据集上训练了一个帧解码器，将 V-JEPA 2 的表征映射为人类可解读的像素。具体而言，我们使用冻结的 V-JEPA 2 视频编码器处理 4 帧片段，通过解码器网络对每一帧分别进行解码，然后使用均方误差（L2）像素重建损失来更新解码器的权重。解码器是一个前馈网络（完全确定性的回归模型，内部不使用任何采样），输出维度为 $256 \times 256 \times 3$，参数化为 ViT-L。我们使用 AdamW 优化器对解码器进行 150000 步训练，采用固定的




**(a) Grasp Cup**



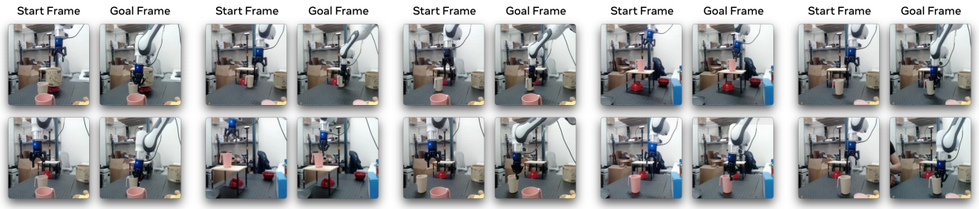

**(b) Reach with Cup**

**(c) Pick and Place Cup**

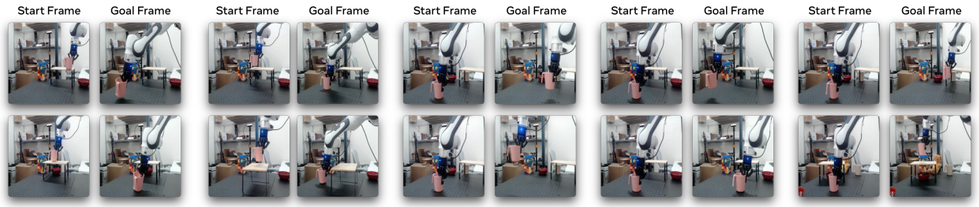
**Figure 14 Prehensile Manipulation Task Definition.** Start and goal frames for prehensile manipulation tasks with a cup in Lab 1. For the *grasp* and *reach with object* tasks the model is shown a single goal image. For the *pick-and-place* tasks we present two sub-goal images to the model in addition to the final goal. The first goal image shows the object being grasped, the second goal image shows the object in the vicinity of the goal position. The model first optimizes actions with respect to the first sub-goal for 4 time-steps before automatically switching to the second sub-goal for the next 10 time-steps, and finally the third goal for the last 4 time-steps.

weight decay of 0.1, gradient clipping of 1.0, and a batch size of 1024 frames. We linearly warmup the learning rate for 2000 steps to a peak value of $5 \times 10^{-4}$ and then decay it following a cosine schedule. For inference, we take the decoder trained on the V-JEPA 2 encoder and apply it off-the-shelf to the representations produced by the V-JEPA 2-AC predictor. The decision to only use use a simple feedforward architecture and






***(a)*** 抓取杯子

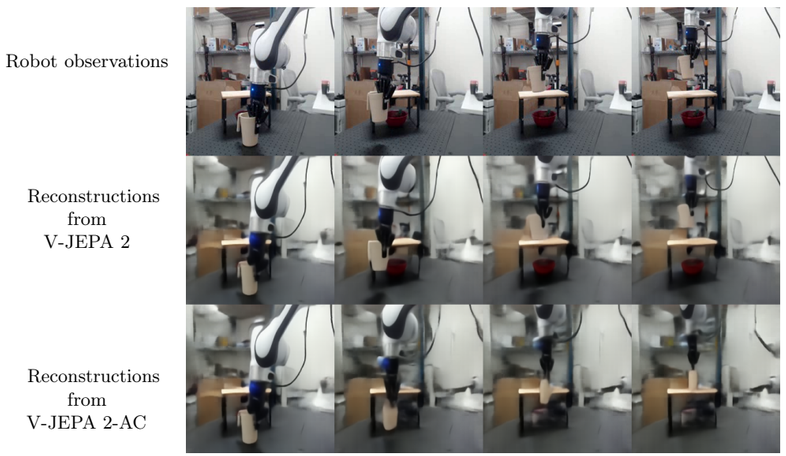



***(b)*** 用杯子够物

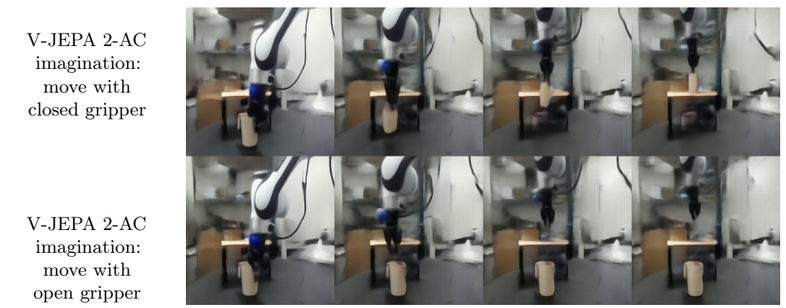
***(c)*** 拾取并放置杯子

图 14 抓取式操作任务定义。Lab 1 中杯子抓取式操作任务的起始帧与目标帧。对于抓取和使用物体够物任务，模型仅接收单张目标图像。对于拾取并放置任务，除最终目标外，我们向模型提供两张子目标图像。第一张目标图像展示物体被抓住的状态，第二张目标图像展示物体位于目标位置附近的状态。模型首先针对第一子目标优化动作 4 个时间步，随后自动切换至第二子目标进行后续 10 个时间步的优化，最后针对第三目标优化最后 4 个时间步。

权重衰减为 0.1、梯度裁剪为 1.0、批大小为 1024 帧。我们对学习率进行线性预热 2000 步至峰值 $5 \times 10^{-4}$，然后按照余弦调度进行衰减。在推理时，我们将在 V-JEPA 2 编码器上训练的解码器直接应用于 V-JEPA 2-AC 预测器产生的表征。仅使用简单的前馈架构这一决定以及



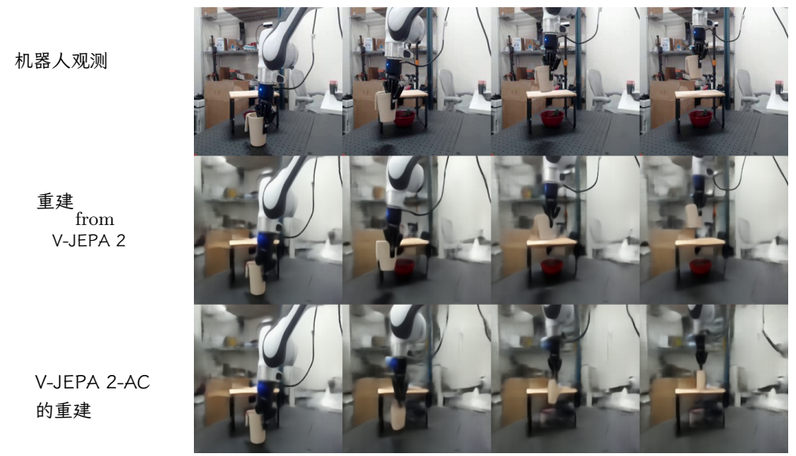

**(a) Comparing accuracy of predictions to ground truth trajectory. (Top Row)** Video frames of a ground-truth trajectory from a robot in our lab. **(Middle row)** Each frame is encoded by V-JEPA 2 encoder, and then decoded using the feedforward frame decoder. Reconstructions of the V-JEPA 2 representations show that the encoder captures the salient parts of the scene necessary for vision-based control; blurry background generation can be partially attributed to the low-capacity of our feedforward frame decoder. **(Bottom Row)** Autoregressive rollout produced by V-JEPA 2-AC world model using the ground-truth action sequence given first frame as context, and then decoded using feedforward frame decoder. Reconstructions of the V-JEPA 2-AC rollout show that the action-conditioned world model successfully animates the robot while keeping the background and non-interactated objects (e.g., the shelf) unaffected. However, we do observe error accumulation as the world model predicts the location of the cup to be slightly lower than that of the real trajectory in the final frame.



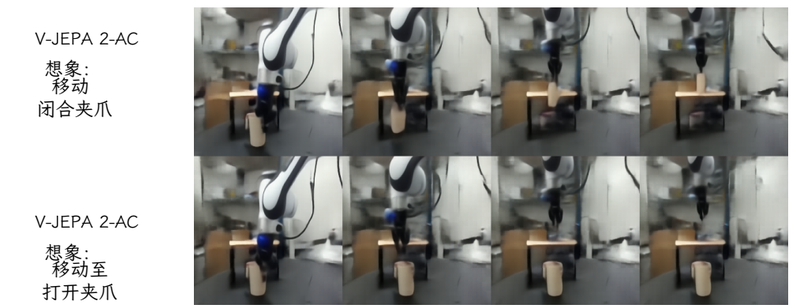

**(b) Ablating predictions with open versus closed gripper.** We explore how the V-JEPA 2-AC predictions change when driving the model with identical action sequences, but in one cause using a closed gripper **(top row)** and in the other with an open gripper **(bottom row)**. The world model predicts the location of the cup to be unchanged across time steps when using an open gripper action sequence, suggesting a reasonable understanding of intuitive physics (e.g., object constancy, shape constancy, and gravity).

**Figure 15 Decoding representations.** To visualize the model's predictions, we train a frame decoder on the Droid dataset that maps the V-JEPA 2 representations to human-interpretable pixels. The decoder is a feedforward network (fully deterministic regression model that does not use any sampling internally) trained with a mean-squared error pixel reconstruction loss. By applying the frame decoder, trained on the V-JEPA 2 encoder, to the representations produced by the V-JEPA 2-AC predictor, we can visualize world model rollouts for various action sequences.






(a) 比较预测轨迹与真实轨迹的准确性。（顶行）实验室中机器人真实轨迹的视频帧。（中间行）每一帧由 V-JEPA 2 编码器编码，然后使用前馈帧解码器进行解码。V-JEPA 2 表示的重建表明，编码器捕捉到了基于视觉的控制所必需的场景显著部分；模糊的背景生成可部分归因于我们前馈帧解码器的低容量。（底行）由 V-JEPA 2-AC 世界模型生成的自回归推演，使用以第一帧为上下文的真实动作序列，然后使用前馈帧解码器进行解码。V-JEPA 2-AC 推演的重建表明，动作条件世界模型成功驱动了机器人的动画，同时保持背景和非交互物体（例如架子）不受影响。然而，我们确实观察到误差累积，因为世界模型在最后一帧中预测的杯子位置略低于真实轨迹的位置。




(b) 使用打开与闭合夹爪对预测进行消融。我们探索了当使用相同的动作序列驱动模型时，V-JEPA 2-AC 的预测如何变化，其中一种情况使用闭合夹爪（上排），另一种使用打开夹爪（下排）。当使用打开夹爪的动作序列时，世界模型预测杯子在不同时间步的位置保持不变，这表明其对直觉物理（例如物体恒常性、形状恒常性和重力）具有合理的理解。

图 15 解码表示。为了可视化模型的预测，我们在 Droid 数据集上训练一个帧解码器，将 V-JEPA 2 表示映射为人类可解释的像素。该解码器是一个前馈网络（完全确定性的回归模型，内部不使用任何采样），以均方误差像素重建损失进行训练。通过将由 V-JEPA 2 编码器上训练的帧解码器应用于 V-JEPA 2-AC 预测器生成的表示，我们可以可视化各种动作序列的世界模型推演。

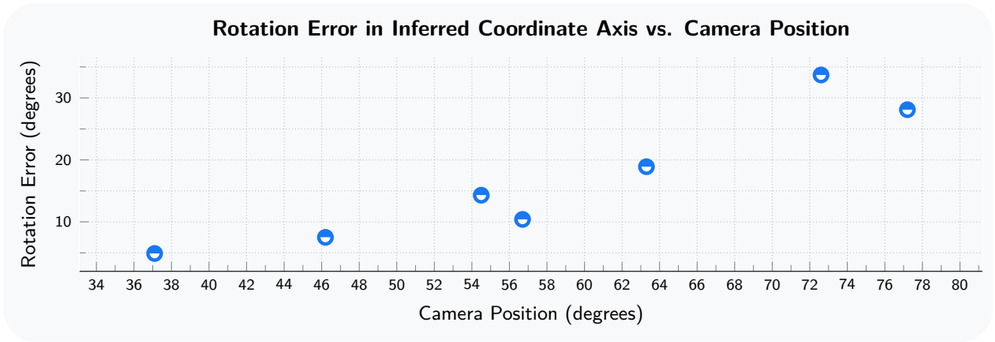



**Figure 16 Sensitivity to camera position.** Rotation error (in the x-y plane) of the action coordinate axis inferred by V-JEPA 2-AC as a function of camera position, with 0 degrees corresponding to a camera located at the robot base, and 90 degrees corresponding to a camera located left of the robot base. While ideally, the model's inferred coordinate axis would be invariant to camera position, here we observe that the model's inferred coordinate axis is sensitive to the camera position.

decode representations at the frame level (as opposed to video level), is to better leverage the decoder as an interpretability tool to analyze the V-JEPA 2-AC rollouts for a set of robot action sequences.

In Figure 15a, we show the video frames of a ground-truth trajectory from a robot in our lab (top row), the decoded V-JEPA 2 encoder representations of each frame (middle row), and the decoded V-JEPA 2-AC world model rollout using the ground-truth action sequence and a single starting frame as context (bottom row). Reconstructions of the V-JEPA 2 representations (middle row) show that the encoder captures the salient parts of the scene necessary for vision-based control; blurry background generation can be partially attributed to the low-capacity of our feedforward frame decoder. Reconstructions of the V-JEPA 2-AC rollout show that the action-conditioned world model successfully animates the robot while keeping the background and non-interactated objects (e.g., the shelf) unaffected. We also see that, with a closed gripper, the model correctly predicts the movement of the cup with the arm, suggesting a reasonable understanding of intuitive physics (e.g., object constancy, shape constancy, and gravity), but we do observe error accumulation as the world model predicts the location of the cup to be slightly lower than that of the real trajectory in the final frame. In Figure 15b we explore how the V-JEPA 2-AC predictions change when driving the model with identical action sequences, but in one cause using a closed gripper (top row) and in the other with an open gripper (bottom row). The world model predicts the location of the cup to be unchanged across time steps when using an open gripper action sequence.

### B.4 Assessing Sensitivity to Camera Position

In practice, we manually tried different camera positions before settling on one that worked best for our experiments; then the camera is kept in the same location for all experiments, across all tasks. In this section, we conduct a quantitative analysis of the V-JEPA 2-AC world model's sensitivity to camera position. While ideally, the model's inferred coordinate axis would be invariant to camera position, here we observe that the model's inferred coordinate axis is sensitive to the camera position; this is problematic as large errors in the inferred coordinate axis can degrade success rate on downstream tasks.

We sweep several camera positions around the robot base, which we describe as a clockwise angular position around the center of the table, with 0 degrees being located at the robot base, and 90 degrees being left of the robot base. Since we train on the left exocentric camera views from the Droid dataset, we sweep camera positions between roughly 35 degrees and 85 degrees. Next, for each camera position, we collect a 201 step trajectory of random robot movements within the horizontal x-y plane. For each pair of adjacent frames in this 201 step trajectory, we compute the optimal action inferred by V-JEPA 2-AC, i.e., the action that minimizes the energy function in eq. (5) given a 1-step rollout. This allows us to construct a dataset for



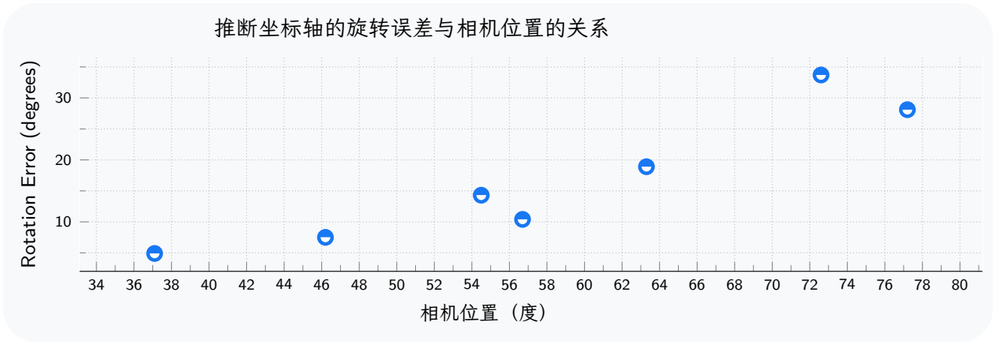



图 16 对相机位置的敏感性。V-JEPA 2-AC 推断的动作坐标轴的旋转误差（在 x-y 平面内）随相机位置的变化关系，其中 0 度对应相机位于机器人基座处，90 度对应相机位于机器人基座左侧。虽然理想情况下模型推断的坐标轴应对相机位置保持不变，但在此我们观察到模型推断的坐标轴对相机位置较为敏感。

解码表示可解释性工具 在帧级别（而非视频级别）进行，是为了更好地利用解码器作为用于分析 V-JEPA 2-AC 在一组机器人动作序列上的推演结果。

在图15a中，我们展示了实验室中机器人真实轨迹的视频帧（顶行）、每一帧经解码的V-JEPA 2编码器表征（中间行），以及使用真实动作序列和单个起始帧作为上下文、经解码的V-JEPA 2-AC世界模型推演结果（底行）。V-JEPA 2表征的重建（中间行）表明，编码器捕捉到了基于视觉的控制所必需场景中的显著部分；背景生成模糊的部分原因可归咎于我们前馈帧解码器的低容量。V-JEPA 2-AC推演的重建表明，动作条件世界模型成功地让机器人动了起来，同时保持背景和未交互物体（如架子）不受影响。我们还看到，在夹爪闭合的情况下，模型正确地预测了杯子随机械臂的运动，这表明模型对直觉物理（如物体恒常性、形状恒常性和重力）有合理的理解，但我们也确实观察到误差累积，世界模型预测的杯子位置在最后一帧中略低于真实轨迹中的位置。在图15b中，我们探究了当用相同的动作序列驱动模型时V-JEPA 2-AC预测的变化，但一种情况使用闭合夹爪（顶行），另一种使用张开夹爪（底行）。当使用张开夹爪的动作序列时，世界模型预测杯子的位置在各时间步上保持不变。

### *B.4* 评估对相机位置的敏感性

在实践中，我们在确定最适合实验的相机位置之前手动尝试了不同位置；随后在所有任务的所有实验中，相机均保持在同一位置。在本节中，我们对 V-JEPA 2-AC 世界模型对相机位置的敏感性进行了定量分析。理想情况下，模型推断的坐标轴应对相机位置保持不变，但在此我们观察到模型推断的坐标轴对相机位置较为敏感；这一问题值得关注，因为推断坐标轴中的较大误差会降低下游任务的成功率。

我们围绕机器人底座扫描了若干相机位置，将其描述为围绕桌子中心的顺时针角位置，其中0度位于机器人底座处，90度位于机器人底座左侧。由于我们使用Droid数据集中的左侧外中心相机视角进行训练，因此我们在约35度到85度之间扫描相机位置。接下来，对于每个相机位置，我们收集一条在水平x-y平面内随机机器人运动的201步轨迹。对于这条201步轨迹中的每对相邻帧，我们计算由V-JEPA 2-AC推断出的最优动作，即在给定1步展开的情况下使公式(5)中能量函数最小化的动作。这使我们能够构建一个数据集，用于

each camera position consisting of *real action* versus *inferred action* pairs. We only focus on the $\Delta x$ and $\Delta y$ cartesian control actions (first two dimensions of the action vector) for our analysis. Let $A \in \mathbb{R}^{200\times 2}$ denote the inferred actions and $B \in \mathbb{R}^{200\times 2}$ denote the ground truth actions. Based on this, we can solve a linear least squares problem to identify the linear transformation $W^\star \in \mathbb{R}^{2\times 2}$ that maps inferred actions $A$ to real actions $B$,

$$W^\star = \underset{W\in\mathbb{R}^{2\times 2}}{\operatorname{argmin}} \|AW - B\|_2.$$

The mean absolute prediction error for all camera position is roughly 1.6cm (compared to a ground truth delta pose of roughly 5cm), suggesting that the error is systematic. In addition, we observe that for each camera position, the matrix $W^\star$ has condition number $\approx 1.5$, i.e., modulo a fixed scalar coefficient, $W^\star$ is approximately a rotation matrix, and thus we can compute the rotation error in the inferred coordinate axis by using

$$W^\star \approx \overline{W}^\star = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

with $\overline{W}^\star := UV^\top$ where $U$ and $V$ are the left and right singular vectors of $W^\star$, respectively.

Figure 16 shows the camera position plotted against the rotation error in the V-JEPA 2-AC inferred coordinate axis. We observe that the rotation error in the inferred coordinate axis is almost a linear function of the camera position. We can most clearly see the effects of rotation errors in the inferred coordinate axis in our single-goal reaching experiments in Figure 8. While the model is always able to move the arm within 4 cm of the goal based on visual feedback from the monocular RGB camera, rotation errors in the inferred coordinate axis result in relatively suboptimal actions at each planning step, yielding a non-maximal, albeit monotonic, decrease in the distance to goal at each step.

Interestingly, since errors in the inferred coordinate axis are primarily rotation-based, one can use this approach to "calibrate" their world model by simply rotating all inferred actions by $W^\star$, and thereby introduce the desired invariance to camera position. Such an unsupervised calibration phase would involve the robot performing random actions, solving a linear least squares problem by comparing its inferred optimal actions to the actual actions it executed, and then multiplying its inferred actions by the rotation matrix before sending them to the controller during task execution. While such an approach is interesting, we emphasize that we *do no such calibration* in our experiments.

# C Visual Classification

We describe in more detail the evaluation procedure used for the classification tasks described in Section 5.

## C.1 Hyperparameters

**Probe Architecture.** We train an attentive probe on top of the frozen encoder output using the training data from each downstream task. Our attentive probe is composed of four transformer blocks, each using 16 heads in the attention layer. The first three blocks use standard self-attention; the final block uses a cross-attention layer with a learnable query token. The output of the cross-attention layer in the final block is added back to the query token as a residual connection before applying the rest of the block (LayerNorm, followed by MLP with a single GeLU activation). The transformer blocks are followed by a final linear classifier layer.

**Evaluation setup parameters.** All models follow the same evaluation protocol and use a resolution of $256 \times 256$, except our V-JEPA 2 ViT-$g_{384}$. For video evaluations, we sampled multiple clip segments from each input video. During validation, we also extracted three spatial views from each segment (instead of one view during training). The number of clip segments, frame step parameter, and global batch size vary for each eval; parameters used for each evaluation can be found in Table 15. By default, we use $16 \times 2 \times 3$ inputs for SSv2 (16 frames clip, 2 temporal crops, 3 spatial crops), $16 \times 8 \times 3$ for K400, $32 \times 8 \times 3$ for COIN, and $32 \times 4 \times 3$ for Diving-48 and Jester. V-JEPA 2 ViT-$g_{384}$ uses a higher resolution of $384 \times 384$ for K400, COIN, Diving-48, and Jester, $512 \times 512$ for ImageNet and $384 \times 384$ with $64 \times 2 \times 3$ inputs for SSv2.



每个相机位置由真实动作与推断动作对组成。在我们的分析中，我们仅关注 $\Delta x$ 和 $\Delta y$笛卡尔控制动作（动作向量的前两个维度）。令 $A \in \mathbb{R}^{200\times 2}$ 表示推断动作， $B \in \mathbb{R}^{200\times 2}$ 表示真实动作。在此基础上，我们可以求解一个线性最小二乘问题，以确定将推断动作 $A$ 映射到真实动作 $B$的线性变换 $W^\star \in \mathbb{R}^{2\times 2}$，

$$W^\star = \underset{W\in\mathbb{R}^{2\times 2}}{\operatorname{argmin}} \|AW - B\|_2.$$

所有相机位置的平均绝对预测误差约为1.6cm（相比约5cm的真值增量位姿），表明该误差是系统性的。此外，我们观察到对于每个相机位置，矩阵 $W^\star$的条件数为 $\approx 1.5$，即模一个固定标量系数， $W^\star$近似为一个旋转矩阵，因此我们可以通过下式计算推断坐标轴中的旋转误差：

$$W^\star \approx \overline{W}^\star = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

with $\overline{W}^\star := UV^\top$ 其中 $U$和 $V$分别是 $W^\star$的左奇异向量和右奇异向量。

图16展示了相机位置与V-JEPA 2-AC推断坐标轴中旋转误差的关系图。我们观察到推断坐标轴中的旋转误差几乎是相机位置的线性函数。在单目标到达实验中（图8），我们可以最清楚地看到推断坐标轴中旋转误差的影响。虽然模型始终能够基于单目RGB相机的视觉反馈将手臂移动到距目标4cm以内，但推断坐标轴中的旋转误差导致每个规划步骤的动作相对次优，使得到目标的距离呈现非最大的、尽管是单调的减少。

有趣的是，由于推断坐标轴中的误差主要源于旋转，人们可以利用这一方法通过将所有推断动作旋转 $W^\star$来"校准"其世界模型，从而引入对相机位置所期望的不变性。这种无监督校准阶段将涉及机器人执行随机动作，通过将其推断的最优动作与实际执行的动作进行比较来求解线性最小二乘问题，然后在任务执行期间将推断动作乘以旋转矩阵后再发送给控制器。虽然这种方法很有趣，但我们强调在实验中并未进行此类校准。

# C 视觉分类

我们更详细地描述第5节中所述分类任务所使用的评估流程。

## C.1 超参数

探针架构。我们在冻结的编码器输出之上，使用每个下游任务的训练数据训练一个注意力探针。我们的注意力探针由四个Transformer块组成，每个块在注意力层中使用16个注意力头。前三个块使用标准自注意力；最后一个块使用带有可学习查询令牌的交叉注意力层。在应用该块的其余部分（LayerNorm，随后是带单个GeLU激活的MLP）之前，最后一个块中交叉注意力层的输出通过残差连接加回查询令牌。Transformer块之后是一个最终的线性分类器层。

评估设置参数。所有模型遵循相同的评估协议并使用 $256 \times 256$ 的分辨率，除了我们的 V-JEPA 2 ViT-$g_{384}$。对于视频评估，我们从每个输入视频中采样了多个片段。在验证期间，我们还从每个片段中提取了三个空间视图（而非训练期间的一个视图）。片段数量、帧步长参数和全局批次大小因每次评估而异；每次评估使用的参数可在表 15 中找到。默认情况下，我们对 SSv2 使用 $16 \times 2 \times 3$ 输入（16 帧片段，2 个时间裁剪，3 个空间裁剪），对 K400 使用 $16 \times 8 \times 3$，对 COIN 使用 $32 \times 8 \times 3$，对 Diving-48 和 Jester 使用 $32 \times 4 \times 3$。V-JEPA 2 ViT-$g_{384}$ 对 K400、COIN、Diving-48 和 Jester 使用更高的 $384 \times 384$ 分辨率，对 ImageNet 使用 $512 \times 512$，对 SSv2 使用 $384 \times 384$ 及 $64 \times 2 \times 3$ 输入。

**Table 15 Visual Classification eval params.** Default parameters used for the visual classification evaluations, with non-default values for each eval (* denotes default). All attentive probes use 4 transformer blocks with 16 heads.

| Parameter | Default (K400) | ImageNet | SSv2 | COIN | Jester/Diving-48 |
|---|---|---|---|---|---|
| Number of frames | 16 | 16 | 16 | 32 | 32 |
| Segments / Clip | 8 | 1 | 2 | 8 | 4 |
| Views / Segment | 3 | 1 | * | * | * |
| Frame Step | 4 | n/a | * | * | 2 |
| Epochs | 20 | * | * | * | 100 |
| Batch Size (global) | 256 | 1024 | * | 128 | 128 |
| Resolution | $256 \times 256$ | * | * | * | * |
| Classifier Heads | 20 (4x5) | * | * | * | 3 (3x1) |
| Classifier Learning Rates | [5e-3 3e-3 1e-3 3e-4 1e-4] | * | * | * | [1e-3 3e-4 1e-4] |
| Classifier Weight Decay | [.8 .4 .1 .01] | * | * | * | [.8] |

**Table 16 Input layers for Jester/Diving-48.** For each encoder size, indices of the four encoder layers whose tokens are used as input to the linear classifier in the Jester and Diving-48 evaluations.

| Encoder | # Layers | Attended Layers |
|---|---|---|
| **ViT-L** | 24 | 17, 19, 21, 23 |
| **ViT-H** | 32 | 25, 27, 29, 31 |
| **ViT-g** | 40 | 24, 29, 34, 39 |

**ImageNet evaluation.** For ImageNet, we repeat each input image to produce a 16-frame video clip. We also use a larger global batch size (1024 instead of 256 or 128), and do not use multiple clips or views per sample.

**Jester and Diving-48 evaluation.** Our Jester and Diving-48 action classification evaluation tasks differ from the other understanding evaluations in several ways, primarily in that we employ a multilayer strategy. Instead of attending to the tokens from only the last layer of the encoder, we extract tokens from four encoder layers (the last layer and three intermediate layers) and attend to all of them. (Table 16 shows the layers we used for each encoder size.) We also train the probes for these two evaluations with only three classification heads (instead of 20 for the other evaluations), but train for 100 epochs (instead of 20) as these evaluations benefit from longer training. We use a global batch size of 128 for both evaluations.

**Optimization.** For each evaluation, we simultaneously train multiple classifier heads with different hyperparameters (learning rate and weight decay), reporting the accuracy of the best-performing classifier. For most of our evaluations (Kinetics, SSv2, COIN, and ImageNet), we train for 20 epochs and use 20 heads, each using one of five learning rate values and four weight decay values, and the learning rate decays according to a cosine schedule. We provide a summary of all hyperparameters in Table 15.

## C.2 Additional Results

**Probe Size.** Since we use a four-layer attentive probe for these evaluations, we investigate whether using a smaller probe impacts evaluation performance. We re-run our six understanding evaluations (for two model sizes, ViT-L and ViT-g) with a smaller probe consisting of a single cross-attention block using 16 attention heads. Unlike in Section 5, we use 16 frames for all evaluations, including Diving-48 and Jester. See Table 18 for classification performance—we confirm that our four-layer probe outperforms a single-layer attentive probe across all understanding evaluations (except for Jester), by an average of +1.4 points accuracy for ViT-L and +1.0 points for ViT-g.

**Impact of encoder multilayer.** We study the impact of feeding tokens from multiple layers from the encoder to the attentive probe during evaluation. Table 17 shows that Diving-48 and Jester strongly benefit from information from deeper layers of the encoder.



表 ***15*** 视觉分类 评估参数。用于视觉分类评估的默认参数，以 h
及各评估的非默认值 （* 表示默认值）。所有注意力探针均使用 4 个 transformer 块，16 个头。

| 参数 | 默认值 ***(K400)*** | ***ImageNet*** | SSv2 | COIN | ***Jester/Diving-48*** |
|---|---|---|---|---|---|
| 帧数 | 16 | 16 | 16 | 32 | 32 |
| 片段数 / 剪辑 | 8 | 1 | 2 | 8 | 4 |
| 视图数 / 片段 | 3 | 1 | * | * | * |
| 帧步长 | 4 | n/a | * | * | 2 |
| 轮次 | 20 | * | * | * | 100 |
| 批次大小（全局） | 256 | 1024 | * | 128 | 128 |
| 分辨率 | $256 \times 256$ | * | * | * | * |
| 分类器头 | 20 (4x5) | * | * | * | 3 (3x1) |
| 分类器学习率 | [5e-3 3e-3 1e-3 3e-4 1e-4] | * | * | * | [1e-3 3e-4 1e-4] |
| 分类器权重衰减 | [.8 .4 .1 .01] | * | * | * | [.8] |

表 16 Jester/Diving-48 的输入层。对于每种编码器大小，列出了在 Jester 和 Diving-48 评估中其 token 被用作线性分类器输入的四个编码器层的索引。

| 编码器 | # 层 | 参与层 |
|---|---|---|
| ***ViT-L*** | 24 | 17, 19, 21, 23 |
| ***ViT-H*** | 32 | 25, 27, 29, 31 |
| ***ViT-g*** | 40 | 24、29、34、39 |

ImageNet 评估。对于 ImageNet，我们将每张输入图像重复以生成 16 帧的视频片段。我们还使用更大的全局批大小（1024 而非 256 或 128），并且每个样本不使用多个片段或视图。

Jester 和 Diving-48 评估。我们的 Jester 和 Diving-48 动作分类评估任务在多个方面与其他理解评估不同，主要在于我们采用了多层策略。我们不是仅关注编码器最后一层的 token，而是从四个编码器层（最后一层和三个中间层）提取 token 并关注所有这些 token。（表 16 显示了每种编码器大小所使用的层。）我们还仅使用三个分类头（其他评估为 20 个）来训练这两个评估的探针，但训练 100 个 epoch（而非 20 个），因为这些评估受益于更长的训练。两个评估均使用 128 的全局批大小。

优化。对于每个评估，我们同时训练多个具有不同超参数（学习率和权重衰减）的分类器头，并报告表现最佳的分类器的准确率。对于我们的大多数评估（Kinetics、SSv2、COIN 和 ImageNet），我们训练 20 个 epoch 并使用 20 个头，每个头使用五个学习率值和四个权重衰减值中的一个，学习率按余弦调度衰减。我们在表 15 中提供了所有超参数的摘要。

## *C.2* 附加结果

探针大小。由于我们在这些评估中使用四层注意力探针，我们研究了使用更小的探针是否会影响评估性能。我们使用由单个交叉注意力块（含16个注意力头）组成的更小探针，重新运行了六项理解评估（针对两种模型规模：ViT-L和ViT-g）。与第5节不同，我们在所有评估中均使用16帧，包括Diving-48和Jester。分类性能见表18——我们确认，在所有理解评估中（Jester除外），我们的四层探针均优于单层注意力探针，ViT-L平均高出 +1.4 点准确率，ViT-g平均高出+1.0 点。

编码器多层特征的影响。我们研究了在评估期间将编码器多个层的token输入到注意力探针的影响。表17显示，Diving-48和Jester从编码器更深层的信息中获益显著。

**Table 17 Encoder Multilayer Ablation.** We vary the number of encoder layers fed to the attentive probe. We report the classification performances of attentive probes trained on top of V-JEPA 2 with 16 frames at $256 \times 256$ resolution.

| Model | Encoder Layers | Diving-48 | Jester |
|---|---|---|---|
| ViT-g | 1 | 82.9 | 96.1 |
| ViT-g | 4 | 86.7 | 97.6 |

**Table 18 Probe Size Ablation.** We vary the number of layers in the attentive probe. We report the classification performances of attentive probes trained on top of V-JEPA 2 with 16 frames at $256 \times 256$ resolution.

| | | *Motion Understanding* | | | | *Appearance Understanding* | | |
|---|---|---|---|---|---|---|---|---|
| Model | Probe Layers | Avg. | SSv2 | Diving-48 | Jester | K400 | COIN | IN1K |
| ViT-L | 1 | 84.0 | 72.0 | 83.2 | 97.7 | 83.3 | 85.9 | 81.8 |
| ViT-L | 4 | 85.6 | 73.6 | 87.1 | 97.7 | 85.1 | 86.8 | 83.5 |
| ViT-g | 1 | 86.0 | 74.8 | 85.3 | 97.8 | 85.6 | 88.9 | 83.5 |
| ViT-g | 4 | 87.0 | 75.6 | 86.7 | 97.6 | 86.6 | 90.7 | 84.6 |

# D Action Anticipation

We provide additional details, results, and ablations related to the Epic-Kitchen 100 action anticipation evaluation of Section 6.

## D.1 Hyperparameters

**Probe Architecture.** Our probe architecture for action anticipation follows the architecture of our classification probe described in Appendix C.1, consisting of four transformer blocks, including a last cross-attention layer with a set of learnable query tokens, followed by a final linear classifier layer for each query token.

**Evaluation setup parameters.** We use a focal loss (Lin et al., 2017) with a $\alpha = 0.25$ and $\gamma = 2.0$ when training the probe; this loss is more suited for training with long-tailed imbalanced class distributions. We use a context of 32 frames with a frame-rate of 8 frames per second at resolution $256 \times 256$ for V-JEPA 2 ViT-L, ViT-H, and ViT-g; and resolution $384 \times 384$ for V-JEPA 2 ViT-$g_{384}$. During probe training, we randomly sample an anticipation time between 0.25 and 1.75 seconds, and an anticipation point between 0.0 and 0.25. The *anticipation point* identifies a point in the action segment from which to perform anticipation; i.e., an anticipation point of 0 means that we predict the representation of the first frame in the action segment using our V-JEPA 2 predictor before feeding it to the probe, whereas an anticipation point of 1 means that we predict the representation of the last frame in the action segment before feeding it to the probe. The validation anticipation time is set to 1 second and the validation anticipation point is set to 0. We provide a summary of the hyperparameters, including the optimization parameters in Table 19.

## D.2 Additional results

**Impact of Architecture.** Table 20 investigates the impact of providing the output of the V-JEPA 2 encoder, predictor, or both, to the action anticipation probe. Using encoder outputs already leads to competitive performance on the EK100 task. Adding the predictor provides a small but consistent improvement across action, verb, and object categories. In addition, using predictor outputs yields a non-trivial performance, but still at a much lower point compared to using the encoder, showing that the EK100 task mostly requires strong semantic understanding, as opposed to forecasting capabilities.

**Impact of Input Resolution.** We report in Figure 17, the impact of input resolution and frame sampling parameters. In summary, V-JEPA 2 benefits from a longer context, a higher frame rate, and higher resolution,



表 17 编码器 Mu 层消融实验。我们改变输入到注意力探测器的编码器层数。我们报告 t 分类性能 在 $256 \times 256$ 分辨率下，基于 V-JEPA 2 并使用 16 帧训练的注意力探测器的表现 .

| 模型 | 编码器层数 | Diving-48 | Jester |
|---|---|---|---|
| ViT-g | 1 | 82.9 | 96.1 |
| ViT-g | 4 | 86.7 | 97.6 |

表 18 探针尺寸消融。我们改变注意力探针中的层数。我们报告了在 V-JEPA 2 之上训练的注意力探针在 $256 \times 256$ 分辨率下使用 16 帧的分类性能。

| | | 运动理解 | | | | 外观理解 | | |
|---|---|---|---|---|---|---|---|---|
| 模型 | 探测层 | Avg. | SSv2 | Diving-48 | Jester | K400 | COIN | IN1K |
| ViT-L | 1 | 84.0 | 72.0 | 83.2 | 97.7 | 83.3 | 85.9 | 81.8 |
| ViT-L | 4 | 85.6 | 73.6 | 87.1 | 97.7 | 85.1 | 86.8 | 83.5 |
| ViT-g | 1 | 86.0 | 74.8 | 85.3 | 97.8 | 85.6 | 88.9 | 83.5 |
| ViT-g | 4 | 87.0 | 75.6 | 86.7 | 97.6 | 86.6 | 90.7 | 84.6 |

# D 动作预期

我们提供了与第6节Epic-Kitchen 100动作预测评估相关的额外细节、结果和消融实验。

## D.1 超参数

探针架构。我们用于动作预期的探针架构遵循附录 C.1 中描述的分类探针架构，由四个 transformer 模块组成，包括最后一个带有可学习查询 token 集合的交叉注意力层，随后是每个查询 token 的最终线性分类器层。

评估设置参数。我们在训练探针时使用焦点损失（Lin 等，2017），其中 $\alpha = 0.25$ 和 $\gamma = 2.0$；该损失更适合在长尾不平衡类别分布下进行训练。对于 V-JEPA 2 ViT-L、ViT-H 和 ViT-g，我们使用 32 帧的上下文，帧率为每秒 8 帧，分辨率为 $256 \times 256$；对于 V-JEPA 2 ViT-$g_{384}$，分辨率为 $384 \times 384$。在探针训练期间，我们在 0.25 到 1.75 秒之间随机采样一个预期时间，并在 0.0 到 0.25 之间随机采样一个预期点。预期点标识动作片段中执行预期的位置；即，预期点为 0 意味着我们使用 V-JEPA 2 预测器预测动作片段中第一帧的表示，然后将其输入探针，而预期点为 1 意味着我们预测动作片段中最后一帧的表示，然后将其输入探针。验证预期时间设置为 1 秒，验证预期点设置为 0。我们在表 19 中提供了超参数的摘要，包括优化参数。

## D.2 补充结果

架构的影响。表20研究了将V-JEPA 2编码器、预测器或两者的输出提供给动作预判探针所产生的影响。仅使用编码器输出即可在EK100任务上取得有竞争力的性能。加入预测器后，在动作、动词和物体类别上均带来小幅但一致的提升。此外，使用预测器输出可产生非平凡的性能，但仍远低于使用编码器时的水平，这表明EK100任务主要需要强大的语义理解能力，而非预测能力。

输入分辨率的影响。我们在图17中报告了输入分辨率和帧采样参数的影响。总之，V-JEPA 2 受益于更长的上下文、更高的帧率和更高的分辨率，

**Table 19 Action Anticipation Evaluation params.** Default parameters used for the EK100 Action Anticipation evaluation.

| Parameter | EK100 |
|---|---|
| Train Anticipation time | 0.25s – 1.75s |
| Train Anticipation point | 0.0 – 0.25 |
| Val Anticipation time | 1s |
| Val Anticipation point | 0.0 |
| Number of frames | 32 |
| Frames per second | 8 |
| Epochs | 20 |
| Warmup epochs | 0 |
| Batch Size (global) | 128 |
| Classifier Heads | 20 (4x5) |
| Classifier Learning Rates | [5e-3 3e-3 1e-3 3e-4 1e-4] |
| Classifier Weight Decay | [1e-4 1e-3 1e-2 1e-1] |

**Table 20 EK100: Impact of Anticipation Probe Inputs.** We investigate the impact of providing the outputs of the V-JEPA 2 encoder, predictor, or both, to the action anticipation probe. Using encoder outputs already leads to competitive performance on the EK100 task. Adding the predictor provides a small but consistent improvement across action, verb and object categories.

| | | *Action Anticipation* | | |
|---|---|---|---|---|
| **Encoder** | **Predictor** | Verb | Noun | Action |
| ✓ | | 61.3 | 57.0 | 39.1 |
| | ✓ | 48.7 | 34.7 | 20.2 |
| ✓ | ✓ | 63.6 | 57.1 | 39.7 |

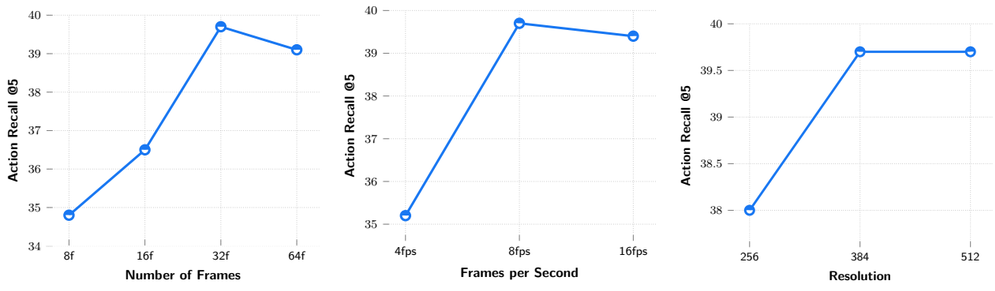



**Figure 17 Protocol ablation for action anticipation on EK100. (Left)** Performance with respect to the number of context frames used for action anticipation. **(Middle)** Performance with respect to the frame rate (fps) used for inference; number of context frames fixed at 32. **(Right)** Performance with respect to the spatial resolution (height and width) of the context frames used for action anticipation.

up to a point where the performance saturates or slightly decreases. The optimal performance is obtained by training with a 32-frames context length, a frame rate of 8 and a resolution of $384 \times 384$.

**Longer-term Prediction.** We report in Figure 18 (Left), the impact of predicting at a longer horizon, by varying the anticipation time between (1s, 2s, 4s, 10s). For each anticipation time, we report recall at several values (1, 5, 10, 20). The results show that the performance sharply decreases as the anticipation time increases, which is expected since forecasting the future in EK100 is a non-deterministic task.

**Analysis of Failure Cases.** We report in Figure 18 (Right), the distribution of failure and success prediction, on the EK100 validation set, between each configuration of success/failure for verb, noun, and action. The model performs very well, and the most represented configuration is, therefore, a full success



表 ***19*** 动作预测 评估参数。用于 EK100 动作预测评估的默认参数。 n
评估。

| 参数 | ***EK100*** |
|---|---|
| 训练预期时间 | 0.25秒 – 1.75秒 |
| 训练预期点 | 0.0 – 0.25 |
| 验证预期时间 | 1s |
| 验证预期点 | 0.0 |
| 帧数 | 32 |
| 每秒帧数 | 8 |
| 轮次 | 20 |
| 预热轮次 | 0 |
| 批大小（全局） | 128 |
| 分类器头 | 20 (4x5) |
| 分类器学习率 | [5e-3 3e-3 1e-3 3e-4 1e-4] |
| 分类器权重衰减 | [1e-4 1e-3 1e-2 1e-1] |

表 20 EK100：预期探针输入的影响。我们研究了将 V-JEPA 2 编码器、预测器或两者的输出提供给动作预期探针所产生的影响。使用编码器输出即可在 EK100 任务上获得有竞争力的性能。加入预测器后，在动作、动词和对象类别上均带来了小幅但持续的提升。

| | | 动作预判 | | |
|---|---|---|---|---|
| 编码器 | 预测器 | Verb | Noun | 动作 |
| ✓ | | 61.3 | 57.0 | 39.1 |
| | ✓ | 48.7 | 34.7 | 20.2 |
| ✓ | ✓ | 63.6 | 57.1 | 39.7 |

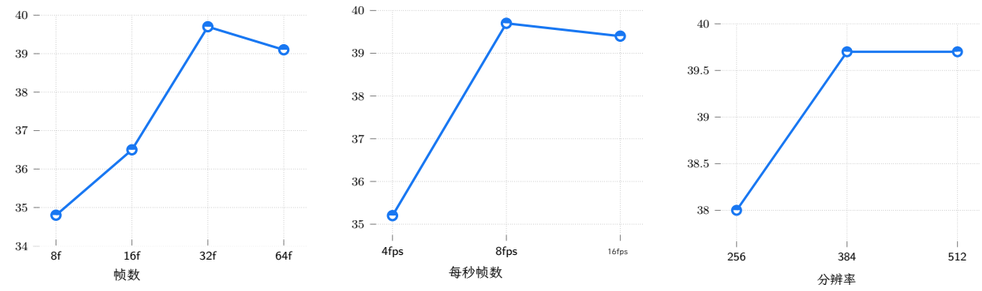



图17 在EK100上进行动作预判的协议消融实验。(左) 用于动作预判的上下文帧数对性能的影响。(中) 推理所用帧率(fps)对性能的影响；上下文帧数固定为32。(右) 用于动作预判的上下文帧空间分辨率（高度和宽度）对性能的影响。

直到某一点后性能饱和或略有下降。最佳性能通过使用32帧上下文长度、8的帧率和 $384 \times 384$的分辨率进行训练获得。

更长时程的预测。我们在图18（左）中报告了通过在(1秒、2秒、4秒、10秒)之间变化预判时间来预测更长时程的影响。对于每个预判时间，我们报告了多个取值(1、5、10、20)下的召回率。结果表明，随着预判时间的增加，性能急剧下降，这是意料之中的，因为在EK100中预测未来是一项非确定性任务。

失败案例分析。我们在图18（右）中报告了在EK100验证集上，动词、名词和动作各自成功/失败配置之间的失败与成功预测分布。模型表现非常好，因此最具代表性的配置是完全成功

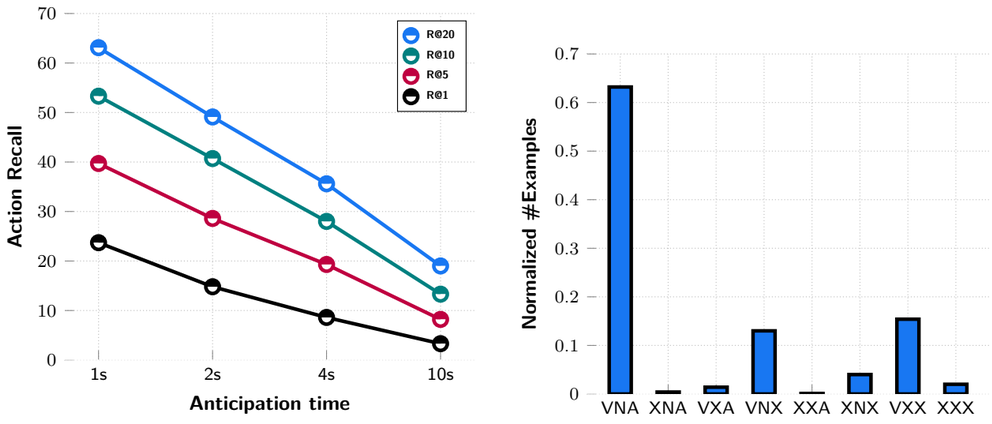



**Figure 18 (Left): Impact of longer-term anticipation times.** Performance on EK100 action anticipation, at several recall values and anticipation times. **(Right): Distribution of success and failure cases of V-JEPA 2.** Calculated on the validation set of EK100. VNA means that verb, noun and action are all correctly classified by the model. An X symbol means that the corresponding attribute is not correctly classified by the model. Note: the probe is composed of 3 independent classifiers for verb, noun and action, hence why the model can have a different prediction for the action and for the verb/noun pair.

across verb noun and action. The most represented failure configurations all include a failure to find the action.

## E Video Question Answering

In this section, we provide details on training a V-JEPA 2 Multi-modal Large Language Model (MLLM). We follow the LLaVA framework (Liu et al., 2024a) to train the MLLM, where the vision backbone is using V-JEPA 2, and the LLM backbone can be any off-the-shelf pretrained LLM, akin to the *non-tokenized early fusion* (Wadekar et al., 2024) setup. The MLLM ingests the output embeddings of the vision encoder, which are projected to the hidden dimension of the LLM backbone using a *projector* module. The *projector* is typically a 2-layer MLP. The MLLM is trained using a mix of image-text and video-text paired data, in a series of progressive training steps.

To understand the impact of data scale, we use a dataset of 88.5 million image-text and video-text pairs, similar to what was used for training PerceptionLM (Cho et al., 2025). As mentioned in Section 7, we investigate two setups: (a) *controlled*, where we train with 18M image and video-text pairs, and we evaluate V-JEPA 2 and other encoders on the exact same MLLM training setup, and (b) *scaling*, where we take V-JEPA 2 VITg$_{384}$ and use the full aligment dataset. To further test the versatility of V-JEPA 2, we use Qwen2-7B-Instruct (Yang et al., 2024a) as the language backbone for the controlled experiments, and Llama 3.1 8B Instruct (Grattafiori et al., 2024) for the scaling experiments. We describe the training details in the following sections.

### E.1 Processing Images and Videos as Input

Since video question answering uses video instead of image inputs, the number of output visual tokens increases significantly compared to image question answering. If required, we can use pooling methods to reduce the number of visual tokens. Popular pooling methods involve adaptive 2x2 pooling (Cho et al., 2025), Perceiver Sampler (Jaegle et al., 2021), Attentive Pooling (Bardes et al., 2024), etc.

Additionally, we observed that learning from image-text pairs is crucial for high performance in downstream



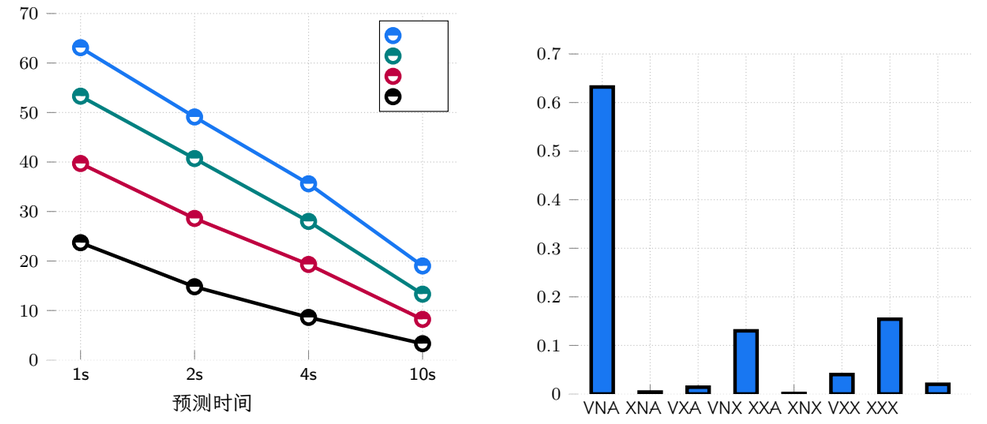



图18（左）：更长预测时间的影响。在EK100动作预测任务上，不同召回值和预测时间下的性能。（右）：V-JEPA 2成功与失败案例的分布。在EK100验证集上计算。VNA表示模型正确分类了动词、名词和动作。X符号表示模型未能正确分类对应属性。注意：探针由3个独立的分类器组成，分别用于动词、名词和动作，因此模型对动作和动词/名词对的预测可能不同。

跨越动词、名词和　　动作。最具代表性的失败配置均包含未能找到t
该动作。

## *E* 视频问答

在本节中，我们提供了训练 V-JEPA 2 多模态大语言模型（MLLM）的详细信息。我们遵循 LLaVA 框架（Liu et al., 2024a）来训练 MLLM，其中视觉骨干网络使用 V-JEPA 2，而 LLM 骨干网络可以是任何现成的预训练 LLM，类似于非tokenized早期融合（Wadekar et al., 2024）的设置。MLLM 接收视觉编码器的输出嵌入，这些嵌入通过投影器模块被投影到 LLM 骨干网络的隐藏维度。投影器通常是一个2层 MLP。MLLM 使用图像-文本和视频-文本配对数据的混合进行训练，采用一系列渐进式训练步骤。

为了解数据规模的影响，我们使用了一个包含8850万图像-文本和视频-文本对的数据集，类似于用于训练 PerceptionLM（Cho et al., 2025）的数据集。如第7节所述，我们研究了两种设置：(a) 受控设置，我们使用1800万图像和视频-文本对进行训练，并在完全相同的 MLLM 训练设置下评估 V-JEPA 2 和其他编码器；(b) 扩展设置，我们采用 V-JEPA 2 VITg$_{384}$ 并使用完整的对齐数据集。为了进一步测试 V-JEPA 2 的多功能性，我们在受控实验中使用 Qwen2-7B-Instruct（Yang et al., 2024a）作为语言骨干网络，在扩展实验中使用 Llama 3.1 8B Instruct（Grattafiori et al., 2024）。我们在以下各节中描述训练细节。

### *E.1* 将图像和视频作为输入进行处理

由于视频问答使用视频而非图像作为输入，与图像问答相比，输出视觉token的数量会显著增加。如有需要，我们可以使用池化方法来减少视觉token的数量。常用的池化方法包括自适应2x2池化（Cho et al., 2025）、Perceiver采样器（Jaegle et al., 2021）、注意力池化（Bardes et al., 2024）等。

此外，我们观察到从图像-文本对中学习对于在下游

benchmarks. In order to train with images, a simple approach is to repeat the given image for $k$ frames, where $k$ is the maximum amount of frames supported by V-JEPA 2. However, during our initial experiments we find this strategy is ineffective at improving downstream performance, as it does not allow the model to extract fine-grained information. Therefore, we employ a modified *Dynamic* $S^2$ strategy introduced by Liu et al. (2024d) to provide V-JEPA 2 higher resolution granularity during training. This method adaptively processes an image at native resolution with different aspect ratios to preserve their original resolution, by creating a sequence of tiles of maximum size supported by V-JEPA 2. In case of videos, we choose to train with a fixed number of frames $f_n$, by balancing the number of visual tokens with compute budget.

## E.2 Controlled Setup

**Training details** For the controlled setup, we follow the LLaVA-NEXT framework (Liu et al., 2024a; Zhang et al., 2024b), where we use Qwen2-7B-Instruct (Yang et al., 2024a) as the base LLM for all encoders. To reduce the number of visual tokens, we employ an attentive pooler with a factor of 4-16, depending on the compute budget and the number of visual patches. See Table 21 for more details.

Our training setup follows the LLaVA-NeXT pipeline (Li et al., 2024b), which consists of multiple staged training phases. Concretely, the stages consist of: a) aligning the attentive pooler with image captioning data (Stage 1), b) training the full model on high quality image captioning (Stage 1.5), and c) training the full model on large scale image question answering (Stage 2). We add an extra stage to train on large scale video captioning and question answering (Stage 3). We use 18 million image and video-text aligned data. The LLM progressively improves its understanding of the visual tokens after multiple staged training, with the biggest improvement in video question answering tasks after Stage 3.

We explore *frozen* and *finetuned* encoder alignment setups. In both setups, full parameters of the LLM and projector are trained, and in the latter, the V-JEPA 2 parameters are additionally unfrozen. To reduce the number of visual tokens and keep the MLLM context length fixed, we employ an attentive pooler as the projector to reduce the number of visual tokens by a factor of 4, unless otherwise denoted. The implementation used for this controlled study is based on the Llava-NEXT codebase,[4] and uses Pytorch 2.5.1, Transformers 4.46.0, Flash attention 2 and DeepSpeed 0.14.4 for model implementation, faster training and multi-gpu model sharding respectively. We train all models using 128 H100 GPUs with an effective batch size of 256 across all stages. We perform all optimizations using AdamW with 0 weight decay. For Stages 1 and 1.5, we use learning rate of 1e-5 with cosine decay, and for Stages 2 and 3 we use constant learning rate of 5e-6. In all stages, we use linear warmup for the first 3% of training steps. Training hyperparameters are listed in Table 21.

**Baselines.** To assess the ability of V-JEPA 2 to capture spatiotemporal details for VidQA, we compare to leading off-shelf image encoders. Specifically, we compare to DINOv2 (Oquab et al., 2023), SigLIP2 (Tschannen et al., 2025), and Perception Encoder (Bolya et al., 2025). DINOv2 is a self-supervised image model, while SigLIP2 and Perception Encoder are both trained with language supervision using noisy image-text captions. We apply all image encoders at their "native" pretrained resolution, which is 518px, 384px, and 448px, respectively, on each video frame independently.

We keep all training details the same, except that we increase the attentive pooling ratio to 16 to keep the number of image tokens relatively similar among models. See Table 21 for details.

**Evaluation.** To evaluate the capability of V-JEPA 2 to understand the world through video and language, we select popular evaluation datasets built to test spatio-temporal reasoning abilities. To ensure reproducible evaluation, we utilize the `lmms-eval` library (Li et al., 2024a; Zhang et al., 2024a) to conduct our experiments, which is a vision model enabled fork of `llm-eval-harness` (Gao et al., 2024), which is a popular evaluation library for evaluating LLMs on text-based tasks. In the controlled setup, for each model and dataset, we evaluate by using uniform frame sampling mechanism, and choosing 128 frames during inference. For PerceptionTest, we further train the model for 5 epochs on the training set.

[4] https://github.com/LLaVA-VL/LLaVA-NeXT



基准测试中取得高性能至关重要。为了使用图像进行训练，一种简单的方法是将给定图像重复 $k$ 帧，其中 $k$ 是 V-JEPA 2 支持的最大帧数。然而，在我们最初的实验中，我们发现这种策略在提升下游性能方面效果不佳，因为它不允许模型提取细粒度信息。因此，我们采用了 Liu 等人（2024d）引入的改进版动态 $S^2$ 策略，以在训练期间为 V-JEPA 2 提供更高的分辨率粒度。该方法通过创建 V-JEPA 2 支持的最大尺寸的瓦片序列，以原生分辨率和不同的宽高比自适应地处理图像，以保持其原始分辨率。对于视频，我们选择使用固定数量的帧 $f_n$ 进行训练，通过在视觉 token 数量和计算预算之间进行平衡。

## *E.2* 受控设置

训练细节 对于受控设置，我们遵循 LLaVA-NEXT 框架（Liu 等，2024a；Zhang 等，2024b），其中我们使用 Qwen2-7B-Instruct（Yang 等，2024a）作为所有编码器的基础 LLM。为减少视觉 token 的数量，我们采用因子为 4-16 的注意力池化器，具体取决于计算预算和视觉图块的数量。更多细节见表 21。

我们的训练设置遵循 LLaVA-NeXT 流程（Li 等，2024b），该流程包含多个分阶段训练阶段。具体而言，这些阶段包括：a) 使用图像描述数据对齐注意力池化器（阶段 1），b) 在高品质图像描述上训练完整模型（阶段 1.5），以及 c) 在大规模图像问答上训练完整模型（阶段 2）。我们增加了一个额外阶段，在大规模视频描述和问答上进行训练（阶段 3）。我们使用了 1800 万图像和视频-文本对齐数据。经过多个分阶段训练后，LLM 逐步提升对视觉 token 的理解，其中在阶段 3 之后视频问答任务的提升最大。

我们探索了冻结编码器与微调编码器两种对齐方案。在两种方案中，LLM 和投影器的全部参数均参与训练，而在后者中，V-JEPA 2 的参数额外解冻。为减少视觉 token 数量并保持 MLLM 上下文长度固定，除非另有说明，我们采用注意力池化器作为投影器，将视觉 token 数量缩减为原来的 1/4。本控制实验的实现基于 Llava-NEXT 代码库，[4] 并分别使用 Pytorch 2.5.1、Transformers 4.46.0、Flash attention 2 和 DeepSpeed 0.14.4 进行模型实现、加速训练和多 GPU 模型分片。我们使用 128 块 H100 GPU 训练所有模型，所有阶段的有效批大小均为 256。我们使用 AdamW 进行所有优化，权重衰减为 0。对于阶段 1 和阶段 1.5，我们使用 1e-5 的学习率并采用余弦衰减；对于阶段 2 和阶段 3，我们使用 5e-6 的恒定学习率。在所有阶段中，我们在前 3% 的训练步数内使用线性预热。训练超参数列于表 21。

基线。为了评估 V-JEPA 2 在 VidQA 中捕捉时空细节的能力，我们与领先的现成图像编码器进行比较。具体而言，我们对比了 DINOv2（Oquab 等，2023）、SigLIP2（Tschannen 等，2025）和 Perception Encoder（Bolya 等，2025）。DINOv2 是一种自监督图像模型，而 SigLIP2 和 Perception Encoder 均使用带噪声的图像-文本描述通过语言监督进行训练。我们对所有图像编码器分别独立地在每个视频帧上应用其"原生"预训练分辨率，分别为 518px、384px 和 448px。

我们保持所有训练细节不变，只是将注意力池化比率提高到 16，以使各模型的图像 token 数量保持相对接近。详情见表 21。

评估。为了评估 V-JEPA 2 通过视频和语言理解世界的能力，我们选择了专为测试时空推理能力而构建的常用评估数据集。为确保评估的可复现性，我们使用 lmms-eval 库（Li et al., 2024a; Zhang et al., 2024a）进行实验，该库是 llm-eval-harness（Gao et al., 2024）的视觉模型支持分支，而后者是一个用于在基于文本的任务上评估 LLM 的常用评估库。在受控设置中，对于每个模型和数据集，我们采用均匀帧采样机制进行评估，并在推理时选择 128 帧。对于 PerceptionTest，我们进一步在训练集上对模型训练了 5 个 epoch。

[4] https://github.com/LLaVA-VL/LLaVA-NeXT

**Table 21** **Hyperparameters for controlled comparison of vision encoders. We use each vision encoder with its native pretrained input resolution.**

| Model | Pooling Ratio | Vision Tokens | BS | Stage 1/1.5 LR | Stage 2/3 LR | WD | Frames |
|---|---|---|---|---|---|---|---|
| V-JEPA 2 ViT-$L_{256}$ | 4 | 4096 | 256 | 1e-5 | 5e-6 | 0.0 | 128 |
| V-JEPA 2 ViT-$H_{256}$ | 4 | 4096 | 256 | | | | |
| V-JEPA 2 ViT-$g_{256}$ | 4 | 4096 | 256 | | | | |
| V-JEPA 2 ViT-$g_{384}$ | 4 | 9216 | 256 | | | | |
| V-JEPA 2 ViT-$g_{512}$ | 8 | 8192 | 256 | | | | |
| $DINOv2_{518}$ | 16 | 10952 | 256 | 1e-5 | 5e-6 | 0.0 | 128 |
| $SigLIP2_{384}$ | 16 | 5832 | 256 | | | | |
| $PE_{448}$ | 16 | 8192 | 256 | | | | |

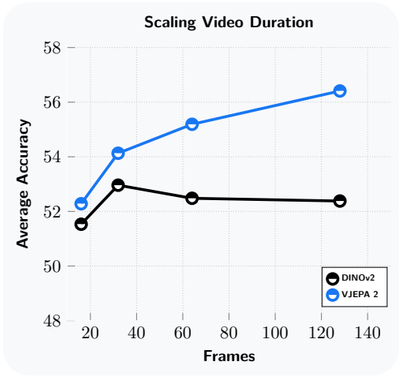



**Figure 19** **Impact of video duration during visual instruction tuning.** We investigate the effect of increasing the number of frames during visual instruction tuning, where the encoder is frozen. We observe that with more frames, V-JEPA 2 performance linearly increases compared to DINOv2, an SSL-based image encoder, showing the potential of V-JEPA 2 to scale with more frames.

**Impact of Video Duration** In the controlled setup, we perform an analysis to understand V-JEPA 2's capability in long-form video understanding. We train MLLMs on V-JEPA 2 and DINOv2, keeping the encoders frozen, and by increasing the number of frames we use in training and testing. We observe as the number of frames increases, performance on downstream tasks *linearly* improves for V-JEPA 2, but decreases and remains flat in case of DINOv2 (Figure 19). This highlights the potential of video encoders such as V-JEPA 2 to understand long-form videos with natural language queries, via adapting an LLM using V-JEPA 2 as the visual encoder.

## E.3 Data scaling setup

**Training details** In the scaling setup, we follow the framework used by Cho et al. (2025) to train Perception LM 8B. Specifically, we utilize the released codebase, which is based on Lingua (Videau et al., 2024). We modify the code to use V-JEPA 2 encoder, and we use the Llama 3.1 8B Instruct (Grattafiori et al., 2024) as the backbone LLM. Unlike Cho et al. (2025), we do not use pooling, instead we train V-JEPA 2 VIT-$g_{384}$ using MLP projector, leading to 288 tokens per frame. The training setup also consists of three progressive stages: Stage 1: aligning the MLP pooler with image captioning data; Stage 2: training on a mix of image-text captioning and QA data; and Stage 3) training on video-text captioning and QA data. We scale up the data size to 88.5 million samples. Our setup uses Pytorch 2.5.1 and Perception LM training code,[5] modified with the V-JEPA 2 encoder. We train on 512 H100 GPUs for Stage 2 and Stage 3 with a global batch size of 2048 and 1024 respectively. Details of the training hyperparams are provided in Table 22.

[5] https://github.com/facebookresearch/perception_models



表 ***21*** 超参数及其原生预 用于视觉编码器受控比较的参数。我们使用每个视觉编码器 r 训练输入分辨率。

| 模型 | 池化比率 | 视觉Token数 | BS | 阶段1/1.5学习率 | 第2/3阶段学习率 | WD | 帧 |
|---|---|---|---|---|---|---|---|
| V-JEPA 2 ViT-$L_{256}$ | 4 | 4096 | 256 | 1e-5 | 5e-6 | 0.0 | 128 |
| V-JEPA 2 ViT-$H_{256}$ | 4 | 4096 | 256 | | | | |
| V-JEPA 2 ViT-$g_{256}$ | 4 | 4096 | 256 | | | | |
| V-JEPA 2 ViT-$g_{384}$ | 4 | 9216 | 256 | | | | |
| V-JEPA 2 ViT-$g_{512}$ | 8 | 8192 | 256 | | | | |
| $DINOv2_{518}$ | 16 | 10952 | 256 | 1e-5 | 5e-6 | 0.0 | 128 |
| $SigLIP2_{384}$ | 16 | 5832 | 256 | | | | |
| $PE_{448}$ | 16 | 8192 | 256 | | | | |

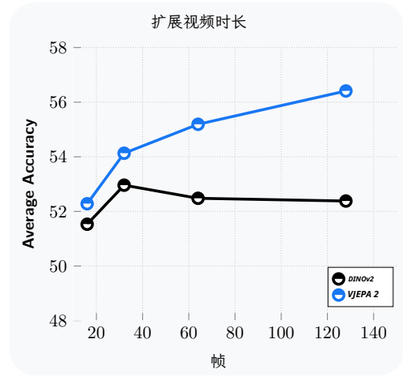



图 19：视觉指令微调过程中视频时长的影响。我们研究了在视觉指令微调过程中增加帧数的效果，其中编码器处于冻结状态。我们观察到，随着帧数增加，V-JEPA 2 的性能相比 DINOv2（一种基于自监督学习的图像编码器）呈线性提升，这表明 V-JEPA 2 具有随帧数扩展的潜力。

视频时长的影响 在受控设置中，我们进行了一项分析，以了解 V-JEPA 2 在长视频理解方面的能力。我们在 V-JEPA 2 和 DINOv2 上训练 MLLM，保持编码器冻结，并增加训练和测试中使用的帧数。我们观察到，随着帧数的增加，V-JEPA 2 在下游任务上的性能呈线性提升，而 DINOv2 的性能则下降并保持平坦（图 19）。这凸显了 V-JEPA 2 等视频编码器通过使用 V-JEPA 2 作为视觉编码器适配 LLM，以自然语言查询理解长视频的潜力。

## *E.3* 数据扩展设置

训练细节 在扩展设置中，我们遵循Cho等人（2025）用于训练Perception LM 8B的框架。具体而言，我们使用了已发布的代码库，该代码库基于Lingua（Videau等人，2024）。我们修改了代码以使用V-JEPA 2编码器，并使用Llama 3.1 8B Instruct（Grattafiori等人，2024）作为骨干LLM。与Cho等人（2025）不同，我们不使用池化，而是使用MLP投影器训练V-JEPA 2 VIT-$g_{384}$，每帧产生288个token。训练设置同样包含三个渐进阶段：阶段1：使用图像描述数据对齐MLP池化器；阶段2：在图像-文本描述和QA数据的混合数据上训练；阶段3：在视频-文本描述和QA数据上训练。我们将数据规模扩大到8850万个样本。我们的设置使用Pytorch 2.5.1和Perception LM训练代码，[5] 并修改为使用V-JEPA 2编码器。我们在512块H100 GPU上分别以2048和1024的全局批量大小训练阶段2和阶段3。训练超参数的详细信息见表22。

[5] https://github.com/facebookresearch/perception_模型

**Table 22** **Data scaling training parameters.**

| Parameter | Values |
|---|---|
| *Common parameters* | |
| Crop Size | 384 |
| Video Frames per Second | 1 |
| Sampling method | Uniform |
| Seed | 777 |
| *Stage 1* | |
| Steps | 16000 |
| Warmup Steps | 96 |
| Batch Size (global) | 128 |
| Learning Rate | 1e-4 |
| Final Learning Rate | 1e-6 |
| Weight Decay | 0.05 |
| Max sequence length | 1920 |
| *Stage 2* | |
| Steps | 35000 |
| Warmup Steps | 200 |
| Batch Size (global) | 2048 |
| Learning Rate | 4e-5 |
| Final Learning Rate | 4e-7 |
| Weight Decay | 0.05 |
| Max sequence length | 6400 |
| Image tiles | 16 |
| Video frames | 16 |
| *Stage 3* | |
| Steps | 28000 |
| Early stopping step | 22000 |
| Warmup Steps | 168 |
| Batch Size (global) | 2048 |
| Learning Rate | 1e-5 |
| Final Learning Rate | 1e-7 |
| Weight Decay | 0.05 |
| Max sequence length | 12800 |
| Image tiles | 32 |
| Video frames | 32 |

**Baselines.** We compare our scaling runs with Qwen2VL (Wang et al., 2024a), Qwen2.5VL (Qwen Team et al., 2025), InternVL-2.5 (Chen et al., 2024), and PerceptionLM 8B (Cho et al., 2025). Baseline numbers are sourced directly from the papers, except for MVP which we run ourselves.

**Evaluation.** We follow similar evaluation pipeline as reported in the controlled setup, using `lmms-eval` library. We report our model evaluations on 32 frames.



表 ***22*** 数据缩放训练参数。

| 参数 | 值 |
|---|---|
| 通用参数 | |
| 裁剪尺寸 | 384 |
| 视频帧率 | 1 |
| 采样方法 | 均匀 |
| Seed | 777 |
| 阶段 1 | |
| 步数 | 16000 |
| 预热步数 | 96 |
| 批大小（全局） | 128 |
| 学习率 | 1e-4 |
| 最终学习率 | 1e-6 |
| 权重衰减 | 0.05 |
| 最大序列长度 | 1920 |
| 第二阶段 | |
| 步数 | 35000 |
| 预热步数 | 200 |
| 批大小（全局） | 2048 |
| 学习率 | 4e-5 |
| 最终学习率 | 4e-7 |
| 权重衰减 | 0.05 |
| 最大序列长度 | 6400 |
| 图像瓦片 | 16 |
| 视频帧 | 16 |
| 阶段 3 | |
| 步数 | 28000 |
| 早停步数 | 22000 |
| 预热步数 | 168 |
| 批大小（全局） | 2048 |
| 学习率 | 1e-5 |
| 最终学习率 | 1e-7 |
| 权重衰减 | 0.05 |
| 最大序列长度 | 12800 |
| 图像块 | 32 |
| 视频帧 | 32 |

基线。我们将我们的扩展运行与Qwen2VL（Wang等人，2024a）、Qwen2.5VL（Qwen Team等人，2025）、InternVL-2.5（Chen等人，2024）以及PerceptionLM 8B（Cho等人，2025）进行比较。基线数值直接来源于原论文，MVP除外，该基线由我们自行运行。

评估。我们遵循与受控设置中报告的相似的评估流程，使用 lmms-eval 库。我们报告了模型在 32 帧上的评估结果。